Space Mission Engineering: The New SMAD

edited by:

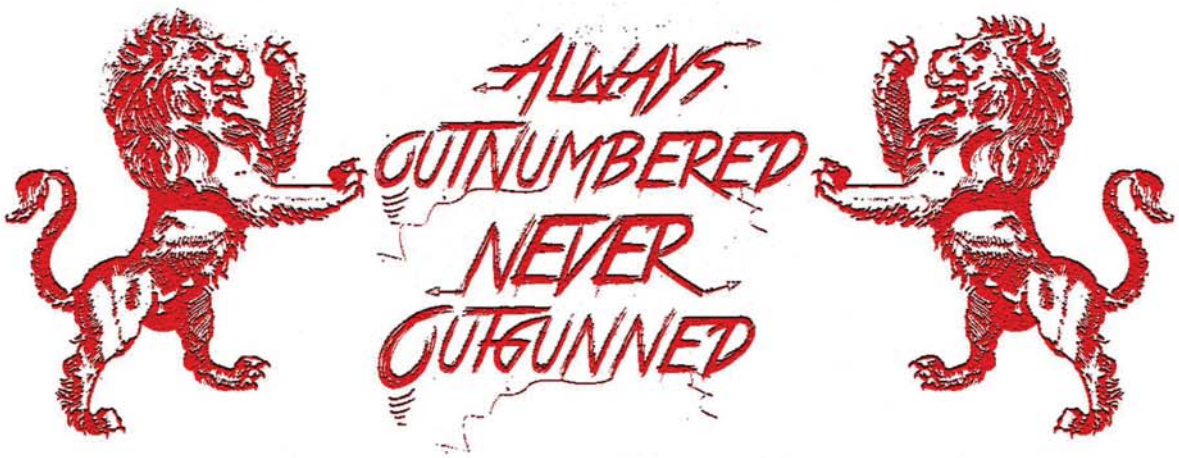
James R. Wertz
Microcosm
University of Southern California

David F. Everett
NASA Goddard
Space Flight Center

Jeffery J. Puschell
Raytheon



Space
Technology
Library



In memory of John

Process Tables and Technical Budgets

Space Mission Engineering	Source	Page
Space Mission Engineering Process	Table 3-1	46
Process for Identifying Alternative Mission Architectures	Table 4-2	63
Defining Mission Concepts	Table 4-7, 4-9	67, 76
Concept Exploration Process	Fig. 4-5	75
Baseline Mission Selection and Evaluating Alternatives	Table 5-11	98
Principal Areas for Key System Trades	Table 5-2	87
Requirements Definition Process	Table 6-1	106
Mission Geometry Evaluation Process	Tables 8-1, 8-2, D-1	149, 150, 973–978
Coverage Formulas	Tables 8-11, 8-12, App. C	184, 186, 963–972
Orbits		
Single Satellite Orbit Selection Process	Table 10-1	236
Constellation Design Process	Table 10-24	270
Keplerian Orbit Formulas	Table 9-4, App. C	205, 963–972
Interplanetary Orbit Design Process	Table 10-30	279
Hohmann Transfer Summary Procedure	Table 10-32	282
Mission Design Considerations for Orbital Debris	Table 7-3	140
Error Budget Development	Table 6-5	115
Cost Estimation Process	Table 11-3	294
Principal Elements of Mission Timeline	Table 4-8	74

Spacecraft Design Elements	Source	Page
Budgets		
Delta V Budget	Table 10-7	252
Link Budget	Table 16-12	479
Observational Payload Budget	Table 16-1	455
Mapping and Pointing Error Budgets	Table 8-15	192
Mass Budget	Table 14-18	422
Power Budget	Table 14-20	424
Propellant Budget	Table 14-19	423
Design Drivers	Tables 4-16, 14-2	80, 398
Manufacturing, Integration, and Test Process	Table 23-1	712

Subsystems Design Process	Source	Page
Attitude Determination and Control System	Table 19-1	567
Tracking, Telemetry, and Command System	Table 21-1	628
Computer System	Table 20-1	602
Software Sizing Process	Table 20-8	613
Guidance and Navigation	Table 19-17	592
Mechanisms	Table 22-5	681
Payloads	Table 15-6	446
Communications Payloads	Tables 16-1, 16-4, 16-14	455–481
Observation Payloads	Table 17-6	509
Optical Sensor Design Calculations	Table 17-9	518–520
Power System	Table 21-10	642
Propulsion System	Table 18-1	528
Structures	Table 22-1	663
Thermal System	Fig. 22-18	691

Launch and Operations	Source	Page
Launch Cost Estimation Process	Table 10-10	257
Launch Vehicle Selection Process	Fig. 26-4	830
Launch Operations Timeline	Figs. 27-2, 27-8	864, 874
Mission Operations		
Process Flow	Fig. 29-2	910
Planning and Scheduling Process	Fig. 29-5	925
Development Process	Table 29-4	914
Realtime Flight Operations Process	Fig. 29-6	927
Data Processing and Management Process	Fig. 29-8	930
Anomalies Detection Process	Fig. 29-7	929

Units

Symbol	Measurement
A	ampere
AU	astronomical unit
C	coulomb
C	celsius
dB	decibel
deg	degree (angular)
eV	electron volt
Hz	hertz
g	gram
J	joule
K	kelvin
kg	kilogram
km	kilometer
m	meter
N	newton
Ω	ohm
rad	radian
s	second
sec	second
sr	steradian
Pa	pascal
T	tesla
V	volt
W	watt
yr	year

The Greek Alphabet		
Alpha	A	α
Beta	B	β
Gamma	Γ	γ
Delta	Δ	δ
Epsilon	E	ε
Zeta	Z	ζ
Eta	H	η
Theta	Θ	θ
Opta	I	ι
Kappa	K	κ
Lambda	Λ	λ
Mu	M	μ
Nu	N	ν
Xi	Ξ	ξ
Omicron	O	o
Pi	Π	π
Rho	P	ρ
Sigma	Σ	σ
Tau	T	τ
Upsilon	Y	υ
Phi	Φ	ϕ
Chi	X	χ
Psi	Ψ	ψ
Omega	Ω	ω

Space Mission Engineering: The New SMAD

Edited By

James R. Wertz
Microcosm
University of Southern California

David F. Everett
NASA Goddard Space Flight Center

Jeffery J. Puschell
Raytheon



Space Technology Library

Published by

Microcosm Press

Hawthorne, CA

Library of Congress Cataloging-in Publication Data

A C.I.P. Catalogue record for this book is available from the Library of Congress.

ISBN 978-1-881-883-15-9 (pb) (acid-free paper)
ISBN 978-1-881-883-16-6 (hb) (acid-free paper)

Cover photo of Earth from Space: View of Africa and the Indian Ocean taken in Dec. 1972, by Apollo 17, the last of the Apollo missions to explore the Moon. Photo courtesy of NASA.

Cover design by Chris Maruyama

Published by
Microcosm Press
4940 W. 147th Street,
Hawthorne, CA 90250 USA

Sold and distributed worldwide by
Microcosm Astronautics Books
4940 W. 147th Street,
Hawthorne, CA 90250 USA

Printed on acid-free paper

All Rights Reserved
©2011 Microcosm Press
Second Printing, 2015

No part of the material protected by this copyright notice may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying, recording, or by any information storage and retrieval system, without written permission from the copyright owner.

Printed in the United States of America

Table of Contents

List of Authors and Reviewers ix

Preface xiv

Part I—Space Mission Engineering

1. Introduction 1

1.1 What is Space Mission Engineering? 1
1.2 The History of Spaceflight 4
1.3 Spaceflight Technology 10
1.4 Spaceflight Economics 16
1.5 The Wide Range of Space Mission Applications 23
1.6 Sources of More Information 24

2. Space Mission Communities 29

2.1 Multiple Space Communities 29
2.2 Differences and Similarities Between Communities 41
2.3 Changing Missions 42

3. Space Mission Engineering 45

3.1 The Space Mission Engineering Process 45
3.2 FireSat II and the Supplemental Communications System (SCS) 50
3.3 Mission Objectives and Constraints (Step 1) 51
3.4 Principal Players and Program Timescales (Steps 2 and 3) 52
3.5 Preliminary Estimate of Mission Needs, Requirements and Constraints (Step 4) 55

4. Mission Concept Definition and Exploration 61

4.1 Defining Alternative Mission Architectures (Step 5)—Choosing the Pieces 61
4.2 Defining Alternative Mission Concepts (Step 6)—How the Pieces Work Together 66
4.3 Introduction to Concept Exploration 75
4.4 Defining System Drivers and Critical Requirements (Step 7) 80

5. Mission Analysis and Mission Utility 83

5.1 Introduction to Mission Analysis 84
5.2 Studies with Limited Scope 85
5.3 System Trade Studies and Performance Assessments (Step 8) 87
5.4 Mission Utility and Figures of Merit—Is the Mission Worthwhile? (Step 9) 91
5.5 Defining the Baseline Mission Concepts, Revising Requirements and Evaluating Alternatives (Steps 10–12) 97
5.6 Examples: FireSat II and SCS 99
5.7 Deciding Whether a Mission Should Proceed 101

6. Formal Requirements Definition 105

6.1 The Requirements Definition Process 106
6.2 Budgeting, Allocation, and Flow-Down 110
6.3 Introduction to Error Analysis 112
6.4 Specifications and Requirements Documentation 116
6.5 System Engineering Tools 117
6.6 The Role of Standards in Space Systems Development 119
6.7 Are Requirements Needed?—Capability-Based vs. Requirements-Based Systems 123

7. The Space Environment 127

7.1 The Space Environment and Space Weather 127
7.2 The Earth's Magnetic Field 131
7.3 Radiation Belts 134
7.4 Microgravity 137
7.5 Orbital Debris—A Space Hazard 139

8. Space Mission Geometry 149

8.1 Introduction to Space Mission Geometry 150
8.2 Applications 162
8.3 Looking at the Earth from Space 171
8.4 Computing Parameters for a Single Target or Ground Station Pass 180
8.5 Satellite Relative Motion 185
8.6 Mapping and Pointing Budgets 191

9. Orbits and Astrodynamics 197
9.1 Keplerian Orbits 198
9.2 Orbits of the Moon and Planets 205
9.3 Spacecraft Orbit Terminology 208
9.4 Orbit Perturbations, Geopotential Models, and Satellite Decay 209
9.5 Specialized Orbits 217
9.6 Orbit Maneuvers 226
9.7 Summary—The Rules of Practical Astrodynamics 233

10. Orbit and Constellation Design—Selecting the Right Orbit 235
10.1 The Orbit Selection and Design Process 235
10.2 Orbit Performance—Evaluating Earth Coverage and Payload Performance 239
10.3 Orbit Cost—Delta V Budget and the Orbit Cost Function 251
10.4 Selecting Earth-Referenced Orbits 258
10.5 Selecting Transfer, Parking, and Space-Referenced Orbits 265
10.6 Summary of Constellation Design 269
10.7 Design of Interplanetary Orbits 277

11. Cost Estimating 289
11.1 Introduction to Cost Estimating 290
11.2 Estimating Tools 294
11.3 Other Considerations in the Cost Estimate 315
11.4 Example Space Mission Cost Estimates 318

12. Space System Financing and Space Law 325
12.1 Sources of Space Financing 325
12.2 GAAP, Amortization and Return on Investment (ROI) 328
12.3 Law and Policy Considerations 337

13. Reducing Space Mission Cost and Schedule 355
13.1 The Need to Reinvent Space 356
13.2 It's Possible, but It Isn't Easy 362
13.3 Counterproductive Approaches to Reducing Cost 365
13.4 Cost vs. Reliability: Focusing on the Mission Objectives 366
13.5 Principal Methods for Reducing Cost and Schedule 375
13.6 Avoiding Cost and Schedule Overruns 390

Part II—Spacecraft and Payload Design

14. Overview of Spacecraft Design 397
14.1 The Spacecraft Design Process 397
14.2 Spacecraft System Design Drivers 398
14.3 Spacecraft Configuration Alternatives 401
14.4 Partitioning Spacecraft into Subsystems 411
14.5 Creating Preliminary Spacecraft Budgets 421
14.6 Design Evolution 426
14.7 Examples 432
14.8 Future of Spacecraft Design 437

15. Overview of Payload Design 439
15.1 Types of Space Payloads 439
15.2 Mission System Concept or Subject Trades—What is the System Measuring or Working With? 441
15.3 Payload Design 445
15.4 The Electromagnetic Spectrum 448
15.5 Examples 450

16. Communications Payloads 455
16.1 Space Mission Communications Architectures 455
16.2 Communication Link Analysis 466
16.3 Communications Payload Design 481
16.4 Sample Missions 491

17. Observation Payloads 493
17.1 Observation Payload Design 494
17.2 Observation Payload Sizing 508
17.3 Sample Mission—VIIRS 523
17.4 The Evolution of Observation Payloads 525

18. Spacecraft Subsystems I—Propulsion 527
18.1 Basic Rocket Equations 529
18.2 Staging 532
18.3 Chemical Propulsion Systems 533
18.4 Plume Considerations 540
18.5 System Design Elements 541
18.6 Electric Propulsion 546
18.7 Alternative Propulsion Systems for In-Space Use 553
18.8 Examples 560

19. Spacecraft Subsystems II—Control Systems 565

19.1 Spacecraft Attitude Determination and Control Systems 565
19.2 Spacecraft Trajectory Navigation and Control Systems 591

20. Spacecraft Subsystems III—On-Board Processing 601

20.1 Computer System Baseline 602
20.2 Preliminary Design 611
20.3 FireSat II Example 619
20.4 Modular Approaches to Processing 622

21. Spacecraft Subsytems IV—Communications and Power 627

21.1 Telemetry, Tracking, and Command (TT&C) 627
21.2 Power 641

22. Spacecraft Subsystems V—Structures and Thermal 663

22.1 Spacecraft Structures and Mechanisms 663
22.2 Spacecraft Thermal Control 685

23. Space Logistics and Manufacturing 701

23.1 LEO Communications Constellations .. 701
23.2 LEO Monolithic vs. Distributed Architectures 704
23.3 Spacecraft Manufacturing Integration and Test 711
23.4 System Mission Verification and Validation 718
23.5 Multi-Spacecraft Manufacturing 726
23.6 Alternative Approaches to Space Manufacturing 737
23.7 Intangible Factors in Manufacturing 743

24. Risk and Reliability 753

24.1 Reliability 753
24.2 Space System Risk Analysis 768

25. Alternative Spacecraft Designs 781

25.1 Space Tethers 781
25.2 Inflatable Structures 786
25.3 SmallSats 791
25.4 CubeSats 803
25.5 Differences Between International Approaches to Space 821

Part III—Launch and Operations

26. Launch Vehicles 829

26.1 Launch Vehicle Selection 830
26.2 History Prior to 2010 841
26.3 Basic Mechanics of Launch 843
26.4 Launch Environments 847
26.5 Available Vehicles 858

27. Launch Operations 861

27.1 Worldwide Launch Sites and Launch Restrictions 861
27.2 Launch Site Preparations 863
27.3 Readiness Reviews and Mission Dress Rehearsals 870
27.4 Launch Site Access 871
27.5 Launch Site Training 871
27.6 Transporting the Spacecraft to the Launch Site 872
27.7 Launch Site Processing 872
27.8 Launch Day 874
27.9 Post Launch and Early Orbit Operations 875
27.10 Modernizing Launch Operations 876
27.11 Common Mistakes to Avoid 877

28. Ground System Design 879

28.1 Antenna Services 880
28.2 Data Accounting and Distribution Services 892
28.3 Ground System Driving Requirements and Sizing 896
28.4 Mission Examples 898
28.5 Technology Trends 899
28.6 Summary 902

29. Mission Operations 903

29.1 Mission Planning and Operations Development 909
29.2 Mission Execution 923
29.3 Mission Termination and Post-Mission Activities 931
29.4 Mission Operations Process Improvement and Best Practices 932
29.5 The Future of Mission Operations 934

30. End of Mission Considerations 937

30.1 Inter-Agency Space Debris Coordination Committee (IADC) End of Mission Guidelines 938

30.2 Low Earth Orbit (LEO) Disposal Options 939

30.3 Non-LEO Disposal Options 941

30.4 Passivation 942

30.5 Disposal Planning 944

30.6 FireSat II and SCS Examples 945

Appendices

A Mass and Power Distribution for Spacecraft 947

B Physical and Orbit Properties of the Sun, Earth, Moon, and Planets 955

C Summary of Keplerian Orbit and Coverage Equations 963

D Mission Geometry Formulas 973

E Time and Date Systems 979

F Coordinate Transformations; Vector, Matrix, and Quaternion Algebra 983

G Statistical Error Analysis ..... Web only

H Units and Conversion Factors 985

I Earth Satellite Parameters 1025

Authors

Henry Apgar. President, MCR Technologies, LLC, El Segundo, California. MBA, Northeastern University; BS (Electrical Engineering), Rutgers University. Chapter 11—*Cost Estimating*.

Matthew Ashmore. Senior Mechanisms Engineer. ATK Space Systems and Services, Beltsville, Maryland. BS (Aerospace Engineering), University of Maryland. Chapter 22.1.5—*Mechanisms and Deployables*.

Karl D. Bilimoria. Aerospace Engineer, NASA Ames Research Center, Moffett Field, California. PhD (Aerospace Engineering), Virginia Tech; MS (Aerospace Engineering), Virginia Tech; BS (Aeronautical Engineering), Indian Institute of Technology–Kanpur. Section 19.2—*Spacecraft Trajectory Navigation and Control Systems*.

Robert S. Bokulic. Principal Professional Staff, Johns Hopkins University / Applied Physics Laboratory (JHU/APL), Laurel, Maryland. MS (Electrical Engineering), Johns Hopkins University; BS (Electrical Engineering), Virginia Polytechnic Institute and State University. Section 21.1—*Telemetry, Tracking, and Command (TT&C)*.

Constantine Cassapakis. CEO ex Officio, L'Garde Inc., Tustin, California. PhD (Experimental Particle Physics), The University of New Mexico/Los Alamos Science Lab; MS (Physics), The University of New Mexico; MA (Mathematics), Eastern New Mexico University; BS (Physics and Mathematics), Eastern New Mexico University. Section 25.2—*Inflatable Structures*.

Becky Christofferson. Systems Engineer, Microcosm Inc., Hawthorne, California. BS, Aerospace Engineering, University of Southern California. Appendix I —*Earth Satellite Parameters*.

John T. Collins. Senior System Engineer, Microcosm, Inc., Hawthorne, California. BS (Astronomy), University of Illinois; BS (Aerospace Engineering) University of Illinois. Section 11.4—*Example Space Missions Estimates*.

Christopher C. DeBoy. Group Supervisor, RF Engineering, Space Department, Johns Hopkins University/Applied Physics Laboratory (JHU/APL), Laurel, Maryland. MS (Electrical Engineering), Johns Hopkins University; BS (Electrical Engineering), Virginia Polytechnic Institute and State University. Section 21.1 —*Telemetry, Tracking, and Command (TT&C)*.

Scott W. Enger. Electrical Engineer, Senior Staff, Lockheed Martin, Littleton, Colorado. BS (Electrical Engineering, Colorado State University. Section 21.2— *Power*.

Jaime Esper. Senior Aerospace Engineer, NASA Goddard Space Flight Center, Greenbelt, Maryland. MS (Mechanical [Aerospace] Engineering), The George Washington University; MS (Astronomy), University of Florida; BS (Physics), University of Florida. Section 23.6—*Alternative Approaches to Space Manufacturing*.

John S. Eterno. Program Director for Systems Engineering, Space Science Department, Space Science and Engineering Division, Southwest Research Institute, San Antonio, TX. PhD (Aeronautics and Astronautics), Massachusetts Institute of Technology; MS (Aeronautics and Astronautics), Massachusetts Institute of Technology; BS (Fluid, Thermal, and Aerospace Sciences), Case Western Reserve University. Section 19.1—*Spacecraft Attitude Determination and Control Systems*.

David F. Everett. Senior Systems Engineer, NASA Goddard Space Flight Center, Greenbelt, Maryland. MS (Electrical Engineering), University of Maryland; BS (Electrical Engineering), Virginia Polytechnic Institute and State University. Chapter 14—*Overview of Spacecraft Design;* Appendix A—*Mass and Power Distribution for Spacecraft*.

Stuart Eves. Principal Engineer, Surrey Satellite Technology Limited, Guildford, Surrey, United Kingdom. PhD (Constellation Design), Cranfield University; MSc (Astrophysics), London University; BSc (Physics with Astrophysics), London University. Section 23.1 —*LEO Communications Constellation*; Section 23.2 —*LEO Monolithic vs. Distributed Architectures*.

Rick Fleeter. Associate Professor, Brown University, Providence, RI; Visiting Professor, University of Rome. PhD (Thermodynamics), Brown University; MSc (Astronautics), Stanford University; AB (Engineering and Economics), Brown University. Section 25.3 —*SmallSats*.

Donald W. Gates, Jr. Senior Systems Engineer, Boeing, Lanham, Maryland. BS (Information Systems Management), University of Maryland. Chapter 27—*Launch Operations*.

Paul Graven. Director of Technology Programs, Microcosm, Inc., Hawthorne, California. M.P.P. (Science & Technology Policy and National Security Policy), Harvard University, JFK School of Government; MS (Aeronautics & Astronautics), Stanford University; BS (Engineering & Applied Science and Economics), California Institute of Technology. Section 20.4—*Modular Approaches*.

Mike Gruntman. Professor of Astronautics and Professor of Aerospace Engineering, Viterbi School of Engineering, University of Southern California, Los Angeles, California. PhD (Physics), Space Research Institute (IKI), USSR Academy of Sciences; MS (Physics), Moscow Physical-Technical Institute. Section 1.2—*The History of Spaceflight*.

L. Jane Hansen. President, HRP Systems, Inc., Rolling Hills Estate, California. MBA, Pepperdine University; BS (Applied Mathematics), California Polytechnic State University at San Luis Obispo. Chapter 20— *Spacecraft Subsystems III—On-Board Processing*.

William A. Hargus, Jr. Micro Propulsion Program Manager, Air Force Research Lab, Spacecraft Propulsion, Edwards AFB, California. PhD (Mechanical), Stanford University; MS (Aerospace), Pennsylvania State University; BS (Aerospace), University of Texas; Associate Fellow of the American Institute of Astronautics and Aeronautics; Associate Editor of the JANNAF Journal of Propulsion and Energetics. Section 18.6—*Electric Propulsion*.

Brian Harvey. Spaceflight Writer, Dublin, Ireland. M.A. (Economic and Social History), University College, Dublin; BA (History and Political Science), Dublin University (Trinity College). Section 25.5—*Differences Between International Approaches to Space*.

Herbert Hecht. Chief Engineeer, SoHaR Inc., Culver City, California. PhD (Engineering), University of California Los Angeles; MS (Electrical Engineering), Polytechnic Institute of Brooklyn; BS (Electrical Engineering, City University, New York. Section 24.1—*Reliability*.

Myron J. Hecht. Senior Project Leader, The Aerospace Corporation, El Segundo, California. J.D., University of California Los Angeles; MBA, University of California Los Angeles; MS (Nuclear Engineering), University of California Los Angeles; BS (Chemistry), University of California Los Angeles. Section 24.1—*Reliability*.

Robert Hoyt. President, CEO, and Chief Scientist, Tethers Unlimited, Inc., Bothell, Washington. PhD (Aero/Astro), University of Washington; B.A (Physics), Williams College. Section 25.1—*Space Tethers*.

Scott Hull. Orbital Debris Engineer, NASA Goddard Space Flight Center, Greenbelt, Maryland. BS, (Materials Engineering), Drexel University. Chapter 30—*End of Mission Considerations*.

Michael S. Hurley, Jr. Spacecraft Development Secction Head, Naval Research Lab, Washington, DC. M.B.A, George Mason University; ME (Mechanical Engineering), Virginia Tech; BS (Engineering), Virgina Tech. Section 13.4—*Cost vs. Reliability—Focusing on the Mission Objectives*.

John M. Jurist. Adjunct Professor of Space Studies, University of North Dakota, Grand Forks; PhD (Biophysics), University of California, Los Angeles Medical School; MS (Biophysics and Nuclear Medicine), University of California, Los Angeles Medical School; AB (Physics), University of California, Los Angeles. Section 1.4—*Spaceflight Economics*; Section 12.1—*Sources of Space Financing*; Section 12.2—*GAAP Amortization and Return on Investment*.

Christopher Lashley. Mechanisms Group Manager, ATK Spacecraft Systems and Services, Beltsville, Maryland. MS University of Maryland; BS University of Maryland. Chapter 22.1.5 — *Mechanisms and Deployables*.

David E. Lee. Program Manager and Lead, Integrated Concept Development Facility, Northrop Grumman Aerospace Systems, Redondo Beach, California. PhD (Mechanical Engineering), University of California, Los Angeles; MS (Manufacturing Engineering), University of California, Los Angeles; BA (Mathematics), University of California, Los Angeles. Section 23.3—*Spacecraft Integration and Test*.

Ivett A. Leyva. Lead, Combustion Devices Group, Aerophysics Branch, Air Force Research Laboratory, Edwards Air Force Base, California. PhD (Aeronautics), Caltech; MS (Aeronautics), Caltech; BS (Engineering and Applied Science), Caltech. Chapter 18—*Spacecraft Subsystems I—Propulsion*.

David Livingston. Host of The Space Show and Executive Director of One Giant Leap Foundation, Inc., Tiburon, California. DBA, Golden Gate University; MBA, (International Business Management), Golden Gate University; BA (Political Science), University of Arizona; Adjunct Professor of Space Policy, Odegard School of Aerospace Sciences, University of North Dakota. Section 1.4—*Spaceflight Economics*; Section 12.1—*Sources of Space Financing*; Section 12.2—*GAAP Amortization and Return on Investment*.

Ron Madler. Professor and Dean of Engineering, Embry-Riddle Aeronautical University, Prescott, Arizona. PhD, MS, BS (Aerospace Engineering Science), University of Colorado. Section 7.5— *Orbital Debris*.

Benjamin Malphrus. Chair, Department of Earth and Space Sciences, Morehead State University Space Science Center, Morehead, KY; EdD (Physics and Astronomy), West Virginia University; MAT (Physics), University of South Carolina-Columbia; BA (Art), University of South Carolina-Columbia. Section 25.4—*CubeSats*.

Joseph K. McDermott. Senior Manager, Electrical Power Systems, Lockheed Martin Space Systems Company, Denver, Colorado. ME (Engineering Management), University of Colorado; BS (Chemistry), Loras College. Section 21.2—*Power*.

Wade Molnau. Senior Engineering Manager, General Dynamics Advanced Information Systems, Bloomington, Minnesota. PhD, MS, BS (Industrial Engineering), Arizona State University. Section 23.5—*Multi-Spacecraft Manufacturing*.

Ron Noomen. Assistant Professor, Aerospace Engineering, Delft University of Technology, Delft, The Netherlands. MSc (Aerospace Engineering), Delft University of Technology. Section 10.7—*Design of Interplanetary Orbits*.

John E. Oberright. Senior Systems Engineer, Goddard Space Flight Center, Greenbelt, Maryland. BME (Mechanical Engineering), Catholic University of America. Section 22.2—*Spacecraft Thermal Control*.

Jean Olivieri. Senior Director of Structures Production, SpaceX, Hawthorne, CA. MSE, Arizona State University; BSE, University of Arizona; PMP, George Washington University. Section 23.5—*Multi-Spacecraft Manufacturing*.

David Parsley. SBIRS Deputy Director/Program Manager HEO Payloads, Northrop-Grumman Electronic Systems, Azusa, California. MS, BS (Electrical Engineering), University of Utah. Section 23.4—*System Mission Verification and Validation*.

Ed Pentaleri. Principal Systems Engineer, Space Systems/Loral, Palo Alto, California. MS (Physics), San Jose State University; BS (Mechanical and Nuclear Engineering), University of California, Berkeley. Chapter 16—*Communications Payloads*.

Kent M. Price. Principal Engineer, Space Systems/Loral, Palo Alto, California. MS (Physics), University of Sydney, Australia; BS (Physics), Yale University. Chapter 16—*Communications Payloads*.

Michael Pryzby. Sr. Systems Engineer/Program Manager, ATK Aerospace, Springfield, Virginia. MSE (Systems Engineering), Johns Hopkins University; BS (Mechanical Engineering), Virginia Polytechnic Institute and State University. Section 22.1—*Spacecraft Structures and Mechanisms*.

Bill Purdy. Mechanical Engineer, Purdy Engineering, Poolesville, Maryland. BSME University of Maryland. Chapter 13.4—*Cost vs. Reliability—Focusing on Mission Objectives.*

Lloyd Purves. Senior Systems Engineer, NASA Goddard Space Flight Center, Greenbelt, Maryland. BS (Electrical Engineering), Union College; Graduate courses (Aerospace Engineering), University of Maryland. Section 6.5—*System Engineering Tools*.

Jeffery J. Puschell. Principal Engineering Fellow, Raytheon Space and Airborne Systems, El Segundo, California. PhD (Astrophysics), University of Minnesota; AB with honors (Physics), The University of Chicago. Chapter 6—*Formal Requirements Definition;* Chapter 15—*Overview of Payload Design*; Chapter 17—*Observation Payloads*.

James Rendleman. Consultant/Attorney, Rendleman & Associates, Colorado Springs, Colorado. BS (Chemistry), University of Chapel Hill; MBA, MPA, Golden Gate University; JD, Whittier Law School; LLM, University of San Diego School of Law. Chapter 12.3—*Space Law and Policy Considerations.*

Markus Rufer. Executive Vice President, COO, Scorpius Space Launch Company, Hawthorne, California. MS (Mechanical Engineering), Federal Technology Institute; MBA, Alameda University. Section 23.7—*Intangible Factors in Manufacturing.*

Nicola Sarzi-Amade. Vice President, Scorpius Space Launch Company, Hawthorne, California. Systems Engineer, Microcosm, Inc., Hawthorne, California. PhD (Astronautical Engineering), University of Southern California; MS (Aerospace Engineering), University of California, Irvine; Laurea (Aerospace Engineering), Politecnico di Milano. Section 7.3—*Radiation Belts*; Appendices; Inside Front Cover.

Jeremiah P. Schneider. Electrical Engineer, Lockheed Martin Space Systems Company, Littleton, Colorado. MS (Systems Engineering), University of Denver; BS (Electrical Engineering), University of Colorado at Denver. Section 21.2—*Power*.

Rabindra (Rob) Singh. Executive Director, Communications Systems Space Systems Loral, Palo Alto, California. MASc (Electrical Engineering), University of Waterloo, Canada; BASc (Systems Design Engineering), University of Waterloo, Canada. Chapter 16—*Communications Payloads*.

Frederick A. Slane. Executive Director, Space Infrastructure Foundation, Colorado Springs, Colorado. MBA (International Business), University of Colorado; MS (Physics), University of New Mexico; BS (Aeronautics Engineering), Air Force Institute of Technology; BA (Physics/Mathematics), Willamette University. Section 6.6—*The Role of Standards in Space System Development*.

Geoffrey Smit. Senior Engineering Specialist, The Aerospace Corporation, El Segundo, CA. PhD (Mechanical Engineering/Aerospace Sciences), Northwester University; MBA, University of Chicago; BS, (Mechanical Engineering), University of Witwatersrand, Johannesburg, ZA; Appendix G—*Statistical Error Analysis.*

Milton "Skip" Smith.. Attorney, Sherman & Howard L.L.C. Colorado Springs, Colorado. PhD, McGill University Institute of Air & Space Law; MS, McGill University Institute of Air & Space Law; JD, George Washington University Law School; BS, United States Air Force Academy. *Section 12.3—Space Law and Policy Considerations.*

Trevor C. Sorensen. Professor/Project Manager, Hawaii Space Flight Laboratory, University of Hawaii, Honolulu, Hawaii. D.E. (Aerospace Engineering), University of Kansas; MS (Aerospace Engineering), University of Kansas; BS (Aerospace Engineering), University of Kansas. Chapter 29—*Mission Operations*.

David B. Spencer. Associate Professor of Aerospace Engineering, The Pennsylvania State University, University Park, Pennsylvania. PhD (Aerospace Engineering Sciences), University of Colorado; MBA, Pennsylvania State University; MS (Aeronautics and As-

tronautics), Purdue University; BS (Mechanical Engineering), University of Kentucky. Section 7.5—*Orbital Debris*.

Scott R. Starin. Senior Aerospace Engineer, Goddard Space Flight Center, Greenbelt, Maryland. MS (Aerospace Engineering), Ohio State University; BS (Physics), North Carolina State University; BS (Human Relations to Technology), North Carolina State University; BS (Aerospace Engineering), North Carolina State University. Section 19.1—*Spacecraft Attitude Determination and Control Systems*.

Craig L. Stevens. Aerospace Engineer, NASA Goddard Space Flight Center, Greenbelt, Maryland. BS, MS Virginia Polytechnic Institute and State University. Section 22.1.3—*Structural Analysis and Sizing*.

W. Kent Tobiska. President and Chief Scientist, Space Environment Technologies, Pacific Palisades, California; Director of Utah State University Space Weather Center, Logan, Utah. PhD (Aerospace Engineering with Solar and Atmospheric Physics), University of Colorado; MS (Aerospace Engineering), University of Colorado. Section 7.1—*Summary of Space Environment and Weather*; Section 7.2—*The Earth's Magnetic Field*; Section 7.3—*Radiation Belts*.

Robert J. Twiggs. Professor of Astronautics, Morehead State University, Morehead, Kentucky. MS (Electrical Engineering), Stanford University; BS (Electrical Engineering), University of Idaho. Section 25.4—*CubeSats*.

Richard E. Van Allen. Vice President and Director, Space Systems Division, Microcosm, Inc., Hawthorne, California. PhD (Aerospace Engineering), Purdue University; MS (Aerospace Engineering), Purdue University; BS (Aeronautical and Astronautical Engineering), Purdue University. Section 18.7—*Alternative Propulsion Systems for In-Space Use*.

Jeff Volosin. Deputy Division Chief, Exploration and Space Communication, NASA/GSFC, Greenbelt, Maryland. Adjunct Professor (Astronautical Engineering), Capitol College; BS (Space Science), Florida Institute of Technology. Chapter 28—*Ground System Design*.

David B. Watson. Sr. Manager, Mechanical Engineer, ATK Space Systems, Beltsville, Maryland. MS, Univsersity of Missouri; BSAOE, Virginia Tech. Chapter 22.1.2—*Design Concepts*.

James R. Wertz. President, Microcosm, Inc., Hawthorne, California. PhD (Relativity and Cosmology), University of Texas, Austin; MS (Administration of Science and Technology), George Washington University; SB (Physics), Massachusetts Institute of Technology. Chapter 1—*Introduction*; Chapter 2—*Space Mission Communities*; Chapter 3—*Space Mission Engineering*; Chapter 4—*Mission Concept Definition and Exploration*; Chapter 5—*Mission Analysis and Mission Utility*; Chapter 8—*Space Mission Geometry—What the Spacecraft Sees*; Chapter 9—*Orbits and Astrodynamics*; Chapter 10—*Orbit and Constellation Design—Selecting the Right Orbit*; Chapter 13—*Reducing Space Mission Cost and Schedule*; Appendix E,—*Time and Date Systems*.

Julie Wertz. Systems Engineer, Jet Propulsion Laboratory, Pasadena, California. PhD (Astronautical Engineering), Massachusetts Institute of Technology; MS (Astronautical Engineering), Massachusetts Institute of Technology; BS (Aeronautical and Astronautical Engineering), Massachusetts Institute of Technology. Section 24.2—*Space System Risk Analysis*.

James S. Wood. Launch Services Program Chief Engineer, Kennedy Space Center, Kennedy Space Center, Florida. BS, Pittsburg State University, Pittsburg. Chapter 26—*Launch Vehicles*.

Marcus Young. Program Manager, Advanced Propulsion Concepts, Air Force Research Laboratory, Edwards Air Force Base, California. PhD (Aerospace Engineering), University of Southern California; MS (Aerospace Engineering), University of Southern California; BS (Aerospace Engineering), University of Southern California. Section 18.7—*Alternative Propulsion Systems for In-Space Use*.

Charles M. Zakrzwski. Systems Engineer, NASA Goddard Space Flight Center, Greenbelt, Maryland. MS (Aeronautical and Astronautical Engineering), Ohio State University; BS (Aerospace Engineering), Pennsylvania State University. Section 18.5—*System Design Elements (Propulsion)*

Advisory Committee

Stuart Eves, *Surrey Satellite Tech Ltd.*
Michael Gruntman, *University of Southern California*
Jane Hansen, *HRP Systems, Inc.*
John E. Kessee, *MIT*
Jin Kang, *United States Naval Academy*
James Longuski, *Purdue University*
Ronald A. Madler, *Embry-Riddle University*
Landis Markley, *NASA Goddard Space Flight Center*
John Martin, *NASA Goddard Space Flight Center*
John C. Mather, *NASA Goddard Space Flight Center*
Carlos Niederstrasser, *Orbital Sciences Corp.*
Ron Noomen, *Delft University of Technology*
Lloyd R. Purves, *NASA Goddard Space Flight Center*
Steven Raque, *West Virginia University*
Aaron Q. Rogers, *Johns Hopkins University; Applied Physics Lab*
Frederick A. Slane, *Space Infrastructure Foundation*
Michael Swartwout, *St. Louis University*
Richard Van Allen, *Microcosm, Inc.*

Reviewers

Viqar Abbasi, *Canadian Space Agency*
Ossama Abdelkhali, *Michigan Tech University*
Stephen P. Airey, *ESA*
Thomas Bauer, *Microcosm, Inc.*
Grant Bonin, *UTIAS Flight Laboratory*
Mario Ciaramicoli, *ISU*
Amanda C. Cordova, *The Boeing Company*
Bernd Dachwald, *Aachen University*
Bruce Davis, *University of Colorado Boulder*
Simon Dawson, *SSI*
Cornelius J. Dennehy, *NASA Goddard Space Flight Center*
Jorge Diaz del Rio, *ESA-ESAC*
Michel Doyon, *Canadian Space Agency*
Don Edberg, *Cal State Polytechnic*
David Geller, *Utah State University*
Doug Griffin, *STFC,RAL,SSTD*
Marcus Hallmann, *DLR-Germany*
Chan Ham, *Southern Polytechnic State University*
Terry Hart, *Lehigh University*
Jeffrey Hoffman, *MIT*
Greg Holt, *NASA Johnson Space Center*
Alexander Jablonski, *Canadian Space Agency*
Julie Jackson, *Microcosm, Inc.*
Tom E. Jones, *NASA Goddard Space Flight Center*
Jin Kang, *United States Naval Academy*
Stephen Kemble, *EADS-Astrium*
James Keravala, *Shackleton Energy Company*
Andrew Ketsdever, *UCCS*
Charles Kilmer, *The Boeing Company*
Robert Klungle, *Aerospace Corporation*
Pam Lindbloom-Esquinca, *Microcosm Astronautics Books*
Ronald A. Madler, *Embry-Riddle University*
Craig A. McLaughlin, *University of Kansas*
Carlos Niederstrasser, *Orbital Sciences Corp.*
Darius Nikanpour, *Canadian Space Agency*
James O'Donnell, Jr., *NASA Goddard Space Flight Center*
Hayrani Oz, *Ohio State University*
David W. Peters, *NASA Goddard Space Flight Center*
Steven Raque, *West Virginia University*
Terry Rector, *Com Dev-USA*
Aaron Q. Rogers, *Johns Hopkins University; Applied Physics Lab*
Michael Safyan, *ISU*
Nicola Sarzi-Amade, *Microcosm, Inc.*
Ryan Sellers, *Cal State Polytechnic*
Anthony Shao, *Microcosm, Inc.*
Wayne Shiroma, *University of Hawaii*
Daniel Showalter, *Canadian Space Agency*
William Sluder, *NASA Goddard Space Flight Center*
David Spencer, *Penn State University*
Amanada Stiles, *X PRIZE*
Mike Swartwout, *St. Louis University*
James Volp, *ISU*
Nestor Voronka, *Tethers Unlimited*
Erika B. Wagner, *X PRIZE Lab@MIT*

Preface

This book is intended as a replacement for *Space Mission Analysis and Design*, known as SMAD to a great many engineers and students. SMAD was first published in 1991 and served as a practical and widely used reference and text for more than 20 years with over 65,000 copies in print. Nonetheless, the worlds of space exploration, space usage, and how books are used and published have all changed in the intervening period, and it is time for a new, unique, and hopefully more useable and useful reference and text in *Space Mission Engineering (SME)*—i.e., defining the parameters and trading on requirements so as to meet the broad, often poorly defined, objectives of a space mission at minimum cost and risk. This is a definition that was introduced in the first edition of SMAD to try and distinguish mission engineering from the more formal process of space systems engineering which had, in many cases, become simply a process of keeping track of all of the requirements and constraints and determining how each was to be met and verified. Certainly both processes are needed, but our emphasis here is on how to go about starting with a blank sheet of paper and **creating** a space mission to meet the needs of the end user as economically and quickly as possible.

It is clear that the technology exists today, or can be created, to do essentially anything we choose to do in space. The major challenges facing us are that space has become dramatically expensive and far too slow to achieve the objectives set for it, and, as a result, is far less useful to the end user than it should be. President Kennedy's call for landing a man on the Moon was followed 8 1/2 years later by the Apollo 11 mission, including developing from scratch the Saturn V launch vehicle—the largest, most reliable, and lowest cost per pound launcher ever built. Today the Saturn V no longer exists and we most likely couldn't finish doing the system design and verification needed for a Moon mission in 8.5 years, even if we could afford to go there.

Fortunately, the space business is beginning to change again, at least in some areas. SmallSats and CubeSats, with dramatically lower cost and shorter schedules, are starting to be taken seriously. Dr. Pedro Rustan [2004], at the time the Head of Advanced Science and Technology for the National Reconnaissance Office, pointed out in 2004 that the goal for space mission engineering should not be just creating low-cost, responsive small satellites, but using the same or comparable techniques to dramatically reduce the cost and schedule for all space systems, including those that are inherently very large and complex.

As a consequence of the need to change the way we do business in space, SME has two complimentary objectives: (a) to explain how and why space missions are built the way they are and (b) to explain how it can be done "faster, better, cheaper." The difficulty of doing this is best captured by the experience of Dan Goldin, the Director of NASA from 1992 to 2001, who introduced this mantra to try to convince the space community that change was critical. After all, almost every other aspect of modern technology has become faster, better, and cheaper in its implementation—including computers, communications, television, automobiles, and aircraft. But culture is very hard to change. Today, if you say "faster, better, cheaper" in a meeting of aerospace engineers, you will almost certainly hear several people respond with "faster, better, cheaper—pick any two." This suggests, of course, that in the view of many space engineers faster, better, and cheaper isn't possible—that space technology is, in effect, as good as it is ever going to get and improvements in one area will mean doing more poorly in some other area. This is the real challenge for SME—to show both where we are today (and how to build space missions according to today's rules) and show clearly that the goals of Dan Goldin, Pete Rustan, and many others can indeed be met as technology advances. It is also our goal to explain how this can be accomplished in as specific and practical terms as possible. Our objective for SME is to satisfy both needs. We would much appreciate feedback from readers about whether we have been successful or how we could do it better. (Send E-mails to bookproject@smad.com or to any of the editors listed below.)

One of the major changes for this volume is that it is both a physical book and a website, www.sme-smad.com. (The password is ASTRODATA and isn't case sensitive.) The website is not intended as a stand-alone site, but as an extension of this volume. It contains articles which go into more detail on many topics, downloadable Excel models for most of the mathematical tables and plots, data tables that can be updated from time-to-time, and a complete listing of all of the references with an annotated bibliography. Nearly all of the references and bibliography have a direct link to where the material can be downloaded for free or, for copyright material, where it can be purchased and downloaded, or where books can be purchased and shipped with next day delivery. Nearly all the sample plots and tables can be customized to your particular space mission. Our goal here is to make the book fit your mission as much as possible and let you find additional details and more in-depth assessments of specific topics within a day or less. We hope it works well and would appreciate your suggestions on how to make it work better.

It is very difficult to express our full appreciation for over 120 authors, reviewers, and advisory committee members, all of whom worked dramatically hard with no contract, no compensation, and no reward other than seeing the work done and a copy of the book when it was complete. These are among the most experienced and busy individuals in the community, giving up free time to provide this material for you, the reader. Without them, it would not have been possible. We are truly

indebted to them for putting in many more hours than most of them had anticipated. Among the reviewers, we would like to especially thank Steve Raque, Mike Swartwout, Jin Kang, Tom Bauer, and Markus Rufer, who reviewed many chapters and made major contributions to the success of the volume. A special thank you goes to Goddard Space Flight Center for supplying many authors and reviewers, and considerable technical data as well.

The work of creating the camera-ready copy fell to the Microcosm Publications Department, under the direction of Chris Maruyama and Donna Klungle. Jenny Greer and Julie Jackson did much of the 2-column formatting. Nearly all of the art work, including the color version of all of the figures and tables for the educator package, was created by Chris Maruyama. Most of the background work, references, bibliography, web references, and Excel models have been created under the direction of Nicola Sarzi-Amade by Nicola, Anthony Shao, Becky Christofferson, and Kyungmo Koo.

As anyone who has undertaken projects of this sort knows, one of the largest contributions comes from the families of the three editors. We greatly appreciate their forbearance and patience with a project that took longer than we hoped and far more evening and weekend hours than any of us would have guessed or wished.

Ultimately, if this book belongs to anyone, it is Pam Lindbloom-Esquinca, the manager of the Microcosm bookstore, and the force behind creating this volume. It was Pam that told us very forcefully to go out and do it, put up several years of bookstore profits to create the camera-ready copy, pushed and cajoled all of the authors, reviewers, and editors, and dove in to help with editing, references, and anything else that needed to be done. The book would not have occurred without you, Pam. You have made a significant contribution to astronautics that is greatly appreciated.

If you have ideas or suggestions on how we can make the book or the associated website better or more useful, please let us know. In many ways, this book is a new experiment and your feedback is both welcome and appreciated.

July, 2011

James R. Wertz
Microcosm, Inc./USC
Hawthorne, CA
jwertz@smad.com

David F. Everett
NASA GSFC
Greenbelt, MD
David.F.Everett@nasa.gov

Jeffery J. Puschell
Raytheon
El Segundo, CA
jpuschell@raytheon.com

Dedication

Microcosm Press and Microcosm Astronautics Books would like to expressly thank the editors for their vision, the authors for their views, the reviewers and researchers for their help and all for their hard work.

We would like to thank NASA, the US Air Force, US Army, US Navy, European Space Agency, Canadian Space Agency and all the universities, industry and commercial companies that support space programs, for their inspiration.

We would also like to thank the professors that will use this text to teach the next generation for giving us purpose. We would like to thank the small companies that keep on living the dream and the large companies that keep the dream alive, all of whom have been great supporters.

In the end, this book is for us all. We are a small community, in the grand scheme of things, and we take pride in the accomplishments of ourselves and each other. The willingness to share these accomplishments and the knowledge we learn from them is one of our greatest attributes.

So this book is dedicated to us all. May we all live long and prosper!!

Pam Esquinca
Chris Maruyama

1 Introduction

James R. Wertz, *Microcosm, Inc./USC*

1.1 What is Space Mission Engineering?

1.2 The History of Spaceflight—*Mike Gruntman*
Humble Beginnings; Great Pioneers; Building the Foundation; The Breakthrough to Space; Spacefaring Civilization

1.3 Spaceflight Technology
Getting to Space is Really Hard; Rockets Are a Lousy Way to Get Anywhere; Space Parts Are All Unique; Technology Readiness Levels

1.4 Spaceflight Economics—*John M. Jurist, David M. Livingston*
Space Budgets; Diversity of Opinions; Overview of Economics of Aerospace Systems; General Accepted Accounting Practices (GAAP) and Government Accounting; Why is Space Expensive; How Might Costs Be Minimized?; Other Launch Cost Issues

1.5 The Wide Range of Space Mission Applications

1.6 Sources of More Information

The goal of this book is to lay out the entire process of analyzing and designing space missions—from an initial vague concept or mission objective to designing, building, launching, operating, and eventually deorbiting one or more spacecraft. In the words of former NASA Administrator Dan Goldin, our objective is to do this "faster, better, cheaper" than it has been done before. Although many throughout the space community will argue that this is effectively impossible ("faster, better, cheaper—pick any two" is a common rejoinder), experience has shown that it is possible and that progress is real. Perhaps most important, we **must** find ways to do the job faster, better, and cheaper if we are to accomplish the missions that we want to accomplish within the budgets that are available and enable some of the remarkable benefits that space can bring about. To do this, we need to understand both how and why things are done as they are today, so that we can make the process better tomorrow.

In a sense, this book is intended as a microcosm of the faster, better, cheaper mission engineering process. Our goal with *Space Mission Engineering* is to greatly improve and shorten the preliminary mission design process by making it easier and faster to apply equations and examples and find reference data in time to be of use in a current project. For example, most of the analytic models, numerical tables, and plots throughout the book are immediately available to you on the book website and can be changed to show your specific problem of interest. Some extended discussions of relevant material that is a bit too long for the printed text is also there. Most of the references cited throughout the book are immediately available on the web and nearly all of the remainder are available within a day or so, although, of course, copyright material must be purchased from the owner or an appropriate reseller. We also have a bibliography for each chapter on the website, both so that you can have immediate access to that information, and also so that it can be updated from time-to-time as new books and professional papers appear. Details of what's on the website, how to use it, and other sources of mission engineering information, are described in Sec. 1.6. Note that the website is an adjunct to the book, and not a substitute.

Looking at the organization of the book, Chaps. 1 and 2 provide the philosophical basis and broad context for determining what space mission engineering is all about. Chaps. 3–6 then provide the basic engineering details of defining a space mission, undergoing system trades to find the best available solution (or at least an acceptable one), quantifying the cost and utility of the mission, and undertaking the formal requirements definition process. Chaps. 7–10 look specifically at the space environment, orbit and mission geometry, and orbit and constellation design. Chapters 11–13 then look at financial and legal issues and the fundamental problem of reducing cost and schedule. Part II (Chaps. 14–25) goes through the process of spacecraft and payload design. Part III (Chaps. 26–31) covers launch, operations, and end-of-life. Finally, the appendices and end-pages summarize the numerical data and basic formulas that you will need for preliminary mission design.

1.1 What is Space Mission Engineering?

***Space Mission Engineering* is the definition of mission parameters and refinement of requirements so as to meet the broad and often poorly defined objectives of a space mission in a timely manner at minimum cost and risk. The goal of this book is to give you the processes and tools to do that.**

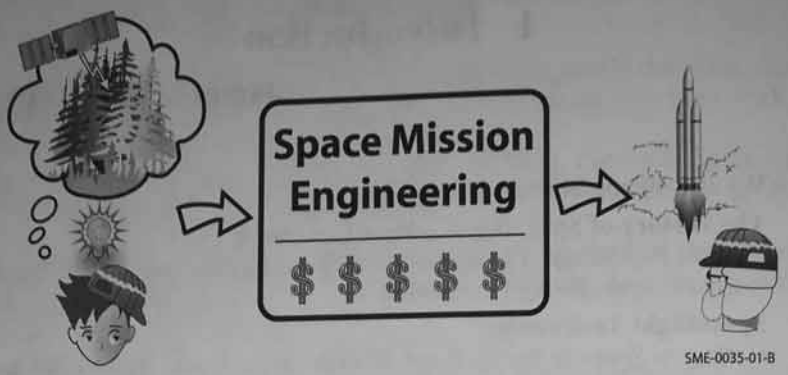


Fig. 1-1. Space Mission Engineering is the Process of Going From Broad Mission Objectives or a Vague Concept to an Operational Mission. Unfortunately many modern space programs perform well, but cost too much and take too long, which is a fundamental problem that is addressed throughout this book.

A good example of the space mission engineering process is the Apollo program that resulted in landing the first humans on the Moon and is one of the major technological achievements of modern times. But how well did it work from an engineering perspective? In its fundamental goal of putting people on the Moon and returning them safely to Earth it was completely successful. In a task that was very high risk, Apollo landed people on the Moon, gave them the time and tools to explore their immediate environment, and returned all of them safely to Earth.* However, our space mission engineering goal is to do this at minimum cost and risk—and we don't really know whether we achieved that. Realistically, we'll settle for reasonable cost and risk.

Like the Apollo program, most modern space missions perform well, and therefore, the space mission engineering job is well done in many ways. However, in another respect, the job is done very badly in that most large space programs today cost too much and take too long (Fig. 1-1). To be more specific, the United States launches approximately 20 space missions per year (including both manned and unmanned, Earth orbiting and interplanetary). The combined US space budget for NASA, DoD, the Intelligence Community, and a few other agencies is on the order of $60 billion/year. (See Sec. 1.5.) This implies that in recent years,

> **Space systems launched by the United States have cost an average of $3 billion/launch. The average time from concept to launch is not well known, but is certainly approaching a decade if not more.**

In my view, that is too much and too long, and fixing it is going to require major changes in how space business is done. Certainly some people within the space community will argue that the cost and time are necessary and appropriate and we will return to this question in Chap. 13. However, irrespective of whether you believe we are doing the job well or badly or, perhaps some combination of both, the goal of this book is to allow you to do it **better**—to create new missions, defined by you, the reader, that are safer, better, more reliable, with greater performance, done more quickly and at lower cost than has been the case in the last decade. In other words, we want space missions to evolve, just as other products in the marketplace evolve.

A part of this evolution includes updating the space mission engineering process from when it was presented 20 years ago in the first edition of *Space Mission Analysis and Design*, known as SMAD [Wertz and Larson, 1991]. The process presented here differs from that in SMAD by adding a timeliness requirement and identifying the people and groups involved, often called the *stakeholders*. There are specific reasons for these changes as discussed in Sec. 3.2.

There is another respect in which the Apollo program was a good example of space mission engineering. Initially, the objectives were broad and had very little definition. President Kennedy, in his proclamation in support of Apollo, did not specify how many people would go on each mission, or how long they would stay, or how much Lunar material they would bring back. The Apollo program itself defined the more detailed objectives, largely by determining what was reasonably feasible within the constraints of the technology. The successive Apollo missions continued to evolve and each landing became more complex and more successful in terms of the areas explored and the material returned. As presented in some detail by Johnson [2002], Apollo was one of the first major examples of systems management and systems engineering at a massive level. Apollo took a vague, poorly defined objective (as it should be) and, using 100's of thousands of people, built the largest, most reliable, and lowest cost per kg rocket ever built, and used that rocket to have 2 people step onto the surface of the Moon only 8.5 years after Kennedy expressed his initial vision. As Johnson calls it, the "Secret of

*Astronauts Gus Grissom, Ed White, and Roger Chaffee were killed in a fire inside the Apollo 1 Command Module during a ground test prior to any of the operational or test flights.

Apollo" was the organizational and engineering process that was able to achieve that.*

In another respect, Apollo wasn't a good example of space mission engineering. It may or may not have been minimum cost and risk, but it certainly wasn't cheap. (Apollo cost well over $150 billion in today's dollars.) What we would like to do today is to do missions much more substantive than Apollo, and for a lot less money. As we will see in Sec. 1.5 and Chap. 2, there is an enormous potential in space and we have only begun to scratch the surface of what can be accomplished for science, for commerce, for the military, for protecting the environment here on Earth, and for starting to use the resources of space for the benefit of mankind. We would like to change that and have missions become faster, cheaper, more useful, and in far greater numbers. All of this requires thinking seriously about both performance **and** cost and finding new ways to do business in space.

What happened? How did we go from having a competent, robust, very active space program to one that is still very competent but is also, for the number of missions launched, dramatically expensive and very slow to respond? At least a portion of the answer probably lies in the *Space Spiral* shown in Fig. 1-2. The ever-increasing cost of space missions leads to longer schedules and fewer missions. This leads to a demand for higher reliability, which, in turn, leads to higher cost, longer schedules, and fewer missions.

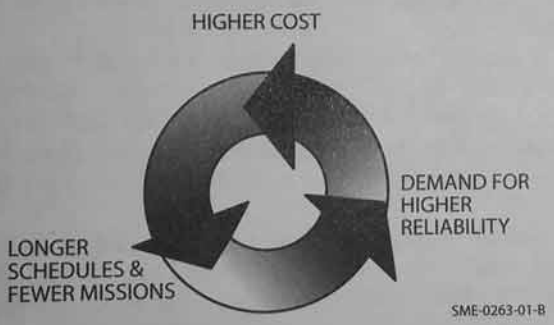


Fig. 1-2. The Space Spiral is a Significant Contributor to Increasing Costs and Longer Schedules.

Chapter 13 provides a detailed assessment of how to solve this problem. A part of the solution is to find ways to reverse the Space Spiral by having a mix of smaller, lower-cost missions and larger, more expensive ones, such that there is less demand for absolute guarantees of success, which are unachievable in the real world. An analogy is a military commander whose troops are on one side of a large canyon and he expects them to be attacked the next day by an enemy force that will move across a bridge over the canyon. The commander sends 3 airplanes to knock out the bridge, when the job could be done by a single plane. It is not that the commander believes that his planes are poorly built or that his pilots are incompetent, it's just that in the real world things go wrong and if you really want that bridge to not be there in the morning, you provide multiple ways of getting the job done. Similarly, in spaceflight, we will achieve a higher level of mission reliability by having multiple, diverse assets available instead of trying to make a single, dramatically expensive asset work perfectly.†

The broad timeline over which space missions are created is described in Sec. 3.4.2. The first phase is Concept Exploration, which is the initial study phase that results in a broad definition of the space mission and an initial estimate of cost, schedule, and performance. While much of this is a vague and "fuzzy" process, it's a critical part of space mission engineering because most of the cost, risk, schedule, and utility of the mission are determined by the end of concept exploration. Therefore, it's important to do this job carefully, thinking about what can be done to reduce the cost and complexity of the mission or increase the utility to the end users and those who will ultimately be asked to pay for the mission. By far the majority of mission concepts die during or at the end of concept exploration. This may be because there is insufficient performance or utility to the end user or because the technology doesn't yet exist to proceed, but it is most often simply because space missions are very expensive and no one is willing to pay to proceed to subsequent, ever more expensive, steps.

Even large, well-defined, active space programs can run into budget overruns and extended schedule delays. The reasons for this, and possible solutions, are described in Secs. 1.3 and 1.4 and in more detail in 13.6. As a result, missions on which hundreds of millions or billions of dollars have been spent are, nonetheless, cancelled from time to time or, when not cancelled, cause major delays in other important programs due to "soaking up" the available resources. This again emphasizes the need for good mission engineering throughout the mission lifecycle. We need to continuously look for ways to constrain cost and schedule growth and to sell the program and keep it sold, if it is still worth doing.

While there are problems with today's space missions, there is also dramatic potential, as described in Sec. 1.5 and Chap. 2. While some missions, such as navigation, communications, and direct broadcast are well-established and routine activities, other missions, such as utilization of space resources, space manufacturing, and

* Many people would argue that the US has lost some key element of the skill or will that was needed to allow Americans to walk on the Moon over 40 years ago and, consequently, the next people to walk on the Moon, and explore it in more depth, will likely be Chinese.

† Both aircraft and ground vehicles are much more reliable than space systems. Launch vehicles are about 90% reliable. We would never use airplanes or automobiles that had a 10% chance of a catastrophic failure that would destroy both the vehicle and its cargo when first used. We can either spend ever more money searching for perfection or find ways to accomplish our space mission objectives in the presence of occasional system failures.

the exploration of space itself, have just begun. These are truly exciting activities for a new generation of space mission engineers who can see further and do more. It's an exciting adventure.

1.2 The History of Spaceflight

Mike Gruntman,
University of Southern California

1.2.1 Humble Beginnings

The heavens had been attracting the imagination of humans for millennia. Some even argue that ancient texts, including the Old Testament, described spaceships in the sky. Reaching the cosmos requires powerful rockets. So, the first steps of the humans toward spaceflight were in rocketry. For centuries an essentially international endeavor of the pursuit of spaceflight attracted people from various lands who advanced the enabling science and technology.

Ancient Greeks observed the principle of jet propulsion more than 2,000 years ago. One thousand years later the first primitive rockets appeared in China and perhaps in India, later rediscovered in many other lands. A combination of charcoal, sulfur, and saltpeter—black powder—propelled the missiles. Natural abundance of saltpeter in China and India facilitated the emergence of the first war rockets in these countries.

Rockets had established a foothold in Europe some time in the 13th century. The word 'rocket' likely originated from the 'rocchetta,' a diminutive of the Italian word 'rocca' for distaff, a staff for holding the bunch of flux or wool from which the thread is drawn by spinning.

The early 19th century witnessed a major step in perfecting the rocket. A British inventor, William Congreve, turned ineffective and erratic missiles into a modern weapon system with standardized and interchangeable parts. These British war rockets, known as the Congreves (See Fig. 1-3), debuted during the Napoleonic wars. Then brought across the Atlantic Ocean, the Congreves bombarded Fort McHenry near Baltimore in 1813. Francis Scott Key immortalized the deadly missiles in his famous line "...And the rockets' red glare..." in the American National Anthem.

War rocketry rapidly proliferated throughout Europe and reached North and South Americas and Asia. The young Chilean republic was among the first to employ the domestically-made rockets—in 1819—in the fight against its former colonial ruler, Spain. Many European countries—particularly Austria, France, and Russia—established large-scale manufacturing of war rockets. The Russian army even built in 1834 an iron-clad submarine with a crew of 10 men which fired missiles from a submerged position.

The Mexican War, 1846–1848, advanced rocketry beyond an occasional experimentation in the United States. In a short period of a few months, the Army and the Navy completed the purchase, evaluation, prototyping, and testing of a new type of spin-stabilized war rockets. (These rockets became known as the Hales, after their inventor William Hale.) The US Army formed the first missile unit, the Rocket and Mountain Howitzer battery. The mass-produced new missiles quickly reached the rocket battery deployed in Mexico with the American expeditionary force. Thus the two military services succeeded in 1840s in a joint procurement and fielding of a new technologically advanced weapon system in less than one year.

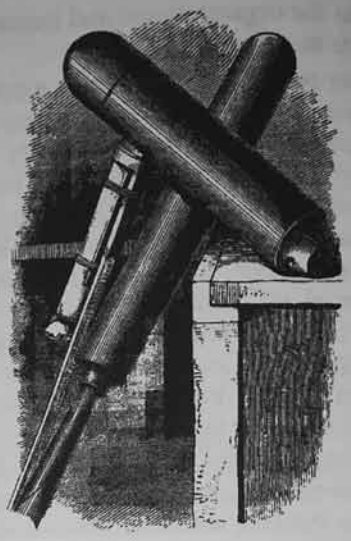


Fig. 1-3. Nineteenth Century Rockets: *Hale* (front), *Congreve* (with the centrally mounted guiding stick), and *skyrocket* (back). [Scoffern, 1859; Gruntman, 2004]

By the end of the 19th century, war rocketry had lost the competition to artillery with the introduction of rifled barrels, breech loading, and the Bessemer steel process. At this time the writers stepped in and replaced the men of sword as keepers of the interest in rocketry and spaceflight.

Nobody captured public imagination in space adventures more than the French writer Jules Verne (See Fig. 1-4). His novels "put on fire" and motivated many young men who would decades later transform a dream of spaceflight into a reality. Jules Verne's classic novel *From the Earth to the Moon* (first published in 1865) became a seminal work on spaceflight.

Early science fiction writers sent main characters on space voyages to satisfy their curiosity, as a bet, or to escape debts. Then, an American author, Edward Everett Hale, published a novel *The Brick Moon* in 1870. The story described a launch of an artificial satellite into orbit along a meridian to help sailors at sea in determining their longitude, in the same way as the Moon aids in determining latitude. It was the first description of an application satellite.

1.2.2 Great Pioneers

The late 19th century brought the realization that until the rocket was perfected there would be no trips through outer space, no landing on the Moon, and no visits to



Fig. 1-4. Jules Verne, *From the Earth to the Moon*—The future express. "Yes, gentleman," continued the orator," in spite of the opinion of certain narrow-minded people, who would shut up the human race upon this globe...we shall one day travel to the Moon, the planets, and the stars..." [Horne, 1911; Gruntman, 2004]

other planets to meet possible inhabitants. A long period followed when isolated visionaries and thinkers, including amateurs, began practical work and sketched out the sinews of the spaceflight concept. Many "intellectuals" of the day and assorted "competent authorities" dismissed the idea of space travel as ridiculous.

A number of outstanding individuals at the end of the 19th century and the beginning of the 20th century laid the foundations of practical rocketry and spaceflight. Four visionaries in 4 countries working under very different conditions became the great pioneers of the space age: the Russian Konstantin E. Tsiolkovsky; the French Robert Esnault-Pelterie; the American Robert H. Goddard; and the German Hermann Oberth. They contributed in unique ways to advancing the concept of spaceflight.

The writings of Konstantin E. Tsiolkovsky (1857–1935) combined development of scientific and technological ideas with the vision of space applications. While he never built rockets, Tsiolkovsky inspired a generation of Soviet rocket enthusiasts, including Sergei P. Korolev and Valentin P. Glushko, who achieved the first satellite.

An engineering graduate of the Sorbonne University, Robert Esnault-Pelterie, 1881–1957, first gained fame as an aviation pioneer who had introduced among other things an enclosed fuselage, aileron, joystick for plane control, four-bladed propeller, and safety belt. His prestige brought the much-needed credibility to the emerging space effort. It was Esnault-Pelterie who first published a spaceflight-related article in a mainstream archival physics journal in 1913; he also introduced the word "astronautics" in the language of science.

With a Ph.D. degree in what we would call today solid-state physics, Robert H. Goddard, 1882–1945, actually demonstrated the first liquid-propellant rocket engine in 1926. Goddard achieved numerous other firsts in rocketry. One of his rockets reached a 9,000 ft (2,700 m) altitude in 1937. Many results of Goddard's work remained largely unknown to contemporary scientists and engineers because of self-imposed secrecy, caused in part by ridicule by the ignorant and arrogant mainstream media.

Hermann Oberth, 1894–1989, published a detailed design of a sophisticated rocket in his book *The Rocket into Interplanetary Space* [Oberth, 1923]. He introduced numerous ideas including staging, film cooling of engine walls, and pressurization of propellant tanks. Oberth played an important role in early practical development of rocketry in Germany and provided inspiration for a generation of European space enthusiasts.

1.2.3 Building the Foundation

Powerful rockets belonged to a category of inherently complex advanced technologies where a lonely creative and gifted inventor could not succeed. Only concerted efforts of numerous well-organized professional scientists and engineers supported by significant resources could lead to practical systems. The totalitarian states were first to marshal the necessary resources and organize a large-scale development of ballistic missiles. In the Soviet Union, the military-sponsored *Jet Propulsion Scientific Research Institute* (RNII) employed 400 engineers and technicians in a sprawling complex in Moscow in the early 1930s. Later in the decade the Soviet program suffered from political purges and resumed its growth after 1944.

The German Army stepped up its rocket effort in 1932 by establishing a dedicated group that included Wernher von Braun. The German program grew immensely and by 1942 produced the first truly modern ballistic missile the A-4, better known as the V-2. The fueled A-4 weighed more than 12.5 metric tons and delivered a 1,000 kg warhead to distances up to 300 km. The German accomplishments also included mass production of the missiles. In a short period, under tremendous difficulties of wartime, the industry built 5,800 A-4's, with 2,000 fired operationally against England and liberated parts of Europe. The rocket manufacturing widely used slave labor from concentration camps, accompanied by atrocities especially during the construction of the underground facilities.

In the United States during WWII, rocketry concentrated on jet assisted take off (JATO) of the airplanes and on barrage solid-propellant missiles. The first American private rocket enterprises Reaction Motors and Aerojet Engineering Corp. were formed in December 1941 and in March 1942, respectively. After the war, several centers of rocketry emerged in the industry and government under sponsorship of the Army, Navy, and Air Force.

The US Army brought a number of captured German V-2 missiles to the United States. Military personnel and industrial contractors launched more than 60 V-2's from the White Sands Missile Range in New Mexico by 1951. Many missiles carried science payloads studying the

upper atmosphere, ionosphere, solar radiation, and cosmic rays. These first rocket experiments gave birth to a vibrant experimental space science. Subsequently, many government and university scientists became energetic advocates of space exploration.

The US Army followed its century-long tradition of the arsenal system with significant in-house engineering capabilities. By the early 1950s, it had concentrated the development of ballistic missiles and emerging space activities at the Redstone Arsenal in Huntsville, AL. The California Institute of Technology (Caltech) managed another important Army rocket center, the Jet Propulsion Laboratory (JPL), in Pasadena, CA. The JPL grew out of pioneering research and development programs from the group of Theodore von Kármán at Caltech.

The Redstone Arsenal became the home to more than 100 "imported" German rocketeers, headed by Wernher von Braun. The Germans had come to work in the United States under contracts through *Operation Paperclip*. While von Braun's rocketeers got the most publicity, the Paperclip program brought to the United States in total more than 600 German specialists in various areas of science and technology. In contrast to the compact von Braun group, the other scientists and engineers were dispersed among various American industrial and research organizations.

The Army, the Air Force, and the Navy were carrying out essentially independent development programs in guided missiles, with some overlap, occasional cooperation, and determined rivalry. In 1956, Secretary of Defense Charles E. Wilson attempted to resolve the problem of duplication by defining the "roles and missions" of the services. Consequently, the Air Force asserted control over intercontinental warfare, with the Army's role reduced to shorter range missiles.

The fateful roles-and-missions decision did not stop a most active leader of the Army's missile program, General John B. Medaris, and von Braun from finding ways to advance their visionary space agenda. In addition to such Army achievements as the development of the operationally-deployed ballistic missiles Redstone and Jupiter in 1950s, they would succeed in launching the first American artificial satellite, Explorer I, to space. Only by the end of 1950s, the Army had finally lost its programs in long-range ballistic missiles and space when the newly formed civilian space agency, the National Aeronautics and Space Administration (NASA), took over and absorbed the JPL and von Braun's team at Redstone.

In contrast to the Army, the Navy and especially the new service Air Force (formed in 1947) relied primarily on the contractors from the aircraft industry in their ballistic missile programs. In late 1940s and early 1950s, the Naval Research Laboratory (NRL) with Glenn L. Martin Co. developed the Viking sounding rocket as a replacement of the dwindling supply of the captured V-2's. This program laid the foundation for Martin's future contributions to ballistic missiles that would include the Titan family of Intercontinental Ballistic Missiles, (ICBM) and space launchers.

In 1946, the Air Force initiated development of a new test missile, the MX-774. The Convair (Consolidated Vultee Aircraft Corp.) team led by Karel J. (Charlie) Bossart introduced many innovations in the MX-774 missiles that reached an altitude of 30 miles. Based on this early experience, Convair later developed the first American ICBM, the Atlas. The Atlas program, including missile deployment became a truly national effort that dwarfed the Manhattan Project of World War II.

Other major ballistic missile programs initiated in 1950s included ICBMs Titan and Minuteman and Intermediate Range Ballistic Missile (IRBM) Thor. The Glenn L. Martin Company, Boeing Company, and Douglas Aircraft Company led the development, as prime contractors, of these missiles, respectively. Aerojet and the Rocketdyne Division of North American Aviation emerged as leading developers of liquid-propellant rocket engines. The Navy selected the Lockheed Aircraft Corporation as the prime contractor for its submarine-launched solid-propellant IRBM Polaris.

The Soviet government made rocket development a top national priority in 1946. The rocketeers first reproduced the German V-2 and then proceeded with building larger and more capable ballistic missiles. Soviet rocket pioneers from the early 1930s Korolev and Glushko emerged as the chief designer of ballistic missile systems and the main developer of the enabling liquid-propellant engines.

Both the Soviet Union and United States pursued development of the ICBMs, R-7, and Atlas. These large ballistic missiles called for new testing sites —the existing American White Sands and the Soviet Kapustin Yar did not meet the requirements of safety and security. Consequently, the United States established a new missile test range at Cape Canaveral in Florida in 1949 and later another site at the Vandenberg Air Force Base in California in 1958. Cape Canaveral would subsequently support space launches into low-inclination orbit while Vandenberg would send satellites into polar orbit, especially important for reconnaissance payloads. The Soviet Union initiated the construction of a new missile test site at Tyuratam (now commonly known as Baikonur) in Kazakhstan in 1955 and another site later in Plesetsk.

1.2.4 The Breakthrough to Space

In the 1950s, spaceflight advocates scattered among various parts of the US government, industry, and academia pressed for the American satellite. The national security policies would shape the path to space.

Rapidly progressing development of long-range ballistic missiles and nuclear weapons threatened devastating consequences should the Cold War turn into a full-scale military conflict. New technologies allowed no time for preparation for hostilities and mobilization and made an intelligence failure such as Pearl Harbor absolutely unacceptable. Therefore, monitoring military developments of the adversary, with accurate knowledge of its offensive potential and deployment of forces,

Table 1-0, Fig. 1-4◀, Eq. 1-0

became a key to national survival and (avoiding a fatal miscalculation,) reduced the risk of war.

Obtaining accurate information about closed societies of the communist world presented a major challenge. The perceived "bomber gap" and later the "missile gap" clearly demonstrated the importance of such information for the national policies. Consequently, President Dwight D. Eisenhower authorized development of overhead reconnaissance programs to be conducted in peacetime. The U-2 aircraft first overflew the Soviet Union in 1956, resolving the uncertainties of the bomber gap. Reconnaissance from space became a top priority for President Eisenhower who considered rare and sporadic U-2 overflights only a temporary measure because of improving Soviet air defenses. In 1956, the Air Force selected Lockheed's Missile Systems Division to build reconnaissance satellites.

The international legality and acceptability of overflights of other countries by Earth-circling satellites—*freedom of space*—was uncertain in the 1950s. The Eisenhower administration considered testing the principle of freedom of space by launching a purely scientific satellite critically important for establishing a precedent enabling future space reconnaissance.

This was the time when scientists in many countries were preparing for the International Geophysical Year (IGY) to be conducted from July 1957–December 1958. They planned comprehensive world-wide measurements of the upper atmosphere, ionosphere, geomagnetic field, cosmic rays, and auroras. Space advocates emphasized that artificial satellites could greatly advance such studies. Consequently, both the United States and the Soviet Union announced their plans of placing into orbit artificial satellites for scientific purposes during the IGY. Both countries succeeded.

President Eisenhower insisted on clear decoupling of American scientific satellites from military applications in order to first assert freedom of space. This national security imperative determined the publicly visible path to the satellite. In 1955, the US government selected the NRL proposal to develop a new space launch vehicle and a scientific satellite, both known as the Vanguard. The choice of the new system was made over a more mature technology of the Project Orbiter advocated by Army's Medaris and von Braun. The Army proposed to use the Jupiter C, an augmented Redstone ballistic missile. In fact, a test launch of the Jupiter C on September 20, 1956, could have put a simple satellite into orbit had the Army been permitted to use a solid-propellant missile — as it would later do launching the Explorer I—instead of an inactive fourth stage.

John P. Hagen led the Vanguard program with Glenn L. Martin Co. as the prime contractor of the launch vehicle and with NRL providing technical direction. The Vanguard program also built scientific satellites and established a process of calling for proposals and selecting space science experiments. In addition, it deployed a network of the Minitrack ground stations to detect and communicate with the satellites which laid the foundation for the future NASA Spaceflight Tracking and Data Network (STDN). Many optical stations around the world would also observe the satellites by the specially designed Baker-Nunn telescope tracking cameras.

The Soviet Union focused its resources on demonstrating the first ICBM. After the R-7 had successfully flown for the full range, Korolev launched the world's first artificial satellite, Sputnik, into orbit on October 4, 1957. Ironically, this Soviet success had finally resolved the lingering issue of the space overflight rights that so concerned President Eisenhower: no country protested the overflight by the Soviet satellite, thus establishing the principle of freedom of space (see Fig. 1-5).

The second, much larger Soviet satellite with the dog Laika aboard, successfully reached orbit on November 3, 1957. The Vanguard program had been steadily progressing but was not ready for launch yet. On November 8, the Secretary of Defense gave the permission to the eager Army team led by Medaris and von Braun to also attempt launching satellites. On January 31, 1958, the Army's modified Jupiter C missile successfully placed the first American satellite Explorer I into orbit.

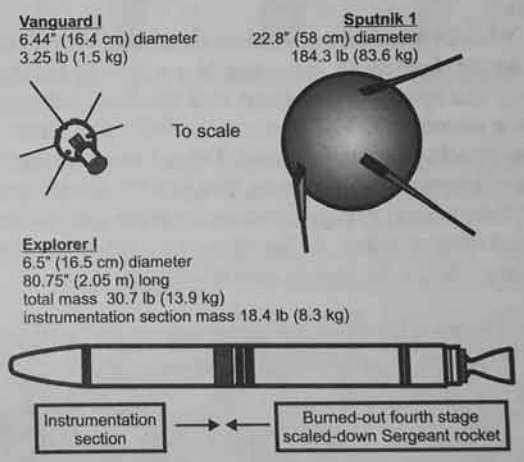


Fig. 1-5. Comparative Sizes and Masses of the Earth Satellites Sputnik 1, Explorer I, and Vanguard I [Gruntman, 2004].

Subsequently the Vanguard launch vehicle deployed the Vanguard I satellite into orbit on March 17, 1958. Popular sentiments in the United States have sometimes blamed the Vanguard program for losing the competition to the Soviet Union. It is grossly unfair. The Vanguard program demonstrated a record fast development of a new space launcher, with only 30 months from the vehicle authorization in August 1955 to the first successful launch in March 1958. The Vanguard spacecraft remains today the oldest man-made object in orbit, and it will reenter the atmosphere in a couple hundred years. We have time to find funding to bring the satellite back to the planet Earth for a place of honor in a museum.

There was no technological gap between the Soviet Union and the United States in the beginning of the space age. Being the first in launching the satellite was a matter of focus and national commitment. Fourteen months

after the launch of Sputnik, the United States had placed spacecraft into orbit by 3 entirely different launchers developed by 3 different teams of government agencies and industrial contractors. (The Air Force's Atlas deployed the first communications satellite SCORE in December 1958.)

The last years of the Eisenhower administration shaped the structure of the American space program. The president established a new Advanced Research Projects Agency (ARPA, the predecessor of DARPA), to fund and direct the growing national space effort. The security-conscious president resisted expansion of the government programs but always supported advancement of spaceflight in the interests of national security.

Bending to powerful political forces Eisenhower reluctantly agreed to establish a new government agency responsible for a civilian effort in space. The president signed the National Aeronautics and Space Act into law which formed NASA on October 1, 1958. Within a short period of time, NASA subsumed the National Advisory Committee for Aeronautics (NACA), Army's Jet Propulsion Laboratory and major elements of the ballistic missile program in Huntsville, and NRL's Vanguard group.

NASA vigorously embarked on scientific exploration of space, launching increasingly capable spacecraft to study the space environment and the Sun and creating space astronomy. The missions to flyby the Moon and, later, nearby planets followed. These first space missions began a new era of discovery that laid the foundation for the flourishing American space science and planetary exploration of today. At the same time, NASA embarked on preparation for human spaceflight.

Rocketry Industry "Namescape"

Mergers and acquisition have significantly changed the "namescape" of rocket industry. Titan's prime contractor, the Martin Company, merged with Marietta in 1961, forming Martin Marietta. Convair became Space System Division of General Dynamics in 1954, known as General Dynamics—Astronautics. Martin Marietta acquired General Dynamics' Space System Division in 1995 and then merged in the same year with Lockheed, forming The Lockheed Martin Corporation. Thus both, the Atlas and the Titan families of space launchers ended up under the same corporate roof. Another important component of Lockheed Martin's rocket assets is the submarine-launched solid-propellant Tridents. Boeing added to its Minuteman missiles the Delta family of space launchers after acquiring McDonnell-Douglas in 1997.

Gruntman [2004], p. 253

At the same time the military space program focused on communications, early warning, command and control, and support of military operations. The Air Force led this effort with the Navy engaged in selected important programs, such as space based navigation. The Army preserved the responsibility for major elements of missile defense.

Another national security program dealt with space reconnaissance and was directed jointly by the intelligence community and the military. In 1960, President Eisenhower established a special office in the Department of Defense (DoD), staffed by military officers and government civilians, to direct space reconnaissance, separated from military procurement and hidden by an extra protective layer of secrecy. This organization would become the National Reconnaissance Office (NRO) overseen by the Air Force and the CIA. The image intelligence satellite Corona achieved the first successful overflight of the Soviet Union in August 1960, returning images that effectively resolved the uncertainties of the perceived missile gap.

President Eisenhower handed over to his successor in the White House a structure of the national space program that has essentially survived in its main features until the present day. NASA leads the civilian space effort. National security space consists of two main components. The services are responsible for military space while the intelligence community and military directs gathering and processing of the intelligence information from space. While these 3 programs are sometimes viewed as separate, they all had originated from the early military space effort and they all have been interacting to varying degrees during the years.

1.2.5 Spacefaring Civilization

The heating up competition in space with the Soviet Union erupted into the public focus when the first man, Soviet cosmonaut Yuri Gagarin, orbited the Earth on April 12, 1961. President Kennedy responded by challenging the nation to land "a man on the Moon and returning him safely [back] to the Earth." The resulting Apollo program culminated with astronauts Neil Armstrong and Edwin (Buzz) Aldrin taking man's first steps on the Moon in July 1969.

The late 1950s and early 1960s witnessed emerging commercial applications in space. The first transatlantic telephone cable had connected Europe and North America in 1956 to meet the increasing demand in communications. Space offered a cost-competitive alternative, and industrial companies showed much interest and enthusiasm for it, especially AT&T, RCA, General Electric, and Hughes Aircraft. The DoD supported the development of space communications on the government side. It was not clear at the time whether satellites in low, medium, or geostationary orbits would offer the best solution. While geostationary satellites provided excellent coverage, the technical challenges of building and deploying such satellites and their control had not yet been met.

Initially, the industry invested significant resources in the development of space communications. The situation drastically changed when President Kennedy signed the Communications Satellite Act in 1962. Now government, including NASA, became a major player in

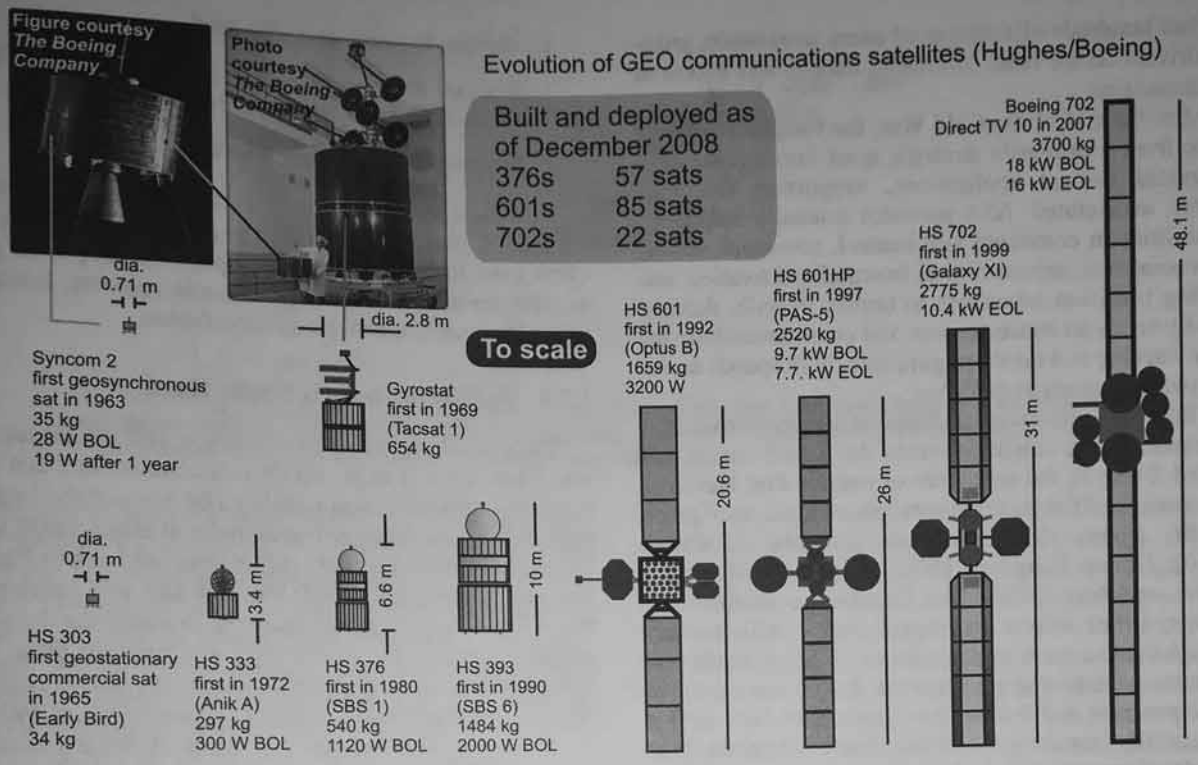


Fig. 1-6. Spectacular Growth of Communication Satellite Capabilities. Example of satellites developed by Hughes/Boeing [Gruntman, 2008].

commercial space communications, with the authority to regulate and to a significant extent dictate the development. Consequently, the Communications Sateliite (Comsat) Corporation was formed in 1963 to manage procurement of satellites for the international communications consortium Intelsat.

The Hughes Aircraft Company demonstrated a practical geostationary communication satellite with launches of 3 test spin-stabilized Syncom satellites in 1963–1964. As the technology progressed, several companies introduced 3-axis stabilized geostationary satellites. Since the beginning of the space age, satellite communications have been dominating commercial space, with most of the activities today concentrated in the direct-to-home TV broadcasting and fixed satellite services. Figure 1-6 demonstrates the astounding increase in capabilities of geostationary communication satellites with the example of one family of satellites built by Hughes, now part of the Boeing Company.

Military and reconnaissance satellites provided critically important capabilities essential for national survival. NASA missions, especially manned missions, were highly visible and reflected on the nation's international prestige, so important in the Cold War battles. As a result, National Security Space (NSS) and NASA missions had one feature in common: failure was not an option, which inevitably led to a culture of building highly-reliable systems. Space missions were thus performance driven, with cost being of secondary importance. The consequent high-cost of the space undertaking led, in turn, to increased government oversight which drove the schedules and costs further up. The government-regulated commercial space, dominated by the same industrial contractors, could not develop a different culture.

After landing twelve astronauts on the Moon, NASA brought to us spectacular achievements in space science and in exploration of the Solar system. Numerous space missions advanced our understanding of the Sun's activity and the near-Earth environment. NASA spacecraft visited all planets of the Solar system with the exception of Pluto*—the New Horizons mission is presently enroute to the latter.

The Soviet Union established a permanent space station, Mir, in low-Earth orbit. The American human space flight concentrated on the development of the Space Shuttle and the International Space Station (ISS). The Space Shuttle carried astronauts to low-Earth orbit from 1981 to 2011. The ISS, with a mass of about 400 metric tons, has the opportunity to demonstrate what humans can do in space.

Today, space affects government, business, and culture. Many countries project military power, commercial interests, and national image though space missions. It is a truly high-technology frontier, expensive and government-controlled or government-regulated. Space has become an integral part of everyday lives of people. We are accustomed to weather forecasts based on space-based sensors. Satellites deliver TV broadcasts to individual homes. The Global Positioning System (GPS)

* Pluto is now officially a dwarf planet. See App. B.

reaches hundreds of millions of users worldwide, guiding drivers on the road, aircraft in the air, and hikers in the mountains.

After the end of the Cold War, the transformation of space from a primarily strategic asset into increasingly integrated tactical applications, supporting the warfighter, accelerated. NSS provides critically important capabilities in command and control, communications, reconnaissance, monitoring of international treaties, and guiding precision munitions to targets. Missile defense heavily relies on space sensors and communications for early warning and intercept guidance. NSS spends annually twice as much as NASA.

The space enterprise has become a true international endeavor. Seven countries joined the Soviet Union and United States in the elite club of nations that launched their own satellites on their own space launchers: France (1965), Japan (1970), People's Republic of China (1970), United Kingdom (1971), India (1980), Israel (1988), and Iran (2009). The European countries have combined their efforts and launch their satellites today through the European Space Agency (ESA). Canada also conducts an active space program. Brazil has an active space program and it is only a question of time until it successfully launches its satellite. South Korea also pursues development of space launch capabilities, with Russia initially providing important parts of launch technology. The secretive North Korea tries to launch a satellite. In addition, numerous other countries bought and operate various satellite systems.

Very few countries presently match the American commitment to space exploration and space applications. "Only France (and the old Soviet Union in the past) approaches the US space expenditures in terms of the fraction of the gross domestic product (GDP). Most other industrialized countries (Europe and Japan) spend in space, as fraction of GDP, 4 to 6 times less than the United States." [Gruntman, 2004, p. 462] People's Republic of China and India are expanding their space programs. The highly space-capable Russia is also increasing its space activities after the decline of the 1990s.

For many years, the United States has led the world in space. The health and the future of the American space enterprise depend on the national commitment—there is no limit to what we can do. President Kennedy observed that "for while we cannot guarantee that we shall one day be first [in space], we can guarantee that any failure to make this effort [in space] will make us last..." Gruntman [2004, p.383].

1.3 Spaceflight Technology

Space is exceptionally expensive and nearly all spaceflight hardware is uniquely built for use in space or launch systems. This section addresses the question of why this is the case and how space technology differs from non-space technology. There are several key ideas that are involved:

1. Getting to space is really hard
2. Rockets are a lousy way to get anywhere, but better than any alternative we have available
3. Space parts are nearly all unique

This section addresses each of these and introduces the idea of *Technology Readiness Levels* (TRLs) that are often used to help manage the space development and acquisition process, but may, in some respects, contribute to the problem as well as the solution.

1.3.1 Getting to Space is Really Hard

The easiest place to get to in space, and the closest, is low Earth orbit (LEO), which is the space between the top of the atmosphere at roughly 100 km and the beginning of the Van Allen radiation belts at about 1,000 km. To get there, and stay there rather than fall back to Earth, we need to go up about 200 km, turn left, and accelerate to 7.8 km/s. Taking into account the losses along the way caused primarily by gravity and atmospheric drag, we realistically need to provide a change in velocity, called *delta V*,* of over 9 km/s. This is an extremely high velocity relative to any moving objects we have experience with. Commercial jet aircraft travel at about Mach 0.8 (= 610 mph = 0.34 km/s). The muzzle velocity of a modern rifle with a high speed cartridge is quite a bit higher at around 4,000 fps (= 1.22 km/s), but still far short of the velocity we need of 9 km/s.

Modern launch vehicles have multiple stages, such that the first stage is dropped off after its fuel is used to reduce the mass that is being accelerated, then the second stage is dropped off when it's done, and so on. Consider this in the context of our high-speed rifle bullet at 4,000 fps. To increase the final speed, we'll make our first bullet large enough to hold another gun inside the first bullet, such that after you fire the first bullet, a gun pops out of that bullet and fires again, so that the second bullet is now going at 8,000 fps. This is basically what staging does for us. However, our rifle is so inefficient relative to launch vehicles that we would have to add 6 larger rifles, for a total of 8 consecutive rifles each contained in a bullet fired from a larger rifle in order to get to the velocity we need. That very first rifle is going to have to be pretty big. (See Fig. 1-7.)

Another way to get a feel for the high velocities involved is to consider someone in a vacuum falling freely toward the Earth at 1 g (9.8 m/s^2). It would take 13 min of continuous falling to reach orbital velocity, without any drag, gravity losses, or whatever.

---

*ΔV, using the Greek Δ (delta) is the standard math symbol for a change in V, the velocity. Given the propensity of modern word processors and E-mail programs to mess up Greek letters, many people have found it easier to simply write out the Greek, so a change in velocity in astronautics is often written delta V. In this book we will usually write "delta V" in the text and ΔV in equations.

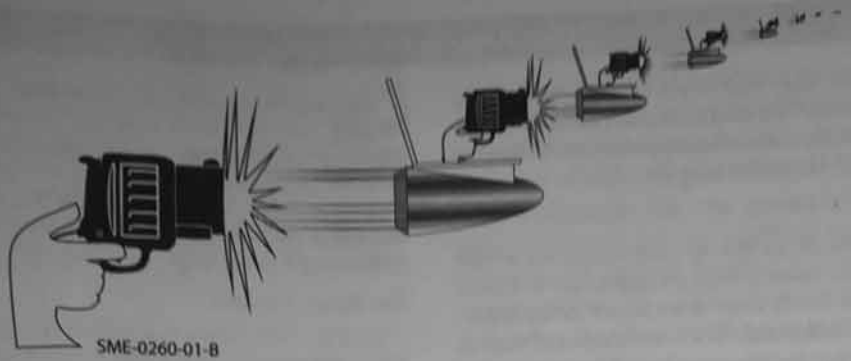


Fig. 1-7. Rocket Stages are Similar to Firing a Rifle with a Bullet that Contains a Rifle with Another Bullet, and So On, for Lots of Times. To get to the lowest energy orbit with a muzzle velocity of only 4,000 fps will require 8 successively larger riles. The first rifle will need to be very large.

Of course, going to higher orbits, the Moon, or planets is even harder, but at least we have more time for things to occur. Once we have gotten to orbital velocity and above the atmosphere, we can spend more-or-less as long as is needed to build up higher velocities to go elsewhere. Nonetheless, it's still a dramatically challenging problem such that virtually all missions are designed around the idea of minimizing the total delta V required to conduct the mission.

As we'll see in Sec. 1.3.2, we build up our velocity of 7.8 km/s (or higher), by throwing burned fuel out the back end of the rocket at 3 to 4 km/s. We can get to orbit by this process, but just marginally. If anything goes wrong there is likely to be a very large explosion and a very expensive payload will become debris in the ocean. If the orbital velocity were less, the problem would be much easier. If we lived on the Moon with a low Lunar orbital velocity of 1.7 km/s, space travel would be easy and we would all have rocketmobiles for heading to the store. If we lived on Jupiter (a bit challenging, since there really isn't a solid surface to live on) with a low Jovian orbital velocity of 42 km/s, there wouldn't be a space program and we would be tied to the (nonexistent) surface of the planet until we could invent something really new that we haven't thought of yet. Here on Earth, we can have a space program, but getting there is really hard.

1.3.2 Rockets are a Lousy Way to Get Anywhere

Not only is getting to orbit very hard, but rockets are a really bad way to do the job, although they're better than anything else we have at the moment. To understand the nature of the problem think of a rocket out in space near the Earth, rather like an ice skater on a frozen lake. To go north on the lake, we throw something, say a frozen turkey, as hard as we can to the south and, in reaction, we head northward. Similarly, in the rocket, to go to Jupiter we throw a lot of propellant out in the direction opposite the way we want to go. The problem for both the skater and rocket is that we're not pushing against anything around us like the Earth or even the air, so the center of mass of the system doesn't go anywhere. To get a kilometer north of where we started, the turkey which is likely lighter than the skater is going to have to be several kilometers south of where we started. (We're assuming the ice is very hard and perfectly smooth such that we can't push against it as skaters normally would.) Similarly, if we are going to get our rocket to Jupiter, there's going to have to be an awful lot of propellant a long ways in the direction opposite of where Jupiter is. In fact, the rocket and the propellant are both in orbit around the Sun and the velocity will change as we get closer to or further from the Sun, but the rules of astrodynamics unfortunately don't change the basic problem of the rocket being inefficient.

There is another way in which our rocket is inefficient. Let's suppose our rocket has a mass of 100 kg without the propellant and has 100 kg of propellant on board for a total mass of 200 kg. We then throw out the propellant as one lump at 3 km/s. (That would be pretty hard, but, again, this is just an example.) Now the relative velocity between them is 3 km/s, the propellant is moving at 1.5 km/s relative to the initial center of mass, and the rocket is going at 1.5 km/s in the opposite direction, again relative to the initial center of mass. If I wanted to get the rocket to 3 km/s, it would appear that I would have to add another 100 kg of propellant to the rocket, but unfortunately it's harder than that. If we start with 300 kg (100 kg for the rocket and 200 kg for the propellant), and throw 100 kg out at 3 km/s, now, relative to the initial center of mass, the propellant mass is moving at 2 km/sec and the rocket (with its 100 kg of extra propellant) is moving at only 1 km/s in the opposite direction. If I then throw the second glob of propellant out at 3 km/s, this will give the rocket an additional 1.5 km/s, such that the final velocity of the rocket is only 2.5 km/s, even though we have doubled the amount of propellant. The basic problem is that as we add propellant, we have to accelerate not only the rocket, but all of the rest of the propellant that we've put in it. As the percentage of propellant becomes higher and higher, we are using up most of the energy simply accelerating the propellant that we are bringing along and the rocket becomes less and less efficient. (Real launch vehicles are typically around 90% propellant.)

The end result of all this is that a launch vehicle is quite a few tons of the highest explosive you can find and

Alternatives to Rockets

Rockets aren't the only way to get to space, but they're the best we have so far. Two alternatives are described here. Once we're in space, there are a few more options such as the space tether (Sec. 25.1) or solar sails (Sec. 18.7)

Gun Launch

If we could build an air gun or something similar that would have a muzzle velocity high enough to get to orbital velocity, then all we would have to accelerate is the spacecraft itself and we could avoid all of the problems of staging and having to accelerate mostly propellant. What's more, the gun itself could stay on the ground and be used multiple times. But can we build such a gun? The answer is yes, but there are a few problems with it.

First, the acceleration has to be pretty high. At a constant acceleration of only 1,000 g's, the gun barrel would have to be over 4 km long, so we probably need a few thousand g's of acceleration. Since people can only withstand 6 or 7 g's, or a bit more, the gun launch will be limited to electronics and similar stuff. But some electronics can withstand that level of loads, so it still may be possible. Second, at the end of the gun, where the capsule is going at over 8 km/s, it's still deep in the atmosphere where drag is very high (As discussed in Sec. 9.4, drag is proportional to the atmospheric density and to V^2), so that's a real problem for both heating and slowing down.

Unfortunately, the largest problem is that we still need some sort of upper stage. As we'll see in Sec. 9.1, orbits are elliptical, which means that after 1 orbit the capsule comes back through the point where it was last accelerated. Since that's at least 90 min later, the gun itself will have moved quite a ways eastward due to the Earth's rotation, but we will still be down essentially at the Earth's surface. To avoid this, we need a small upper stage that can more-or-less circularize the orbit when we get to space. But this means that we need a tank strong enough to hold propellant that weighs several thousand times what it weighs just sitting around. Also, we need a propellant that won't explode when subjected to loads of that magnitude, and that may be even harder to find.

Gun launch to orbit isn't impossible, but it is very challenging, and not yet a practical solution. However, gun launch or, perhaps better, electromagnetic rail launch (similar to many modern carnival rides), may be an excellent solution for getting to orbit from the surface of the Moon where the orbital velocity is only 1.7 km/s and there is no atmosphere to get in the way.

The Space Elevator

The *Space Elevator* is a cable extending 36,000 km from geosynchronous orbit (GEO) to the surface of the Earth. To hold up the cable there must be another large cable or counterweight extending upward from GEO such that the center of mass of the system is in GEO and orbiting with a velocity equal to the rotation rate of the Earth on its axis. People or machinery then climb the cable (i.e., ride up an "elevator") from the Earth's surface to GEO, step off, and they are at orbital velocity in GEO.

As with the gun launch, there are a few practical considerations that make the space elevator challenging to build. Most people have concentrated on is the difficulty of creating a cable strong enough to support its own mass over a length of 36,000 km. This is generally addressed by considering very strong materials, such as carbon nanotubes, and using a cable that is tapered with the thickest part in GEO and coming down to a very small strand at the Earth's surface. Even with this approach, it is probably not buildable with today's technology because strong enough materials aren't readily available or usable.

A second problem to consider is what happens to all of the spacecraft in LEO that might run into the cable at 7.5 km/s. As discussed in Sec. 7.5, the kinetic energy associated with the collision is proportional to the square of the velocity. Thus a 1 kg, 10 cm diameter CubeSat (about the smallest satellite or fragment that is visible in LEO) would have the equivalent kinetic energy of hitting the cable with a bowling ball at 2,000 mph. That would likely not be good for either the cable or the satellite.

While rockets are a very poor means of transportation to and in space, they will be what we have to work with for quite a while.

somebody with a match. (Jet fuel and nearly 100% pure oxygen is one of the less explosive mixtures that is often used.) There are, of course, alternatives to rockets, but none of them seem too promising. Two of the more interesting approaches, the gun launch and the space elevator, are described in the boxed example. We can also find ways to supplement the system in some fashion. For example, some launch systems, like Pegasus, are air launched, where the launch vehicle is given an initial altitude and velocity before heading to space. The problem is that getting a start at, say, 10 km altitude and Mach 2 or less, doesn't really help very much when I need to get to 200 km and Mach 27. In addition, there's a limit to how heavy a rocket a modern aircraft can carry. The value of an air launch is largely to get away from the launch ranges and allow launch to occur out over the ocean where there are far fewer rules, restrictions, and obstacles.

Table 1-0, Fig. 1-7◀, Eq. 1-0

1.3.3 Space Parts are All Unique

Nearly all of the parts used in space systems are built uniquely for that purpose. This seems a bit strange in that rockets and spacecraft are just another approach to transportation, observations, and communications and should therefore, use much of same hardware used for similar functions on Earth. While many people interested in cost reduction are striving to make this happen (see Chap. 13), there are some practical considerations that make it difficult.

The first problem is that many of the parts used in spacecraft don't have any applications on Earth or create problems in space that would never be an issue for Earth equipment. Unfortunately, the spacecraft doesn't have any platform to sit on, such that even very small forces can cause it to move or rotate. Even the pressure of sunlight can cause a spacecraft to rotate or tumble. It's a bit

difficult to envision a typical piece of Earth equipment that would flip over just from light pressure when exposed to sunlight. This implies that the attitude control equipment on a spacecraft is typically far more sensitive than would be of much use in most ground-based equipment. In addition, even trivial moving parts, such as a filter wheel or the moving mirror in a single lens reflex camera, will cause the spacecraft to recoil in the opposite direction. This tends to affect the mechanical parts on a spacecraft.

The second problem has to do with the space environment, which is very well understood, but inherently different than the environment on Earth. Unfortunately, there are two features of the space environment that have a major impact on electronics. First is the higher radiation level that has a particularly strong effect on computers and microelectronics. An even larger problem is the thermal environment. Nearly all electronics on Earth rely on natural convection or forced air to provide cooling and, unfortunately, neither of these work in the 0-g, vacuum environment in space. This means that most pieces of desktop or laboratory electronics will likely work for a only short time in space and then burn out or be stopped by the radiation.

Perhaps the biggest difficulty with using most commercial or industrial parts in space is simply the high cost of transportation. It's typically on the order of $10,000 to $20,000 per kg to LEO, 5 times that to GEO, and even more to Lunar or planetary orbits. This means that the "excess" mass associated with making products robust on Earth is too expensive for space use. Typically, we build a box for an electronics gizmo by taking a piece of sheet metal, folding it a few times, and welding it. For space applications we take a block of aluminum the size of the box and machine away everything that isn't the box. This eliminates the mass of the folded sheet metal and the welds.

These issues are exacerbated by the need to know the detailed history of each part to be **sure** that it will work successfully. When I was working on the TDRS communications satellite program, a metal foundry in West Virginia announced that the quenching (water spray) on their aluminum milling machine hadn't been working correctly for several months and it was possible that there could be soft spots in some of the aluminum milled at that facility. The TDRS builder, TRW, was able to go back, locate, and test every single piece of equipment on TDRS that had any aluminum that had been milled at that plant. No problems were found. Nonetheless, imagine the difficulty and the cost if every piece of aluminum, plastic, or electronics in your car came with a complete history of where all of the material in that component had been mined, milled, manufactured, and integrated into the final product. That requirement in itself would eliminate nearly all commercial and industrial equipment from consideration.

Fortunately, the "rules" by which spacecraft are built are changing somewhat and there is more potential for change in the future. Nonetheless, it remains exceptionally challenging to introduce new technology into the space arena as described in the next subsection.

1.3.4 Technology Readiness Levels

The *Technology Readiness Level* (TRL) is a numerical scale used to express the degree to which any particular technology is ready for application in operational space missions. As shown in Fig. 1-8, TRLs range from 1, in which only the basic principles have been observed,

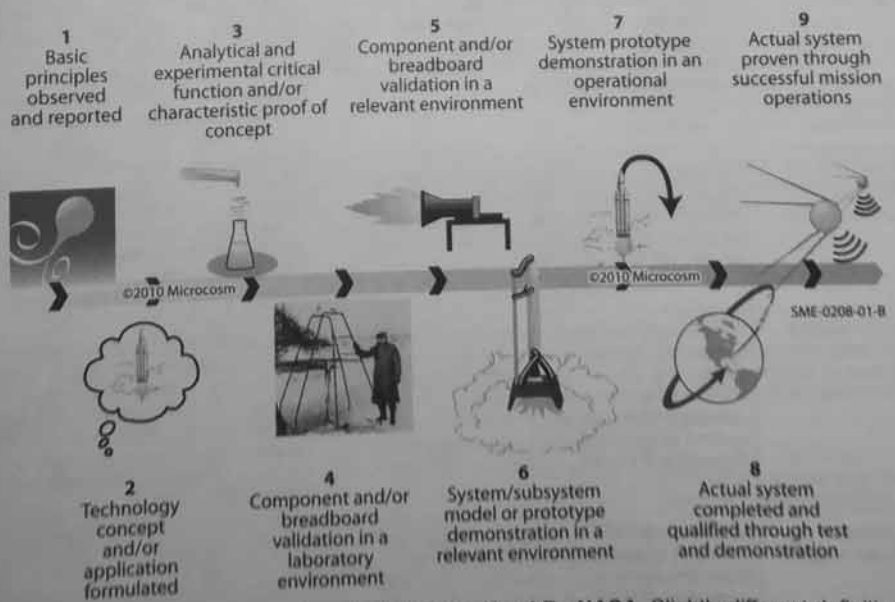


Fig. 1-8. Technology Readiness Levels (TRLs) as Used By NASA. Slightly different definitions are used by DoD as shown in Table 1-1 for hardware and in Table 1-2 for software.

to 9 for a fully mature technology that has been used successfully in mission operations. While different organizations have somewhat different formal definitions, the basic levels and ideas are now reasonably consistent. The verbal descriptions of what is required at each level for DoD are given in Table 1-1 for hardware and in Table 1-2 for software. The use of TRLs in cost modeling is discussed in Sec. 11.3.5

Table 1-1. Technology Readiness Levels (TRLs) for DoD Hardware Development. [DoD, 2009]

TRL	Definition	Description	Supporting Information
1	Basic principles observed and reported.	Lowest level of technology readiness. Scientific research begins to be translated into applied research and development (R&D). Examples might include paper studies of a technology's basic properties.	Published research that identifies the principles that underlie this technology. References to who, where, when.
2	Technology concept and/or application formulated.	Invention begins. Once basic principles are observed, practical applications can be invented. Applications are speculative, and there may be no proof or detailed analysis to support the assumptions. Examples are limited to analytic studies.	Publications or other references that outline the application being considered and that provide analysis to support the concept.
3	Analytical and experimental critical function and/or characteristic proof of concept.	Active R&D is initiated. This includes analytical studies and laboratory studies to physically validate the analytical predictions of separate elements of the technology. Examples include components that are not yet integrated or representative.	Results of laboratory tests performed to measure parameters of interest and comparison to analytical predictions for critical subsystems. References to who, where, and when these tests and comparisons were performed.
4	Component and/or breadboard validation in a laboratory environment.	Basic technological components are integrated to establish that they will work together. This is relatively "low fidelity" compared with the eventual system. Examples include integration of "ad hoc" hardware in the laboratory.	System concepts that have been considered and result from testing laboratory-scale breadboards. References to who did this work and when. Provide an estimate of how breadboard hardware and test results differ from the expected system goals.
5	Component and/or breadboard validation in a relevant environment.	Fidelity of breadboard technology increases significantly. The basic technological components are integrated with reasonably realistic supporting elements so they can be tested in a simulated environment. Examples include "high-fidelity" laboratory integration of components.	Results from testing a laboratory breadboard system are integrated with other supporting elements in a simulated operational environment. How does the "relevant environment" differ from the expected operational environment? How do the test results compare with expectations? What problems, if any, were encountered? Was the breadboard system refined to more nearly match the expected system goals?
6	System/subsystem model or prototype demonstration in a relevant environment.	Representative model or prototype system, which is well beyond that of TRL 5, is tested in a relevant environment. Represents a major step up in a technology's demonstrated readiness. Examples include testing a prototype in a high-fidelity laboratory environment or in a simulated operational environment.	Results from laboratory testing of a prototype system that is near the desired configuration in terms of performance, weight, and volume. How did the test environment differ from the operational environment? Who performed the tests? How did the test compare with expectations? What problems, if any, were encountered? What are/were the plans, options, or actions to resolve problems before moving to the next level?
7	System prototype demonstration in an operational environment.	Prototype near or at planned operational system. Represents a major step up from TRL 6 by requiring demonstration of an actual system prototype in an operational environment (e.g., in an aircraft, in a vehicle, or in space).	Results from testing a prototype system in an operational environment. Who performed the tests? How did the test compare with expectations? What problems, if any, were encountered? What are/were the plans, options, or actions to resolve problems before moving to the next level?
8	Actual system completed and qualified through test and demonstration.	Technology has been proven to work in its final form and under expected conditions. In almost all cases, this TRL represents the end of true system development. Examples include developmental test and evaluation (DT&E) of the system in its intended weapon system to determine if it meets design specifications.	Results of testing the system in its final configuration under the expected range of environmental conditions in which it will be expected to operate. Assessment of whether it will meet its operational requirements. What problems, if any, were encountered? What are/were the plans, options, or actions to resolve problems before finalizing the design?
9	Actual system proven through successful mission operations.	Actual application of the technology in its final form and under mission conditions, such as those encountered in operational test and evaluation (OT&E). Examples include using the system under operational mission conditions.	OT&E reports.

Table 1-2. Technology Readiness Levels (TRLs) for DoD Software Development. [DoD, 2009]

TRL	Definition	Description	Supporting Information
1	Basic principles observed and reported.	Lowest level of software technology readiness. A new software domain is being investigated by the basic research community. This level extends to the development of basic use, basic properties of software architecture, mathematical formulations, and general algorithms.	Basic research activities, research articles, peer-reviewed white papers, point papers, and early lab model of basic concept may be useful for substantiating the TRL.
2	Technology concept and/or application formulated.	Once basic principles are observed, practical applications can be invented. Applications are speculative, and there may be no proof or detailed analysis to support the assumptions. Examples are limited to analytic studies using synthetic data.	Applied research activities, analytic studies, small code units, and papers comparing competing technologies.
3	Analytical and experimental critical function and/or characteristic proof of concept.	Active R&D is initiated. The level at which scientific feasibility is demonstrated through analytical and laboratory studies. This level extends to the development of limited functionality environments to validate critical properties and analytical predictions using non-integrated software components and partially representative data.	Algorithms run on a surrogate processor in a laboratory environment, instrumented components operating in a laboratory environment, laboratory results showing validation of critical properties.
4	Module and/or subsystem validation in a laboratory environment (i.e.,software prototype development environment).	Basic software components are integrated to establish that they will work together. They are relatively primitive with regard to efficiency and robustness compared with the eventual system. Architecture development initiated to include interoperability, reliability, maintainability, extensibility, scalability, and security issues. Emulation with current/legacy elements as appropriate. Prototypes developed to demonstrate different aspects of eventual system.	Advanced technology development, stand-alone prototype solving a synthetic full-scale problem, or stand alone prototype processing fully representative data sets.
5	Module and/or subsystem validation in a relevant environment.	Level at which software technology is ready to start integration with existing systems. The prototype implementations conform to target environment/interfaces. Experiments with realistic problems. Simulated interfaces to existing systems. System software architecture established. Algorithms run on a processor with characteristics expected in the operational environment.	System architecture diagram around technology element with critical performance requirements defined. Processor selection analysis, Simulation/Stimulation (Sim/Stim) Laboratory buildup plan. Software placed under configuration management. Commercial-off-the-shelf/ government-off-the-shelf (COTS/GOTS) components in the system software architecture are identified.
6	Module and/or subsystem validation in a relevant end-to-end environment.	Level at which the engineering feasibility of a software technology is demonstrated. This level extends to laboratory prototype implementations on full scale realistic problems in which the software technology is partially integrated with existing hardware/software systems.	Results from laboratory testing of a prototype package that is near the desired configuration in terms of performance, including physical, logical, data, and security interfaces. Comparisons between tested environment and operational environment analytically understood. Analysis and test measurements quantifying contribution to system-wide requirements such as throughput, scalability, and reliability. Analysis of human-computer (user environment) begun.
7	System prototype demonstration in an operational, high-fidelity environment.	Level at which the program feasibility of a software technology is demonstrated. This level extends to operational environment prototype implementations, where critical technical risk functionality is available for demonstration and a test in which the software technology is well integrated with operational hardware/software systems.	Critical technological properties are measured against requirements in an operational environment.
8	Actual system completed and mission qualified through test and demonstration in an operational environment.	Level at which a software technology is fully integrated with operational hardware and software systems. Software development documentation is complete. All functionality tested in simulated and operational scenarios.	Published documentation and product technology refresh build schedule. Software resource reserve measured and tracked.
9	Actual system proven through successful mission-proven operational capabilities.	Level at which a software technology is readily repeatable and reusable. The software based on the technology is fully integrated with operational hardware/ software systems. All software documentation verified. Successful operational experience. Sustaining software engineering support in place. Actual system.	Production configuration management reports. Technology integrated into a reuse "wizard."

The idea of TRLs was first introduced at NASA by Sadin, et al. [1989] and has since been expanded somewhat by Mankins [1995], also at NASA headquarters. The basic goal of the TRL is to evaluate and ultimately minimize risk on very expensive space programs by requiring that technology is mature before it is applied to a program where immature technology could cause program delays, cost overruns, or on-orbit failures.

When you try to apply the idea of TRLs, several problems arise. Does the TRL apply to the technology broadly or the specific implementation being proposed for a given mission? If 20 m solar arrays were used on a previous mission, what TRL is applicable to 30 m solar arrays for a new mission? Here it would seem unreasonable to go back and require that the longer array be regarded as an entirely new technology. On the other hand, if we have 3 m diameter space-qualified telescope mirror, it might be reasonable to regard a 4.5 m mirror as new technology that has to go through a development and test program before being acceptable for space application, because the difficulty of building precision optics is a very strong function of their size. Similarly, the TRL of a systems is usually regarded as the lowest TRL of any of its components. Yet there is clearly a fundamental difference between a spacecraft that is simply using a new sensor that has never flown in space and one that is using 100 W more power than any previous spacecraft.

While TRLs are a useful guideline, arbitrarily applying rules is rarely a good approach. What we really want to determine about any spacecraft component or spacecraft technology are the following:

- Can it be built in a timely manner at a cost consistent with the needs of the program?
- Will it survive launch and work in the space environment that we want to use it in?
- If it has a limited lifetime, will it last sufficiently long to be useful in the system we want to fly it on?

Ultimately, there is no substitute for thinking carefully about the fundamental question, Will it work in space. This is a main topic in Chap. 7.

1.4 Spaceflight Economics

John M. Jurist, ***University of North Dakota***
David M. Livingston, ***The Space Show***

Historically, getting into space, doing things in the space environment, and returning information or materials back to Earth has often been viewed as outrageously expensive. This section examines some of the factors that contribute to the high costs of space access and utilization. Potential approaches to reducing costs are also discussed.

1.4.1 Space Budgets

According to the Space Foundation [2010], the total global space-related spending in 2008 was about $257.2 billion dollars. Their report showed that global space spending appeared to be rising at a decelerating rate from 2005 to 2008 as illustrated in Fig. 1-9. The world spending on space activities was devoted mostly to commercial satellite services and commercial infrastructure according to their report. Spending on commercial transport services, about $40 million, was so small as to disappear in a graphical representation of the major spending categories (Fig. 1-10).

Commercial space infrastructure spending in 2008 totaled $81.97 billion plus $950 million for insurance and $180 million for independent research and development (R&D) as shown in Fig. 1-11.

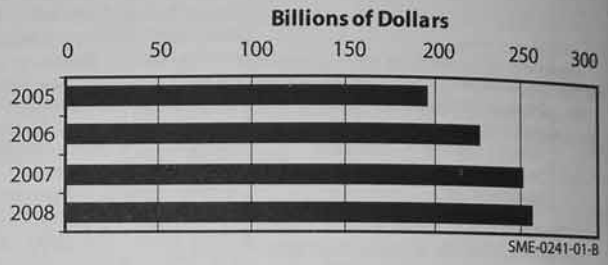


Fig. 1-9. US Aerospace Industry Cost Trends. [Modified from Space Foundation, 2009, Table 1b.]

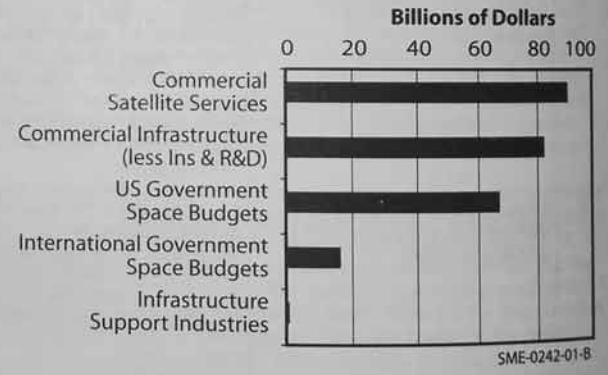


Fig. 1-10. 2008 Global Space Spending By Major Category. Space commercial transportation services at $40 million are invisible on this scale. [Modified from Space Foundation, 2009, Fig. 1c.]

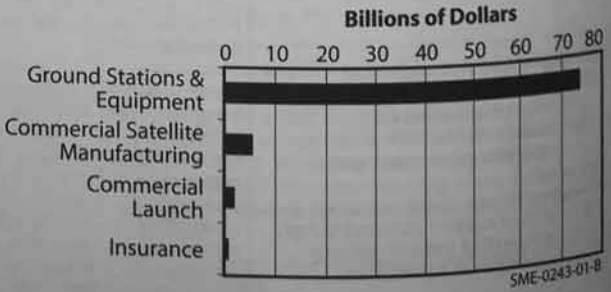


Fig. 1-11. 2008 Commercial Space Infrastructure Spending By Major Category. Independent R&D is invisible at $180 million on this scale. [Modified from Space Foundation, 2009, Fig. 1d.]

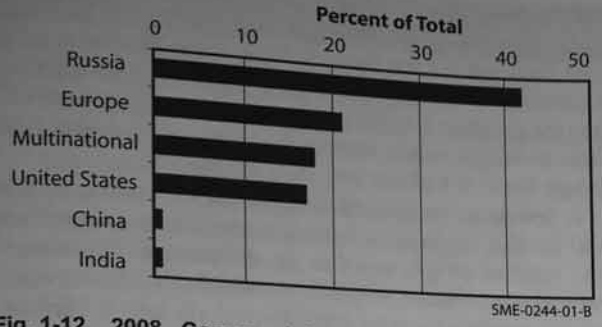


Fig. 1-12. 2008 Commercial Orbital Launch Market in 2004–2008 Based on Number of Commercial Payloads Launched. The US market share was 100% up until the first commercial Ariane launch in 1984. [Modified from Space Foundation, 2009, Fig. 1e.]

Commercial payloads launched in 2008 were dominated by the Russians and the Europeans. Together, they accounted for 63% of the total number of payloads. The United States (US) accounted for 17% as illustrated in Fig. 1-12.

The 2008 US space budget is shown by agency for 2008 in Fig. 1-13. Department of Energy (DOE) and the Federal Aviation Administration (FAA) space spending are invisible on the graph at $30 million and $10 million.

Putting the spending described above into better perspective may result from considering some historical examples of major space projects:

- Apollo Program
- Hubble Space Telescope (HST)
- International Space Station (ISS)

The NASA history website [2004] summarizes the cost of the Apollo program:

"It had been an enormous undertaking, costing $25.4 billion... with only the building of the Panama Canal rivaling the Apollo program's size as the largest non-military technological endeavor ever undertaken by the United States and only the Manhattan Project to build the atomic bomb in World War II being comparable in a wartime setting."

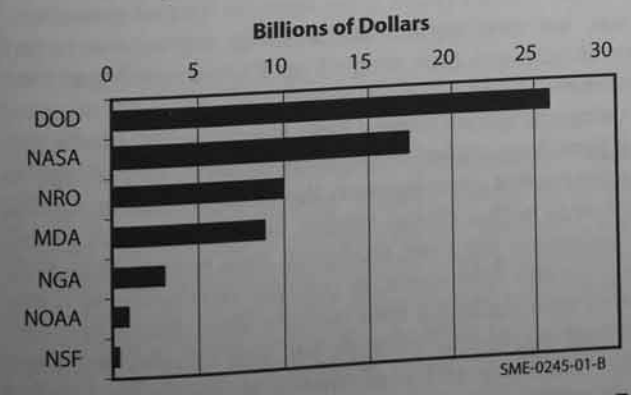


Fig. 1-13. 2008 US Governmental Space Budgets By Agency. [Modified from Space Foundation, 2009, Fig. 1m.]

Unfortunately, this reference does not specify whether this figure includes the predecessor Mercury and Gemini programs, but most authors assume that it was for Apollo only. Also, the money was spent over a number of years, so the annual spending amounts would be needed to accurately adjust the cost to current dollars. The Consumer Price Index (CPI) as an approximate inflation compensation tool is shown in Fig. 12-4. Using the data underlying that discussion, the CPI ranged from 29.9 to 36.7 during the period 1961 to 1969, and was 214.5 in 2009. Significant inflation occurred during the lifespan of the Apollo program and is attributed in part to the war in Vietnam. Thus, correcting the $25.4 billion for inflation requires multipliers ranging from 5.84 to 7.17 and results in values of between $148.5 billion and $182.2 billion in 2009 dollars.

The Hubble Space Telescope (HST) was launched on April 24, 1990 aboard Space Shuttle Discovery (STS-31) and deployed the following day. Its cost was $1.5 billion at launch [Greene, 2010]. This does not include the costs associated with the 5 Shuttle missions dedicated to repair and servicing of HST that were launched in December 1993, February 1997, December 1999, February 2002, and May 2009.

Obtaining reliable cost information for the International Space Station (ISS) is essentially impossible. One Wapedia [2010] article states that:

"The cost of the station has been estimated by ESA as €100 billion over 30 years, and, although estimates range from $35 billion to $160 billion, the ISS is believed to be the most expensive object ever constructed."

In addition to the difficulties in defining reliable cost data for large government-funded multi-year space projects, project managers must satisfy different constituencies with different agendas. The US Congress, for example, may require specific design decisions based on providing or maintaining employment in their members' districts rather than on best available engineering advice. One might speculate that an emphasis on political considerations in addition to management and technical issues influencing the process of decision-making drives up total project costs inordinately. The influence of tightly focused investigator-led missions or of artificially imposed deadlines on total project cost efficiency is unclear, as is the present emphasis on "commercial space development."

1.4.2 Diversity of Opinions

At the current stage of space technology, differing technical approaches to holding costs down are being proffered. For example, it is unclear whether Reusable Launch Vehicles (RLVs) in which vehicle costs can be amortized over many flights would be more or less expensive over their complete life cycle than simple expendable launch vehicles (ELVs). Recent advances in material properties allow parts counts of ELVs to be reduced—thereby potentially reducing their costs [Jurist et al., 2009].

Table 1-2◀, Fig. 1-13, Eq. 1-0

A currently fashionable tenet is that increasing launch rates will drive down launch costs and that RLVs are the way to accomplish that task. Another tenet considers development of RLVs to be considerably more expensive than ELVs with comparable performance.

These tenets are hotly debated in some quarters. There are additional opinions as well:

Clayton Mowry, the president and chairman of Arianespace US, when asked about high launch costs and the high cost of space access recently commented that: "There is no elasticity of demand for satellite launches in the markets we see today" [Mowry, 2010a]. This remark about elasticity was an extension of what he said in a recent Space Show interview [Mowry, 2010b]:

"It's [total program cost] driven by spacecraft and applications in space. What you are able to do when you are up there. To focus on the launch price is a bit misguided. The launch price is generally less than 10% of the overall cost. People will say the satellite will cost $200 million or $300 million and the launch price is $100 million. Therefore [it] is 25% of the cost, or 30% of the cost, or maybe even half the cost with a smaller spacecraft. Maybe the launch costs more than the satellite itself. That is a gross simplification of what takes place with a launching system. When we look at communication satellites, you've got not only the spacecraft itself, the launch price, insurance which is a significant amount of money, the operations, ground systems to fly the satellite, the people it takes to keep something in orbit for 15 years flying properly, and all the overhead that's associated with that. When you do those numbers and you look at the launch price, it ends up being somewhat less than 10% of the overall cost structure. If you took away that piece with 8% or 9% of the overall cost structure and you did not have it, would it mean they would be building and launching more satellites in that class? The answer is simply no. The demand for those spacecraft is driven by the applications. …The launch price is a small portion of that."

Charles Precourt [2010a,b] had a different take on the problem when asked about the factors contributing to high launch costs:

"Fundamentally it does not come down to only the design; a design will cost more if the acquisition strategy of the procuring agency is not efficient. The question is: Can you gain significant efficiencies within government systems acquisition management practices by placing more burden on the contractors to design and manufacture directly from the agency's higher level requirements?"

"This would be in contrast to the manufacturers waiting for blueprints from the agency which detail lower levels of system requirements, which is the current practice in several areas of Constellation. The result of shifting this detailed design burden to the contractors is delivery acceleration of produced materials, hardware, and systems. Slips in delivery schedule are the key driver of cost, not the design itself."

"Carrying a workforce is also costly, so minimizing the time between milestones, lowers total costs. If the time required for the agency to develop blueprint level requirements exceeds what a contractor could create, then the program timeline is stretched out with no payoff. Time to design means money. So again, it's not about the design itself, it's about how you acquire it."

A lesson to be gained from these quotations is that total project or mission cost is a multifactorial problem with relative weightings that are different with commercial and government projects. A multibillion dollar commercial satellite must be reliable over its design lifetime. The associated insurance for the commercial project is a significant cost that may not be directly identified in a government project. The cost of debris mitigation of a commercial satellite is incurred up front even though the mitigation takes place at the end of the satellite lifetime, and the satellite operating corporation assumes some liability even after the satellite's primary mission has ended. Range costs and insurance may not be directly identified in a government space project, and the spacecraft operations and monitoring over its mission lifetime may be performed by university grantees and funded through research and training grants.

1.4.3 Overview of Economics of Aerospace Systems

Getting into space, doing things in the space environment, and returning things back to Earth are all expensive. Returning information back to Earth is cheap. Two primary metrics are generally used to specify launch and payload costs: Cost per unit mass delivered to space or total cost for a payload delivered to space. The former is more appropriately applied for delivery of commodities or multiple payloads and the latter for dedicated launches of specific payloads.

Some of the relevant factors are discussed below. Space programs tend to be over budget and completed late. These problems appear to be partly rooted in starting out with ill-defined projects that require research, development, and testing as part of the early and mid-portions of the program. The space industry and its companies are unique. As such, space companies do not really compare easily to terrestrial businesses, even to those with similar risks.

Other factors relate to the specific type of space business, but most cannot be started in a garage or a tiny warehouse and then ramped up from a small part time business. Component manufacturing for a lot of space businesses cannot benefit from assembly line production or from huge global or regional markets. One exception to the market comparison is the communication satellite but even with a global market, the satellites and launch vehicles are not mass produced, they require exceptionally careful fabrication in pristine environments (clean room manufacturing with positive atmospheric pressure, filtered air, special overalls and hair covers). Additionally, reliability and performance standards that are not commonly found in most terrestrial businesses are a must, given the expenses involved.

Table 1-2◀, Fig. 1-13◀, Eq. 1-0

Comparisons of Various Industries

SPACE INDUSTRY:
- High technical risks—complex systems
- Routine requirements for clean room assembly and testing
- Hostile operating environment
- Labor intensive production
- High entry barriers—expensive R&D and tooling
- Complex and costly integration technology and methodology
- Launchers are finicky—need lots of attention
- Narrow market—few customers
- Few suppliers—limited competition
- Long lead times
- Transportation infrastructure critical
- Extensive testing during and following manufacture and during prelaunch phase
- Heavy regulatory burdens
- Not location centric

MINING:
- Hostile operating environment for men and machines
- Dirty, potentially hostile environment
- Labor intensive
- High entry barriers—expensive essentially hand made equipment
- Equipment needs lots of maintenance
- Limited markets—broader than for space industry
- Limited numbers of mines—geography limits competition
- Long lead times for mines to become productive
- Transportation infrastructure critical
- Heavy regulatory burdens
- Location centric

BIG PHARMA:
- High technical risks—complex equipment and methodologies
- High entry barriers
- Wide markets
- Long lead times
- Transportation infrastructure moderately critical
- Heavy regulatory burdens
- Not location centric

BIOTECH (STARTUPS):
- High technical risks
- Labor intensive
- Low entry barriers—usually evolve from academic laboratories
- Narrow markets—goal is acquisition by Big Pharma
- Long lead times

BIOTECH (STARTUPS) Continued:
- Transportation infrastructure not critical
- Regulatory burdens low until human trials begin
- Usually location centric

AMBULATORY SURGICAL CENTERS:
- Low technical risks
- Labor intensive
- Moderate entry barriers—no more than several million dollars
- Broad markets with favorable demographic trends
- Short lead times—roughly 1 yr from funding to operations with 1–2 months float on receivables
- Transportation infrastructure not critical
- Heavy regulatory burdens
- Extremely high returns on initial investment
- Location centric

EARLY MICROCOMPUTER INDUSTRY (HARDWARE):
- Variable technical risk—ICs developed for other purposes
- Low entry barriers
- Wide market
- Highly competitive market over time
- Variable lead times
- Transportation infrastructure not critical
- Low regulatory burden
- High returns on initial investment
- Not location centric

EARLY MICROCOMPUTER INDUSTRY (SOFTWARE):
- Low technical risk
- Low entry barriers
- Potentially broad market
- Variable competition increasing over time
- Transportation infrastructure not critical
- Low regulatory burden
- Very high returns on initial investment
- Not location centric

SANDWICH SHOP:
- No technical risk
- Low to moderate entry barriers
- Potentially broad market
- Variable competition increasing over time
- Transportation infrastructure not critical
- Low to moderate regulatory burden
- Variable return on investment
- Location centric

The frequent argument that the space industry needs to follow the microcomputer industry model is, in our opinion, invalid. Early microcomputer hardware was frequently designed and produced by a few people working in their spare time. Early software was often developed by one or a few people also keeping their day jobs with the only entry barrier being access to a microcomputer. The high technical, financial, and regulatory risks associated with the space industry are responsible for the cliché "that the way to make a small fortune in space is to start out with a large one."

Perhaps the only other sector with many similarities to space business is mining. Direct comparisons between various economic sectors and industries are difficult and subject to debate. Some simple comparisons between the space industry and various other industries, including small entrepreneurial efforts are made in the boxed section above.

Table 1-2◀, Fig. 1-13◀, Eq. 1-0

Rational commercial endeavors seek to create high returns for investors. Corporate officers are legally required to protect the financial interests of their shareholders, a responsibility known as the *fiduciary relationship*. Deviation from the capitalistic goal of seeking high returns is more common in closely held corporations* and those largely funded by one or a few people with special interests (angel investors—defined in Sec. 12.1.4) and less likely in those funded by venture capitalists (also defined in Sec. 12.1.4) or in publicly held corporations.

Although one might argue that return to the public treasury by government investment is or should be the dominant motivation for expenditure of taxpayer money, other factors are also important. Those factors may include expenditures for national security, creation of new markets to encourage long-term economic growth, and support of existing markets to provide employment stability. The end result is that programs funded by government investment must satisfy multiple constituencies—including Congress. Private corporate managment must satisfy the corporate board of directors and indirectly the shareholders who elect the board.

1.4.4 General Accepted Accounting Practices (GAAP) and Government Accounting

Whenever government funding is involved, corporate vendors must adhere to a variant of General Accepted Accounting Practices. GAAP is also a necessity for compliance with US securities laws. Often, this requires keeping and reconciling multiple sets of financial records. GAAP is discussed in more detail in Sec. 12.2.1.

1.4.5 Why is Space Expensive?

Because design, development, fabrication, testing, and operation of both launch vehicles and payloads are so labor-intensive, labor costs are a dominant factor in space mission design and implementation.

Based on data accumulated by NASA, the total cost of an engineering man-year in 2007 was estimated at $252,000 in the United States (then-year dollars) [Koelle, 2010]. This figure appears to be increased considerably when compared to $208,700 in 2000 and $156,200 in 1990, but inflation during the same period largely parallels the increase. From 1960 to the late 1970s engineering costs outpaced inflation as represented by the Consumer Price Index. Figure 1-14 shows US aerospace industry cost trends as derived from Koelle's data in then-year dollars and in 2007 dollars in order to account for general inflation.



Fig. 1-14. US Aerospace Labor Cost Trend in Then-Year Dollars. Modified from Koelle [2007], Table 1-II.

Koelle also developed relative productivity factors per working hour and effective working hours per year for the United States versus other countries. The productivity models incorporated relative education and standard work week, among other variables.

In order to attain the requisite reliability, launch vehicles, satellites and other payloads are essentially hand built with great attention to post-production testing and maintenance of quality control. Use of clean room technology for fabrication and assembly drives costs up markedly compared to more traditional manufacturing. Transportation of completed hardware to its launch destination also adds to costs because of the attention paid to prevention of shipping damage. Arrival at the destination then necessitates further testing.

In addition to his work described previously, Koelle developed a series of parametric equations for estimating standard aerospace industry costs associated with development of rockets, missiles, and space transportation systems. Koelle's models use linear regression equations relating the logarithm of cost to the logarithm of the mass of the system or subsystem. Individual constants have been derived from historical data for various types of rocket motors, airframes, and other systems. The historical data include suborbital rockets and liquid-fueled missiles. Additional multiplicative factors relate to the experience of the development team, the technical complexity of the development, development time compression requirements, and management structure.

Finally, Koelle developed a similar set of models to estimate the cost of a first production unit (Theoretical First Unit or TFU). This is significant because production costs tend to decline with increased production numbers—the so-called learning curve discussed in Sec. 12.2.3.

Compared to, for example, automobile manufacturing, production runs for launch vehicles and satellites are very small. This precludes taking full advantage of mass production techniques and of progressively working down the learning curve that reduces production costs with increases in production run.

One aspect of maintaining high quality in the manufacture, processing, and assembly of satellites and rocket components, and payload integration is the frequent use of clean room environments. This drives up overall project and mission costs. Attaining the required

* A closely-held corporation is one in which there are few shareholders and which may not meet the standards required for a publicly-held corporation. Different states have different requirements on the maximum number of allowed shareholders before a corporation must be registered as a public corporation. Closely held corporations tend to be smaller in terms of capitalization and are frequently run more informally than public corporations.

high reliability also requires considerable post-production testing and quality assurance both at the subsystem and fully integrated systems level. Long drawn out programs and technical advances frequently result in shortened life cycles for a specific program because of the temptation to initiate development of advanced models (new and improved versions). Low launch rates, and especially launch uncertainties tied to vehicle problems, range and safety considerations, orbital dynamics, and weather drive up costs. This contributes to the so-called "standing army" problem.

Given the fact that on-site repair and refurbishment of a space asset such as a satellite is impossible at worst (as in a planetary probe, for example) and extremely expensive at best (as in repairing the Hubble Space Telescope), reliability is very important. That reliability comes from extensive design and testing at all phases of product development, manufacturing, and delivery to the end user.

Launch vehicles are very complex and have multiple failure modes—many of which can result in loss of mission and loss of vehicle. The potential loss of very expensive satellites and space vehicles drives the need for very high reliability of launch vehicles and high insurance costs to cover associated risks. The reliability problem drives design toward redundancy and fault tolerance.

For example, as stated previously, launch vehicle cost accounts for perhaps 10% of the total mission cost of a satellite or space vehicle placed into space. The remainder comes from amortization of research and development of the launch vehicle, payload integration into the vehicle, pre-launch testing and verification of the vehicle systems, payload insurance against loss, range fees for tracking and range safety, as well as operation and control of the satellite or vehicle. This may result, for example, in a 95% probability of successful payload placement. By cutting the launch vehicle cost in half and reducing launch reliability to 90%, the risk of mission loss is doubled while 5% of the total cost of payload placement and operation is saved.

The high cost of space launch motivates manifesting multiple payloads whenever possible. Launch service personnel must then consider payload integration to assure that multiple payloads do not interfere with one another. This adds another dimension to the cost problem. Payload integration costs are also incurred when assuring that a given payload is compatible with the internal environment of the launch vehicle and assuring that the interaction of the payload or payloads with the vehicle dynamics are within tolerable limits for the specified vehicle. These are labor intensive activities.

Technical advances lead to potential improvements of a given launch vehicle and spacecraft. At some point, incorporation of these advances into production become desirable for various, and often noneconomic, reasons. With low production rates, research and development (R&D) costs cannot be amortized over a large production run and unit costs are high as a result. Keeping vehicle production lines open and waiting for additional demand reduces programmatic costs at the expense of increasing overhead. This tradeoff results in decision-making determined by reimbursement mechanisms associated with the procurement contract.

It is a truism that delays drive up costs. These delays result in increased labor costs, additional management oversight, increased finance costs (interest on loaned capital), and opportunity costs.

1.4.6 How Might Costs Be Minimized?

Extrapolation of that which is not forbidden by science and technology to what may be possible involves a risk that does not exist with already proven technology. That risk usually results in cost overruns and is not limited to space technology, but can occur in non space-related activities. One way to hold down R&D costs is to make use of already-developed technology to the maximum possible extent. This tends to drive evolutionary progress at the expense of revolutionary progress.

Given that a launch system consists of a collection of parts, most of which have non-zero probabilities of failure during a launch, reducing the parts count of a system may improve reliability. In addition, designing for fault tolerance with redundancy or other modalities may improve overall reliability and modestly reduce total launch costs. Design processes should ideally seek the "sweet spot" in the tradeoff of reliability and component cost.

A very important consideration in creating a large scale business with large capital investment is to clearly identify the products or services to be supplied rather than adhering to the "build it and they will come" philosophy of hoping for demand to eventually manifest itself.

As an example of minimizing cost, we will consider range costs which are a large component of the total flyaway costs of launching into orbit. Comprehensive analysis of orbital launch economics shows that cost-effective launch operations are inhibited by high (and perhaps unwarranted) range fees imposed on a launch company for use of a federal range. The adverse economic impact of range fees are only a symptom of the underlying disease. The fundamental problem is badly outdated federal launch and operational systems and an infrastructure that will not support high flight rates and commercial operations. The federal range launch infrastructure was historically developed for missile launches [Muncy, 2005]. This governmental infrastructure is now maintained, operated, and updated by numerous government contracts. For example, the US Air Force has a contract for operations, a certain company has the contract to repair broken equipment, another company handles R&D needs, and still another company handles the procurement of new or replacement equipment. There is no single entity with the authority to make rational economic decisions related to the launch infrastructure. The systems are outdated and they are not user friendly. It is challenging if not impractical for a federal range to cycle more than one launch daily. Therefore, the existing federal range launch infrastructure cannot support the commercial launch needs anticipated by private

launch companies with their commercially designed rockets and their projected markets.

Discounting range fees does not solve the underlying problem although such discounting may allow a more cost effective economic profile for a commercial operation. However, discounting does nothing to modernize the obsolete infrastructure, systems, and management of federal ranges.

If the goal is to have frequent multiple launches, it is essential to run the spaceports or ranges as rational economic entities. It may be possible to privatize and streamline federal range operations, but there is not much interest among the range operating entities or the US Congress to do this. Unless this changes, the federal range will still be operated uneconomically and require taxpayer subsidy.

Pressure to overhaul the federal ranges and launch infrastructure might increase as private spaceports receive licenses for vertical launches. A good example of potential pressure can be seen by examining the developing private Spaceport America in New Mexico. This range plans to offer private rocket companies state of art launch and range services designed to be commercial from the ground up. As it offers launch services, what will happen to the Florida Spaceport which is stuck in the federal range environment? If Florida wants to compete for private launch business, it will have to modernize. Simply put, without comprehensive modernization, the existing Florida launch infrastructure will not easily support competitive commercial launch services. While the Mid-Atlantic Regional Spaceport (MARS), the commercial component of the federal Wallops Island Launch Facility, and Kodiak Island may be able to launch private rockets with lower range costs, their infrastructure is still not suitable for anticipated commercial operations.

Regulatory relief may be as important as range cost relief. Some launch companies have opted to avoid the national ranges entirely by choosing air launch or sea launch. Two NewSpace CEOs have privately indicated that they would not use the national ranges even if they were free because of the regulatory burden.

1.4.7 Other Launch Cost Issues

Major trends in the space industry in the US appear to be:

- Decreasing cost of basic electronics
- Increased capability of software due to more capable, faster processors
- Increasing performance capability of electronic packages of a specific size
- Increasing capability of mission-related software,
- Increasing reliability of basic components
- Increased use of advanced composite technology to reduce structural masses
- Increased use of computer technology for decreasing staffing requirements for launch day operations
- Motivation to use commercial launch providers whenever possible for government projects as well as for commercial projects
- Increasing political pressure to separate NASA from launch services
- Establishment of private spaceports to compete with federal ranges
- Severe budgetary pressure on federal discretionary spending

The trends outlined above result in decreased size and mass of payloads designed to provide specified capabilities. Because of this resultant trend, launch cost per unit mass of payload becomes less important at some point than total launch cost for dedicated missions as the last stage becomes the most massive part of the orbited mass. Additional study of the minimum feasible size of an orbital launch system is indicated to better define where this break point may exist. Decreased mass of specific payload packages and the availability of excess payload mass in existing specific launch vehicle configurations drive a tendency to manifest multiple payloads on a given launcher. That results in entities providing payload brokering to multiple space programs. An example of such an entity is Detachment 12 at Kirtland Air Force Base, New Mexico. Current technological, economic, and regulatory realities forbid payload delivery to LEO for true costs of less than about $2,000/kg [Jurist et al., 2005].

Mowry [2010b] has said that one of the successful strategies employed by Arianespace is to make multiple rockets at one time, all identical. The only alterations needed per rocket deal with specific payload interfacing. Not only does this strategy allow for greater probability of launch success, it maximizes the potential for some cost reductions. This is related to the learning curve discussed in Sec. 12.2.3. The SpaceX Falcon 1 is being replaced by the Falcon 1e, which has yet to fly. The Falcon 9 has flown 2 successful orbital launches at the time of this writing. As of September 2010, their offered prices to LEO are $10,792/kg and $5,358/kg for the Falcons 1e and 9, respectively, exclusive of insurance [SpaceX, 2010a, 2010b].

Creating fleets of reusable launch vehicles (RLVs) is unlikely to solve the launch cost problem unless they are large and have lifetimes of many frequent flights. Of course, this presupposes sufficient demand for space launch. Present technology may permit orbital RLVs, but economic factors currently forbid them. The shuttle Space Transportation System (STS) is only partly reusable and is generally considered to be an economic failure. Nobody has currently demonstrated or convincingly documented the ability to:

- Design, fabricate, and economically fly orbital RLVs
- Finance and then amortize the necessary R&D
- Define a clear, solid business plan to implement the program required to do so

- Demonstrate the market that would support the costs of creating such vehicles
- Develop a viable strategy to go from our present flight rate of expendables to the high flight rates of RLVs projected for the mature industry which would permit achieving low cost flights to LEO

Two of the many major obstacles to attaining this goal are the nontechnical factors of range costs as discussed previously and insurance (discussed in Sec. 12.2.4).

Advances in electronic miniaturization and performance have led to improved capability of unmanned space vehicles. Humans in space, on the other hand, are massive and not subject to miniaturization. They require massive life support systems and large quantities of consumable supplies. There are many unresolved issues related to long-term human space flight, such as the effects of microgravity, health effects of prolonged low level and episodic high level radiation exposures, and degradation of life support systems. The trends toward miniaturized electronic systems therefore drive decision-making processes toward unmanned systems at the expense of human-tended missions and have the paradoxical effect of driving up the costs of human space flight. Further consideration of this issue is beyond the scope of this text.

1.5 The Wide Range of Space Mission Applications

The goal of space mission engineering should be to satisfy the needs of the customer or end user as effectively, as quickly, and at as low a cost as possible. But who is that customer or end user and what is it that they want? The answer differs greatly, depending largely on 3 key issues:

1. The aspect of space that you're exploiting
2. Where you're coming from, i.e., the space community you're in, and
3. What you perceive as the role of space

In Chap. 2, we'll explore the various space communities and what they perceive as the role of space. Here we'll summarize the varying characteristics of space that are exploited in different space missions and look at the very wide range of mission applications that these imply. If our mission doesn't rely on some fundamental and unique characteristic of space, it will likely cost more to do in space than on Earth. For example, there is little reason to manufacture and distribute low-cost consumer goods or manage a restaurant chain in Cleveland from space.

What fundamental characteristics of space make space missions worthwhile? Table 1-3 lists the primary characteristics of space that distinguish it from Earth-based activities. Exploring and utilizing space serves various objectives, from extremely practical telecommunications and weather, to major scientific observatories hoping to understand the Universe better, to advance military applications and exploring and exploiting the Moon and the planets. Space missions are very diverse partly because they use many different characteristics of space. For example, materials processing uses the microgravity and high vacuum of space, without regard to the spacecraft's position over the Earth. Conversely, communications or observation satellites emphasize Earth coverage as the most fundamental space characteristic used to achieve their mission.

Table 1-3 reveals a second important feature—the varying levels at which different characteristics are currently being used in space. For example, many missions use the global perspective of space—for telecommunications, weather, navigation, and other aspects of Earth monitoring. Space-based telecommunications will likely continue to grow, but is already a major and mature industry.

Satellite communications by telephone and television have become a part of everyday life and have helped to bring about a communications revolution that has contributed significantly to our shrinking world. Equally dramatic changes are likely in the future as new applications for space-based communications and navigation continue to emerge and every cell phone, car, and notebook computer will know where it is on the Earth to within a few meters.

In contrast to telecommunications, materials processing and precision manufacturing in the gravity-free environment of space is only in its infancy. Major strides appear possible in pharmaceutical and semiconductor devices that may bring about an entirely new industrial segment. Exploiting space's almost limitless resources is even further removed. Unlimited continuous power and huge, accessible supplies of physical materials may, in the long run, maintain an industrialized society without destroying the Earth's fragile environment. These objectives will require greater vision, and lower launch costs, than is required for the more fully developed areas of communications, resource mapping, and monitoring.

We see from Table 1-3 that we have either not used or only begun to use most of the major characteristics of space, so changes in future space exploration should be far larger than what has occurred to date. To take practical advantage of these characteristics, we must greatly reduce the costs of exploring and exploiting space. Finding ways to lower these costs is a principal objective of this book. (See Chap. 13 and Wertz and Larson [1996].)

Based on Table 1-3, we would expect to find a wide range of space mission applications and this is indeed the case as shown in Fig. 1-15. Note that some missions fall into multiple categories. For example, GPS is a navigation system used by a large number of people worldwide, but is funded, built, and run by US Air Force as a military mission. Similarly, the Apollo program could reasonably fall under exploration or science and, in many respects, there is a great deal of overlap between those two catego-

Table 1-3. Characteristics of Space Exploited by Various Space Missions. Note the wide variety and that many are only beginning to be used. (Spacecraft acronyms are defined in the index.)

Characteristic	Relevant Missions	Degree of Utilization	Sample Missions
Global Perspective	Communications Direct Broadcast Navigation Weather Surveillance	Some are mature industries; major new advances will come with increased onboard processing	IntelSat GeoEye NOAA weather satellites Direct TV FireSat II, SCS
Above the Atmosphere	Scientific observations at all wavelengths	Well developed; space observatories will continue to dramatically change our view of the universe	Hubble Space Telescope Spitzer Chandra X-Ray Observatory
Gravity-free Environment	Materials processing in space	Largely dormant because of high cost; may be many future applications	ISS Wake Shield Facility Space Manufacturing
Abundant Resources	Space industrialization Asteroid mining Solar power satellites	Essentially none	Space colonies Solar power satellites Asteroid mining
Exploration of Space Itself	Exploration of Moon and planets, scientific probes, asteroid and comet missions	Initial flybys and some planet orbitting missions have been done; Some landings done or planned; limited human exploration	Manned Lunar or Martian bases Apollo Mars rovers Space tourism

ries. One category that has been omitted is weapons of mass destruction in space, which are banned by international treaty. (See Sec. 12.3.) This is important in that space is often referred to as the "ultimate high ground." The implication of this is that unilateral control of space represents a very strong position from which to militarily control the Earth below. Fortunately, the major spacefaring nations at the start of the space age, the United States and the Soviet Union, have chosen to peacefully co-exist in space.

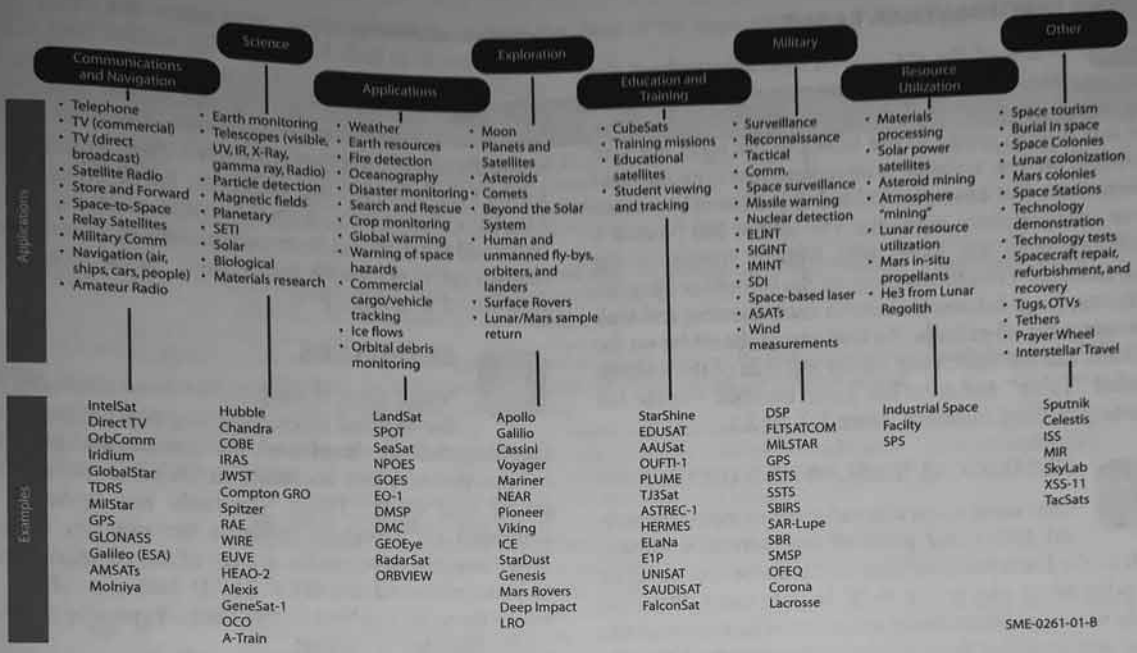
Certainly the most striking feature of Fig. 1-15 is the exceptional variety of mission applications. This is both a challenge and an opportunity for space mission engineering. Most people work in only one, or at most a few, of these many application areas. Our view of "the space program" and how spacecraft are designed, built, and flown is constrained by this limited exposure. It is likely that large weather satellites in GEO, student CubeSats, and spacecraft built for space tourism will have relatively little in common. This makes it hard to generalize about how these missions are or should be built. But it is also an opportunity to bring concepts, hardware, software, and processes developed in one arena into other areas to give us a wider range of tools, equipment, and processes to solve the problems of the end user. Can we, for example, use rapidly advancing CubeSat technology to dramatically lower the cost of other, larger application and surveillance satellites? We will return to this theme in Chap. 2 where we look at the very diverse set of space communities that largely act independently of each other. Ultimately, the main characteristic that nearly all of the application areas and space communities have in common is a desire to achieve their objectives in a timely manner at minimum cost and risk—i.e., to do a good job of space mission engineering.

1.6 Sources of More Information

Text and reference books such as this one often have extensive references and bibliographies. However, the books and papers cited are frequently hard to find and, of course, the lists aren't updated after the book is published. Our goal with Space Mission Engineering is to dramatically change this, such that much of the information cited is immediately available on the web and nearly all of the remainder is available within a day or two, so that you can use the full resources of this book for a project or proposal that is due next week. Here's how we're doing that.

This book has an associated website, www.smad.com. The password for the website (not case sensitive) is ASTRODATA. As discussed below, the website has 6 main types of data:

- ***Reference Material***—all of the references and bibliographic material from throughout the book with direct links to where you can obtain it.
- ***Information Tables***—most of the information tables in the book that can be updated are on the web and are updated from time-to-time as the information changes.
- ***Numerical Tables and Graphs***—Most of the graphs and numerical tables in the book are adjustable for your particular project of interest. The Excel models that created them are on the website for you to download.
- ***Extended Discussion***—Some relevant material that is beyond the scope of the printed text is included on the website.
- ***Standards***—Astronautics standards are regularly updated so references to them are on the website.



Get Info **Fig. 1-15. The Wide Range of Space Mission Applications.** Note that some missions fall into multiple categories. See the index for definition of mission acronyms

- ***Errata***—Sadly, virtually all technical and reference texts are in need of fixing typographical (or mental) errors that inevitably occur. These are also on the website.

Wherever there is material available on the website, it is marked in the text by a "computer button," such as:

Get Ref Get Info Live Calc Get More Get Stnd

REFERENCE MATERIAL

Get Ref

Each chapter in the book has the usual list of references, primarily to professional papers and other books. This list is reproduced on the website with a web reference to where you can obtain each reference. Some of the papers and a small number of books are available for free and can be downloaded immediately. Most professional papers are copyright, typically by one of the professional societies discussed below. In this case, the link from the book website will take you to where that specific paper can be purchased from the society that published it. In most cases, you can get additional information and an abstract or one page of text for no cost and can purchase the full paper, typically for $15 to $30. It can then be downloaded immediately. Books can, of course, be purchased from nearly any online bookseller. In most cases, the link from the book website will take you to the description of that volume on the Microcosm Astronautics Books website, or to a more recent edition of the same book, if one has been published. From there, the book can be purchased and shipped via next day air or normal delivery. The Microcosm bookstore is the publisher of this volume and maintains the book website. It tries to keep all of the book references in stock and available, many with a reduction from the list price.

References are, of course, fixed at the time that the chapter is written. Therefore, the website for each chapter also includes an annotated bibliography of books and some survey papers that are relevant to the chapter, with a 2–3 sentence description of each and a web link to where it can be purchased or downloaded at no cost. The bibliography is only on the website so that it can be continuously updated as new volumes become available. In addition, the bookstore manager is knowledgeable about what is available in astronautics books and can provide suggestions and recommendations. The current contact information is www.astrobooks.com, bookstore@smad.com, 1-888-ASTROBK, or 310-219-2700 extension 105. If this information changes, it will be on the website. Finally, Microcosm attempts to have available on site one copy of each of the items referenced in the bibliography. If you're in southern California, feel free to come by and look at the reference material during normal business hours. We're about 2 miles from LAX at 4940 W. 147th St., Hawthorne, CA 90250. The goal here is to provide as nearly as possible "instant access" to high quality, referenced and reviewed technical information in astronautics, without violating the copyright rights of those who own the material. If you have any suggestions on how we might do this better, please get in touch with us.



INFORMATION TABLES

Much of the material in tables throughout the book—such as hardware manufacturers, lists of missions, or professional societies—changes with time. For this reason, many of the tables in the book are also included on the website with either more or updated information. For example, Table 2-2 is a list of representative interplanetary missions. The author has created a more extensive list, along with website references for each mission. Thus this table is on the website with more missions, more information about each mission, and web references for all of them. To find any of the tables on the website, use the search box on the left side of the website labeled "Table" and enter the Table number. For the list of interplanetary missions, enter 2-2, or 2.2.



NUMERICAL TABLES AND GRAPHS

Like most texts, this volume has many numerical tables and plots of representative values, such as the Earth satellite table in the inside rear cover or the orbit decay plot in Fig. 9-18. In both cases, these are made more useful by being downloadable to your computer and modified for your particular mission. The originals of these tables and plots are done in Excel. The Excel model itself (often with some modifications to make it more widely applicable) is then put on the website. For example, the table of satellite orbits on the inside rear cover is on the website. You can adjust the range of altitudes or other parameters and can also change the central body from the Earth to any other planet. You can even create your own planet if you'd like. It also allows you to plot a number of the variables so that you seen how different parameters vary with altitude. Similarly, Fig. 9-18 allows you to plot the decay of both apogee and perigee over time for essentially any circular or elliptical low-Earth orbit. As with the informational tables, use the search boxes on the left side of the website labeled "Table," or "Figure" and enter the table or figure number, such as H-1 for the satellite parameters on the inside rear cover or 9-18 or 9.18 for the orbit decay plot.

A word of caution—we have tried to make the Excel models very easy to use by clearly distinguishing the user inputs, but we have not prevented you from changing formulas or entering values that would not be meaningful. You can download the Excel model to your computer and change it as you choose at your own risk. If you choose to have gravity fall off as $1/r^3$, you can do so, with perhaps interesting, but likely not very realistic, results. We recommend that after a few experiments of this sort, you download the model again and start over.



EXTENDED DISCUSSION

In some cases, there is an occasional need for a more extended discussion than would be needed by most readers of the book. If this discussion is well covered in one of the outside references, we will simply reference the other books or papers. However, in some cases, alternative sources either don't exist or are inconvenient to find. In these cases, a more extended discussion is included on the web and referred to in this volume with the GET MORE button. For example, vector and matrix arithmetic is well known and covered in many standard texts and college courses. Therefore, we haven't included it here. However, for those that would like a good summary, with consistent notation, we have reprinted the material from an appendix to an older astronautics reference (with permission) on the website and referenced it in App. F.



STANDARDS

Many areas in astronautics have standards similar to those throughout engineering and manufacturing that are developed, maintained, and published by the professional societies in order to reduce both errors and costs. These standards are updated and expanded on a regular basis as the industry evolves. Therefore, we have put the source of the standards on the website and used the GET STND button to show you where there are applicable standards. Typically these can be downloaded at no cost.

ERRATA

While none of us would like to admit that errata are ever necessary, in the rare event that they are required, errata for each chapter are included at the end of that chapter on the website.

The major astronautics professional societies worldwide, listed in Table 1-4, publish a large fraction of the professional data used by the astronautics industry. This is primarily in professional journals, listed in Table 1-5, published by the societies or as a part of conferences. Many individual conferences are arranged on specific topics. However, most of the major conferences, listed in Table 1-6, occur annually or biannually. In addition, several of the societies publish books that are either conference proceedings or individual volumes on specific topics. The individual books can typically be purchased through the normal book retail outlets on the web (such as Amazon, Barnes and Nobel, or Microcosm Astronautics Books), but very few are available in physical bookstores, other than Microcosm Books, because of the relatively low volume. A great deal of both popular and technical material also exists in various sites on the web, although much of the web-based material is not refereed or well-edited, and may be less reliable.



There are also a large number of websites associated with space data and space news. These change rapidly, however, and are listed only in Table 1web-1 on the website. Note that most of the websites associated with space are concerned with news, events, and specific space missions and rarely contain validated technical information. If you have additional suggestions on how to provide better information to the Space Mission Engineering readers, please let us know at bookstore@smad.com or bookproject@smad.com.

Get Info **Table 1-4. Major Astronautics Professional Societies.** Most of the larger professional societies cover fields other than astronautics as well.

Society	Comments	Website
AIAA = American Institute of Aeronautics and Astronautics	The largest American professional society with a major focus on astronautics. Holds national and regional conferences. Publishes both refereed journals and many books. AIAA technical committees create many of the standards used in the space industry. (See Sec. 6.6.)	www.aiaa.org
AAS = American Astronautical Society	The largest of the American societies devoted exclusively to astronautics and space exploration. (Note that another AAS = American Astronomical Society, is the largest astronomical organization is the US.)	astronautical.org
BIS = British Interplanetary Society	One of the oldest professional societies in astronautics. Publishes papers on topics, such as interstellar travel, that are more far-reaching than many of the other professional journals.	www.bis-spaceflight.com
IAF = International Astronautical Federation	The IAF is the principal worldwide organization for astronautics with an annual conference held at various locations throughout the World. The IAF is a collection of organizations, i.e., organizations not people are IAF members. Associated with the IAF, are the ***IAA*** = International Academy of Astronautics, which is a worldwide association of astronautics professionals, and the ***IISL*** = International Institute of Space Law, which focuses on legal and treaty issues in the exploration of space.	www.iafastro.com
IEEE = Institute of Electrical and Electronics Engineers	The world's largest professional technical society. Focused on electrical engineering, but does a great deal of work in astronautics, including publication of many relevant papers.	www.ieee.org
INCOSE = International Council on Systems Engineering	A relatively recent organization, founded in 1990, but with a strong interest in space systems engineering. Intended to be a focal point for the dissemination of systems engineering knowledge.	www.incose.org
SEDS = Students for the Exploration and Development of Space	An independent student-based organization (both graduate and undergraduate) with chapters at various universities around the World. Loosely affiliated with some of the professional space organizations. Runs both conferences and various local activities.	seds.org
NSS = National Space Society	Independent educational organizations formed from the merger in 1987 of the National Space Institute and the L5 Society. Dedicated to the creation of a spacefaring civilization.	www.nss.org

Get Info **Table 1-5. Major Professional Journals in Astronautics.** Most are published by one of the professional societies. Technical journals are those that provide refereed professional articles, typically for specialists in a particular area. Popular journals are monthly magazines with broader appeal and relevant to a broad segment of the astronautics community. Many of the larger aerospace organizations also publish semi-technical journals on work being done within the organization.

	Comments	Abbreviation	Content
AIAA	*Journal of Spacecraft and Rockets*	JSR	Technical
	Journal of Guidance, Control, and Dynamics	JGCD	Technical
	Journal of Aerospace Computing, Information, and Communication	—	Technical
	Journal of Propulsion and Power	JPP	Technical
	Aerospace America	AA	Popular
AAS	*Journal of the Astronautical Sciences* (quarterly)	JAS	Technical
	Space Times (bi-monthly)	ST	Popular
BIS	*Journal of the British Interplanetary Society*	JBIS	Technical
	Spaceflight (includes a running chronology of space missions)	—	Popular
	Aviation Week and Space Technology (weekly)	Av Week	News
	Space News (weekly)	SN	News

Table 1-6. Representative Regularly-Held Conferences in Astronautics. Conferences are typically organized and run by, or on behalf of, one or more of the professional societies. For more information see the website of the sponsoring professional society.

Sponsor	Conference	Location	Date
AIAA	Aerospace Sciences Meeting	Florida	Jan
AIAA, FAA	Commercial Space Transportation Conference	Varied (US)	Feb
AIAA, ASME, ASCE, AHS, ASC	Structures, Structural Dynamics and Materials Conference	Colorado	April
AIAA	Atmospheric Space Environments Conference	Hawaii	June
AIAA, ASME, SAE, ASEE	Joint Propulsion Conference	San Diego	July
AIAA	Guidance, Navigation, and Control Conference	Varied (US)	Aug
AIAA	Modeling and Simulation Technologies Conference	Varied (US)	Aug
AIAA	Space 20(year) (largest of the AIAA conferences)	Varied (US)	Sept
AIAA	AIAA Aerospace Sciences Meeting	Varied (US)	Jan
AAS	Guidance and Control Conference	Colorado	Feb
AAS, AIAA	Space Flight Mechanics Conference	Varied (US)	Feb
AAS	Goddard Memorial Symposium	NASA GSFC	Mar
AAS, AIAA	Astrodynamics Specialist Conference	Varied (US)	July
AIAA (LA & Orange County Sections)	Reinventing Space (= Responsive Space Conference prior to 2011)	Los Angeles	Apr/May
IAF, ISU	International Astronautical Congress (IAC)	Varied (Worldwide)	Oct
Utah State University, AIAA	SmallSat Conference	Logan, UT	Aug
IEEE, AESS, AIAA, PHM Society	IEEE Aerospace Conference	Big Sky, MT	Mar

References

For annotated bibliography of introduction, see the book website.

Department of Defense. 2009. *Technology Readiness Assessment (TRA) Deskbook*. Jul.

Greene, Nick. 2010. "Hubble Space Telescope." *Space/Astronomy*.

Gruntman, Mike. 2004. *Blazing the Trail: The Early History of Spacecraft and Rocketry*. Reston, VA: AIAA.

Gruntman, Mike. 2008. "Space Systems Fundamentals." course notes.

Horne, Charles F. 1911. *Works of Jules Verne*. vol. 3. New York: Parke and Co.

Johnson, Stephen B. 2002. *The Secret of Apollo: Systems Management in American and European Space Programs*. Baltimore, MD: Johns Hopkins University Press.

Jurist, J.M., D.C. Hook, and D.M. Livingston. 2009. "Small Unit Space Transport and Insertion (SUSTAIN): How to Do It and Use It as a Driver for Low-Cost Responsive Orbital Launch." 7th Responsive Space Conf. AIAA-RS7-2009-1002. Apr. 27–30.

Jurist, J.M., S. Dinkin, and D.M. Livingston. 2005. "When Physics, Economics, and Reality Collide: the Challenge of Cheap Orbital Access." AIAA, Space 2005 Conference. AIAA-2005-6620. Long Beach, CA. Aug. 30–Sept. 1.

Koelle, Dietrich E. 2007. *Handbook of Cost Engineering for Space Transportation Systems (Rev 2) with Transcost 7.2*. Ottobrunn, Germany: TransCostSystems.

Koelle, Dietrich E. 2010. *Handbook of Cost Engineering for Space Transportation Systems (Rev. 3) with Transcost 8.0*. Ottobrunn, Germany: TransCostSystems.

Mankins, John C. 1995. "Technology Readiness Levels, A White Paper." Advanced Concepts Office, Office of Space Access and Technology, NASA. Apr. 6.

Mowry, Clayton. 2010a. E-mail to D. Livingston. Sep. 6.

Mowry, Clayton. 2010b. Interview on *The Space Show*. Aug. 3.

Muncy, J. 2005. Interview on *The Space Show*. May 16.

NASA History Program Office. 2004. "Project Apollo: A Retrospective Analysis." Oct. 22.

Oberth, Hermann. 1923. *Die Rakete zu den Planetenräumen*. R. Oldenbourg, Munich and Berlin.

Precourt, C. 2010a. *The Space Show* Interview. Aug. 13.

Precourt, C. 2010b. E-mail to D. Livingston. Sep. 13.

Sadin, S.T., F.P. Povinelli, and R. Rosen. 1989. "NASA Technology Push Towards Future Space Mission Systems." Space and Humanity Conference, Bangalore, India. S39th IAF Congress. *Acta Astronautica*. vol. 20, pp 73–77.

Scoffern, John. 1859. *Projectile Weapons of War and Explosive Compounds*. London, UK: Longman, Brown, and Co. Reprinted in 2009 by General Books LLC.

Space Foundation. 2009. *The Space Report 2009*.

Space Foundation. 2010. *The Space Report 2010*.

SpaceX. 2010a. "Falcon 1 Overview." Website

SpaceX. 2010b. "Falcon 9 Overview." Website.

Wapedia. 2010. "International Space Station." Website.

Wertz, J.R., W.J. Larson. 1991. *Space Mission Analysis and Design. 1st ed.* Dordrecht, The Netherlands: Kluwer Academic Publishers and Microcosm Press

Wertz, J.R. and W.J. Larson. 1996. *Reducing Space Mission Cost*. El Segundo, CA: Microcosm Press and Springer.

2 Space Mission Communities

James R. Wertz, *Microcosm, Inc./USC*

2.1 Multiple Space Communities
Human Spaceflight; Interplanetary; Military and Intelligence; Commercial; Science and Applications; Education; Launch; Smallsat; Alt Space (Alternative Space)
2.2 Differences and Similarities Between Communities
2.3 Changing Missions

We often think of the space enterprise as a monolithic whole with different groups working on different types of spacecraft. In fact, looking at this more closely, we see that the space enterprise really consists of a number of very different communities, each with very different budgets, timescales, and fundamental objectives. Each community tends to regard themselves as the "real" space program and, therefore, to ignore the others.

While the differences are real and substantial, the different communities are more related than is often seen to be the case. Failures and successes in one community tend to affect all the rest. The Iridium bankruptcy had a major impact on all space activity by making it much harder to raise capital, reducing the number of launch options available (and thereby increasing the cost), and increasing the apparent risk of space activities. Significantly reducing launch cost will have a major impact on all of the space activity, in part by reducing cost, but also by changing the way spacecraft are built. Anything that dramatically increases the use of space will impact all areas by driving down cost. CubeSats, intended for the low-cost education and science community are being adapted by others to drive down the cost of a variety of programs. Any major advance in large space telescopes could be used for Earth observation, for planetary monitoring, or military surveillance.

The importance of this for space mission engineering is that it affects how we design and build space missions. We can't realistically expect to find the same technology or same processes used for a million dollar smallsat as for a multi-billion dollar interplanetary mission or military surveillance spacecraft. On the other hand, major breakthroughs in cost and performance can come about by applying technologies or processes from one community to the problems found in another.

Perhaps the most critical issue for space mission engineering is to clearly recognize the basic goals and objectives of the community for which the space mission is being designed as well as a broad sense of the budgets and timescales that are involved. While we are always trying to build a space mission to achieve the basic objectives at minimum cost and risk, what this means in practical terms changes dramatically based on the space mission community in which we are working.

2.1 Multiple Space Communities

Table 2-1 lists the primary space communities and their basic objectives and typical cost. Notice the dramatically wide range of mission cost, going from approximately $20,000 at the low end to nearly $200 billion for the Apollo program. Although it can't be measured quantitatively, the basic goals and objectives of the people who work in these communities are equally diverse, and, therefore, so are the perceptions, methods, and processes that they use.

2.1.1 Human Spaceflight*

For many people, human spaceflight **is** the space program. Launches make the evening news and disasters become known worldwide. Astronauts give a visible face and a human side to what is otherwise viewed as an almost entirely technological enterprise. It is also the most expensive of the space activities. The Apollo program (Fig. 2-1) cost approximately $200 billion in 2010 dollars, but is widely regarded as one of the most significant events of the 20th century. The International Space Station, ISS (Fig. 2-2), cost about half what Apollo cost, but has not captured the imagination of the World in the same sense that landing on the Moon did.

The reasons for having a human spaceflight program, and, therefore, the purpose and objectives of such a program vary significantly among proponents, but fall broadly into 3 categories. First, there are many tasks that

* For the first 40 years of space exploration, the general distinction was made between *manned* spaceflight and *unmanned* spaceflight, depending on whether people were on board the spacecraft, and using "man" in its broad meaning of humankind. In order to be more politically correct, *manned spaceflight* has become *human spaceflight*, but *unmanned spaceflight* is a bit more challenging. *Robotic spacecraft* has become the most widely used term, but conveys the incorrect impression that all spacecraft without people are somehow robots, which certainly isn't the case with any reasonable definition of robot. On the other hand, "inhuman spaceflight" doesn't seem to convey the right impression and "unhuman spaceflight" isn't much better. Throughout this volume we will try to avoid the terminology altogether, but, when that's not possible, we will typically use "human spaceflight" and "unmanned spacecraft" in order to be half politically correct.

Table 2-1. The Principal Space Communities. See text for discussion. NRE (Non-Recurring Engineering) is the development cost.

Community	Basic Objective	Example Missions	Typical Mission Cost
Human Spaceflight	Explore and exploit human capabilities	Apollo, ISS, MIR	$2 billion–$200 billion
Interplanetary	Explore beyond Earth	Galileo, Cassini, Mars Rovers	$1 billion–$5 billion
Military and Intelligence	Give sponsor a military advantage over their adversary	GPS, ComSats, Surveillance Satellites	$1 billion–$10 billion
Commercial	Make money	IntelSat, ComSats, GeoEye, GlobalStar, Iridium	$100 million–$1 billion
Science and Applications	Gain/apply scientific knowledge (Earth, astronomy, space itself)	Weather Sats, Hubble, Chandra, JWST	$100 million–$10 billion
Education	Teach, train, inspire	StarShine, CubeSats, Spheres	$100 thousand–$500 thousand
Launch	Put payloads or people into space	Shuttle, RLVs, Ariane, Atlas, Cosmos, Delta, Pegasus	**NRE:** $100 M–$50 B **Recurring:** $10 M–$1 B
Smallsat	Demonstrate utility of small, low-cost missions	CanX-2, AmSats, Explorers, SurreySats, DMC, Alexis	$20,000–$20 million
Alt Space (Alternative Space)	Advance civilization's expansion into space	Apollo, Lunar Colonies, Human Mars Mission	$200 billion and up (possibly with commensurate income)

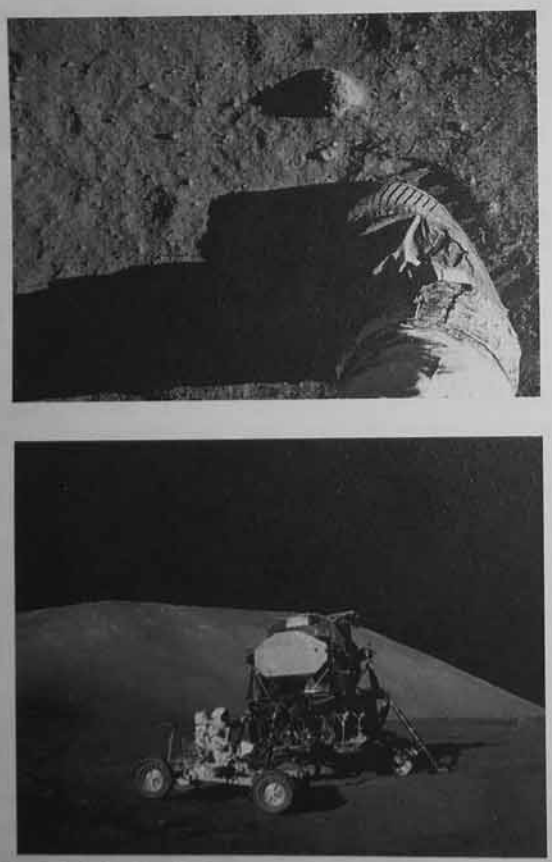


Fig. 2-1. The Apollo Program. Top: Buzz Aldrin's footprint in the Lunar soil on Apollo 11. Bottom: The Lunar Module and Lunar Rover on Apollo 17. On this mission, Harrison Schmitt and Eugene Cernan, the last man on the Moon in the Apollo program, explored the Taurus-Littrow valley in the Montes Taurus mountains. (Photo courtesy of NASA)*

people can do much better and more efficiently than machinery. Simple tasks, such as running a communications relay mission are most easily and economically done by simple machinery. But more complex tasks, such as exploring the surface of the Moon or evaluating and fixing problems that occur, can best be done by having people in space.

Second is the political element. The Soviet Union became a super power, on a par with or superior to the United States, by launching the first satellite and first person into space. The Race to the Moon was a political challenge for superiority here on Earth and the Apollo 11 landing reestablished the superiority of the United States for much of the World.

Finally, for many people the purpose of human spaceflight is to make space exploration a human activity, i.e., to have **people** explore, learn, walk on the Moon, discover new places, and find new things, just as explorers and discoverers have done for centuries. It is in this sense that astronauts are representatives of mankind, turning over a rock on the Moon to find what is under it, and ultimately, turning over many rocks on many moons and planets throughout the Solar System. This view is carried further by the Alt Space community discussed in Sec. 2.1.9.

The human space program is dramatically expensive because spacecraft carrying people are inherently large and heavy and the cost per pound to space is very high, and also because all of the human spacecraft and their equipment must be *man-rated or human-rated*, i.e., tested and certified to be safe to be operated by or near people. This typically means double or triple redundancy in any components where a failure is possible, plus eliminating any sharp edges or possible source of combustion, explosions, or toxic substances. Even if the

Get More

* See the book website for color versions of each of the photos in this chapter. In addition, the Educator Package contains high resolution, color versions of each of these photos and several additional photos representing each of the various space communities.

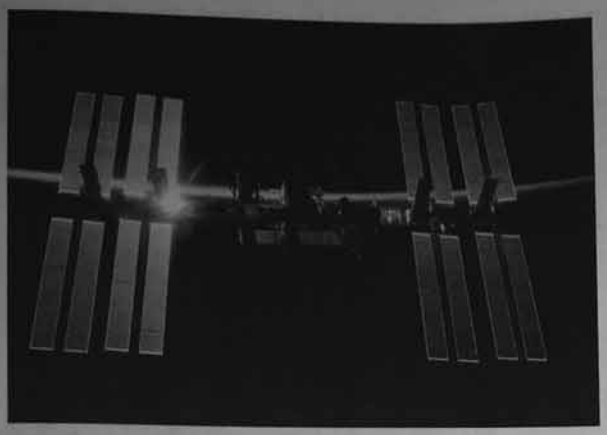


Fig. 2-2. The International Space Station (ISS). Seen here at sunrise. ISS is 110 m long, 50 m wide, and 20 m high and is normally at an altitude of about 350 km. (Photo courtesy of NASA)

equipment is similar or identical to non-human-rated equipment, it will be more expensive because of the higher level of testing, analysis, and certification required.

Because of the high cost of launch and in-space activity for human spaceflight, only a small number of countries are involved in human missions. As of 2010, the only countries doing human space launch are the US, Russia, and China, although Europe and India may do so within a few years. In addition, many countries have astronauts that have flown, primarily on the American or Russian human space missions. In addition, a limited number of *space tourists* have flown on both American and Russian space missions.

This volume does not cover mission engineering for human spaceflight, although of course many aspects are the same for both human and unmanned missions. Orbit properties, the space environment, and the torque available from reaction wheels are the same, irrespective of the payload. There are many good books on human spaceflight in the website bibliography, including excellent ones by Eckart [1996, 2006], Larson and Pranke [1999], Churchill [1997], Clement [2005], Clement et al. [2006], Harrison [2001], and Kanas and Manzey [2003]. Johnson [2002] provides a particularly well-done discussion of the development of systems engineering and systems management that allowed the Apollo program to be successful.

2.1.2 Interplanetary

Interplanetary spaceflight involves sending spacecraft to other planets. A number of well-known interplanetary missions are shown in Table 2-2. The fundamental purpose of interplanetary spacecraft is scientific exploration of all of the elements of the Solar System, both for the purpose of learning more about the solar system and also for learning about the history and possible evolution of the Earth, both past and future. Two central questions are whether there is any potential for life anywhere else in the Solar System, either now or in the past, and how the Solar System formed and evolved. One of the major results of the interplanetary program is that the planets and their satellites are much more individually unique and complex than nearly anyone had anticipated. This, in turn, leads to the objective of both more exploration and more "fine-grained" exploration of the Solar System. The Mars Rovers, Spirit and Opportunity have captured the interest of the public both for their many views of the Martian surface (Fig. 2-3) and their ability to survive for an extended stay in a harsh and dusty environment. They are "the little rovers that could."

Interplanetary programs are typically expensive, ranging from several billion dollars for the large planet observatories such as Galileo or Cassini to a few hundred million for Explorer and Discovery programs. The high cost is driven by several factors—the need to survive for an extended period en route in a high radiation environment and the need to act largely autonomously or with commands stored in advance, due to the long travel time for signals to reach the spacecraft.* The one-way travel time for signals to reach Jupiter ranges from 35 min to over 50 min when Jupiter and the Earth are on opposite sides of the Sun. In addition, they need a substantial propulsion system to either orbit or land on a planet or satellite, precision pointing for very high resolution observations, and larger than typical antennas and solar arrays to provide communications and power back to Earth.

This volume focuses primarily on Earth-orbiting spacecraft, but does include a section on interplanetary orbit design (Sec. 10.7). In addition, most of the web Excel models allow the central body to be any of the Solar System objects, such that you can quickly find all of the relevant orbit and viewing parameters for a Sun synchronous orbit about Mars or Titan. There are quite a few volumes in the website bibliography dealing with the results of specific interplanetary missions and a much smaller number dealing specifically with interplanetary mission design, including Brown [1998], Kemble [2006], and Young [2007].

While interplanetary travel is challenging, practical *interstellar travel*, i.e., going to the stars, is dramatically more challenging. Proxima Centauri, a very small companion to Alpha Centauri, is the nearest star to the Sun at 4.24 light years away. This is 7,000 times further than Pluto. The physics of interstellar travel, including relativistic effects, is summarized by Wertz [2009] Sec. 12.8. Approaches to getting there are discussed by Mallove and Matloff [1989] and in the old, but **very** extensive bibliography (thousands of references) by Forward,

* The NASA *Discovery program* is for missions costing less than $425 million (in FY10$). Over the years, the name "Explorer" has been given to a great many missions. However, generally an "*Explorer mission*" is one operating on a constrained budget, such as *University-Class Explorer* (UNEX)—less than $15 million, *Small Explorer* (SMEX)—less than $120 million, and *Medium-Class Explorer* (MIDEX) —less than $180 million.

Get Info **Table 2-2. Representative Interplanetary Missions.** See the associated website table for additional missions, additional details on these missions, and web references for all of them.

Mission Name	Mission Objective	Launch Date, Vehicle	Comments
Pioneer 10	Explore Jupiter; safely cross asteroid belt	3/3/72, Atlas/Centaur/ TE-364	First to observe Jupiter. First to travel through asteroid belt. Last signal detected on 1/23/03.
Pioneer 11	Explore Jupiter and Saturn (including ring system)	4/6/73, Atlas/ Centaur/ TE-364	First to observe Saturn + ring system up close.
Mariner 10	Observe Mercury and Venus	11/3/73, Atlas/Centaur	First craft to Mercury. First to use gravity assist. First to visit 2 planets.
Viking 1	Mars surface sample analysis; look for life	8/20/75, Titan-IIIE/Centaur	Orbiter+lander. No sign of life detected. First successful Martian Lander.
Voyager 2	Jupiter and outer planet close flyby's	8/20/77, Titan/Centaur	—
Voyager 1	Close Jupiter and outer planet flyby's	9/5/77, Titan-IIIE/Centaur	Most distant man-made object in space. Passed termination shock of heliosphere in 12/04.
Pioneer Venus	Venus orbiter	5/20/78, Atlas/Centaur	—
International Cometary Explorer (ICE)	Study Solar radiation and plasma	8/12/78, Delta	Also known as ISEE-3. First in halo orbit. After primary mission (1981), intercepted comet Giacobini-Zinner.
Vega 1	Study Venus and Halley's Comet	12/15/84, Proton-8K82K	Balloon and lander for Venus. Soviet/French mission.
Sakigake	Observe Halley's Comet; technology testbed	1/7/85, M-3SII-1	First Japanese interplanetary probe. Also named MS-T5.
Giotto	Study Halley's Comet	7/2/85, Ariane 1	ESA's first deep space mission. Passed Halley's Comet at 600 km then flew by comet Grigg-Skjellerup.
Galileo	Detailed exploration of Jupiter system	10/18/89, Shuttle/IUS	Discovered additional moons, Io volcanoes, Jupiter rings. First use of Earth swingby.
Ulysses	Observation of Solar polar regions	10/6/90, Shuttle/IUS/ PAM-S	ESA/NASA mission. Maximum Solar latitude 80.2 deg. Highest escape energy ever used.
Clementine	Study topography and geology of the Moon	1/25/94, Titan IIG	Extension to flyby of asteroid 1620 Geographos failed.
Near Earth Asteroid Rendezvous (NEAR)	Orbit 433 Eros	2/17/96, Delta II-7925	First mission to orbit and land on asteroid. NASA Discovery mission.
Mars Pathfinder (MPF)	Mars orbiter and lander	12/2/96, Delta II-7925	"Faster, better, cheaper" design. Sojourner rover (for 30 SOLs).
Cassini/Huygens	Explore Saturnian system	10/15/97, Titan-IVB/Centaur	Cassini main vehicle. Huygens probe descended in atmosphere.
Lunar Prospector	Mapping Lunar resources	1/7/98, Athena II	NASA Discovery mission.
Stardust	Capture and return dust from comet P/Wild 2.	2/7/99, Delta II	First return of particles from beyond Moon's orbit. NASA Discovery mission.
Genesis	Observe Solar wind and return particles to Earth	8/8/01, Delta II-7326	NASA Discovery mission. Drogue parachutes failed during landing at Earth.
Mars Exploration Rovers (MER)	Study past water activity on Mars	6/10/03, Delta II-7925 (A); 7/3/03, Delta II-7925Heavy (B)	Mars Rovers "Spirit" and "Opportunity," designed for 90 SOLs but still active after 7 years.
SMART 1	Test solar-electric propulsion; Lunar science	9/27/03, Ariane 5	First European satellite to travel around Moon. Lunar impact on 9/3/06. Piggyback launch.
MESSENGER	Observe physical phenomena at Mercury	8/3/04, Delta II-7925Heavy	First Mercury observations since Mariner 10 (1974–1975).
Deep Impact	Study comet Tempel 1	1/12/05, Delta II-2925	Discovery mission. Impactor + flyby.
New Horizons	First mission to Pluto	1/19/06, Atlas V-551	—
Lunar Reconnaissance Orbiter (LRO)	Identify landing sites and resources	6/18/09, Atlas V-401	Accompanied by LCROSS, to observe impact of Centaur upper stage.



Fig. 2-3. Panorama of Mars as Seen By the Mars Rover Spirit. (Photo courtesy of NASA)

listed in the website bibliography for Chap. 10. Because interstellar travel isn't practical in the near term,* it is often included as part of the Alt Space community described in Sec. 2.1.9.

2.1.3 Military and Intelligence

The *military and intelligence space program* is intended to collect data with military utility wherever and whenever it can. (The Outer Space Treaty of 1967 bans weapons of mass destruction in space and requires that the Moon and other celestial bodies be used only for "*peaceful purposes*," defined by the U.S. as "non-aggressive," meaning that defensive measures are permitted. See Sec. 12.3.) Military and intelligence space activity can be broadly divided into categories based on specific objectives, such as:

- *Reconnaissance and Surveillance*, obtaining high-resolution imagery of the ground or air space, typically in the visible or infrared (often called *IMINT*)
- *Signals Intelligence*, intercepting communications (often called *SIGINT*)
- Detection of nuclear explosions
- Military communications
- *Guidance and Navigation*, implemented largely with GPS (Fig. 2-4 and Sec. 19.2.2.3), which is built, launched, and operated by the Air Force
- *Space Situational Awareness (SSA)*, which is obtaining knowledge about what an adversary may be doing in space, or, potentially, about space debris or other space hazards

The military space program is largely pragmatic, i.e., it will use the lowest cost, most effective means to obtain the needed data, irrespective of whether it uses spacecraft, aircraft, high altitude balloons, or terrestrial resources. Similarly, it will at least evaluate the use of space resources in new ways that have the potential of providing military utility. For example, the National Security Space Office (NSSO) has studied using satellite solar power to provide power to remote military outposts in order to reduce the cost and risk of bringing fuel to remote locations. (Typically fuel is one of the biggest items in the military logistics supply chain.)

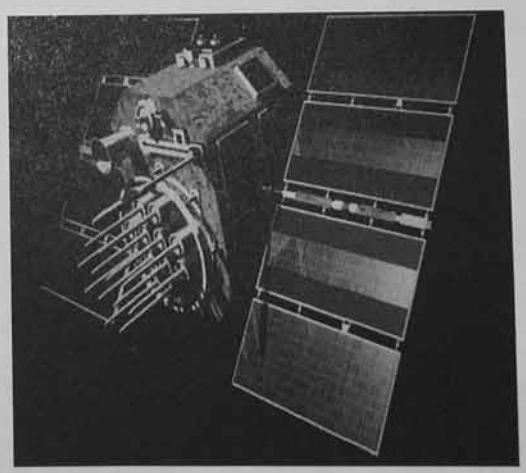


Fig. 2-4. Drawing of GPS Satellite. The GPS constellation, at an altitude of approximately 20,000 km (half-GEO), is run by the Air Force to provide precision navigation services for the military and also for a great many civilian applications.

Because of the wide variety of military and intelligence applications, there is also a wide range of costs. Traditionally, military "*programs of record*," i.e., long-term programs approved by both DoD and Congress, are large and very expensive spacecraft with a long operational life and, unfortunately, an equally long

* As shown in Table 2-2, Voyager I is often regarded as the first interstellar probe because it left the solar system and is exploring interstellar space. However, it isn't headed for any nearby star and will simply continue to drift, sending back data on the interstellar medium.

development cycle. In an attempt to reduce both cost and schedule and meet urgent needs of military commanders, the Operationally Responsive Space Office (ORSO) was created in 2008. In 2009, the National Reconnaissance Office, often regarded as building the best and most expensive observations satellites in the world, began buying extremely low-cost CubeSats, which are 10 cm cubes with equally small, low-cost components. (See Sec. 2.1.8 and Sec. 25.4.) While large, expensive programs will certainly continue, augmenting these larger spacecraft with much smaller, lower-cost and more quickly replaceable satellites is at least a possibility for the future, in part because of continuing pressure on the defense budget to do more with less and reduce the fragility of current systems. (See Wertz [2010], for a discussion of the arguments for low-cost, much smaller military options.) This potential transition is hampered by the lack of an appropriate small, low-cost launch vehicle.

There is also a broader view of military space reflected in the title of the book *Mastering the Ultimate High Ground* [Lambeth, 2003]. From a military perspective, space is the ultimate high ground, such that whoever controls space, controls the Earth. This, in turn, implies the need to defend space assets and be fully aware of what is going on in space, i.e., the space situational awareness mission. Other authors that discuss military space include Chun and Taylor [2006], Klein [2006], Parkinson and Spilker Jr. [1996], and Richelson [2001].

2.1.4 Commercial

Of all of the space communities, the *commercial community* is one of the largest and has the most straightforward objective—to make money. Historically, this has been done primarily with geosynchronous communications satellites, such as IntelSat, shown in Fig. 2-5, and direct broadcast television and radio. The GEO commsat community is now a mature industry with approximately 300 satellites on orbit, substantial competition for building new ones, and an active market in used GEO satellites. (As discussed in Sec. 1.2, the use of GEO for communications was first described by science fiction writer Arthur C. Clarke in 1945.)

So far as the communications portion of the industry is concerned, the basic objective is not to minimize cost or maximize performance, but to minimize the cost per unit of performance (for example, the cost per phone call). There is ample capital available in the communications market to support systems of essentially any size, so long as the cost per call is minimized so that the operator can compete successfully for communications business.

Other elements of the commercial space industry are less mature. There have been only a small number of commercial observations satellites, such as GeoEye-1, shown in Fig. 2-6. Space tourism is underway as a commercial activity, as is one of the more unusual space businesses, burial in space [Chafer and Price, 2002]. In this case, a small vial of cremation ashes is launched into low Earth orbit where it will typically decay and burn up in the atmosphere after a short period.

Table 2-2◀, Fig. 2-6, Eq. 2-0



Fig. 2-5. IntelSat 603. This satellite was launched in 1990 on an unmanned Soviet launcher, but a launch failure left it in a useless low-Earth orbit. In 1992 it was captured (by hand), repaired, and sent successfully on its way to GEO by the crew of STS-49. (Photo courtesy of NASA)

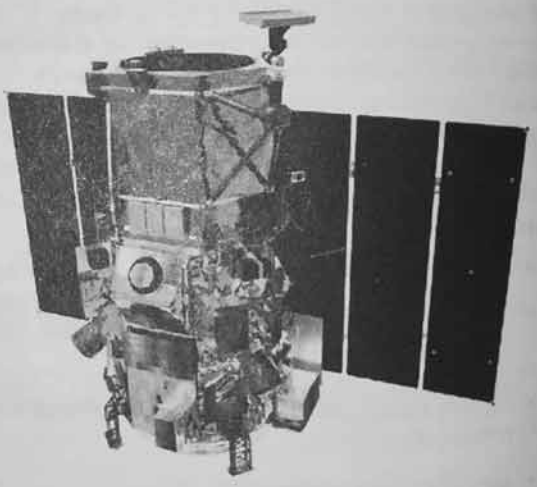


Fig. 2-6. Graphic Rendition of GeoEye-1 Launched into a Sun Synchronous Orbit in 2008. The satellite provides 0.5 m resolution imagery to Google-Earth and slightly better imagery to the government. (Photo courtesy of GeoEye)

The success or lack of success of commercial space activities has a very strong effect on non-commercial space activities. The bankruptcy of the Iridium low-Earth orbit communications constellation in 1999 had a very chilling effect on other aspects of space. There were many similar constellations being planned, had Iridium proven to be financially successful, including the Teledesic "internet in the sky" constellation with potentially as many as 840 satellites. The Iridium bankruptcy made it difficult to obtain financing for commercial space ven-

Get Info **Table 2-3. The NASA Great Observatories.** (See the associated website for references to individual spacecraft and their results.)

Name	Name Before Launch	Spectral Region	Year Launched	End-of-Life	Comments	Follow-On Mission
Hubble Space Telescope, HST	Large Space Telescope, LST	Visible, UV, and Near-IR	1990	—	Launched and serviced by Space Shuttle.	James Webb Space Telescope, JWST (IR and visible)
Compton Gamma Ray Observatory, CGRO	Gamma Ray Observatory, GRO	Gamma Ray	1991	2000	Shuttle-launched	Fermi Gamma-Ray Space Telescope, formerly Gamma-Ray Large Area Space Telescope, GLAST
Chandra X-Ray Observatory, CXO	Advanced X-Ray Astrophysics Facility, AXAF	X-Ray	1999	—	Shuttle-launched; extended delays due to cost of HST	Constellation-X provides greater sensitivity, but lower resolution
Spitzer Space Telescope, SST	Space Infrared Telescope Facility, SIRTF	Infrared	2003	—	Launched on Delta II	JWST observations will replace SST in the near infrared

tures and prevented the significant launch cost reductions that would probably have come about with a major spurt in the number of LEO satellites. GlobalStar and ORBComm LEO communications constellations have been launched, but have contained much smaller numbers of satellites with less ambitious goals than Iridium.

The fundamental impediment to substantial growth in commercial space is the high cost of launch, which in turn, drives up the cost of spacecraft, as discussed in Chap. 13. For example, there are a number of manufacturing processes that can occur in space, that simply can't occur on the Earth's surface. (See Sec. 7.4.) This has led to a long-term interest in space manufacturing, with some experiments done on the Space Station and free-flying payloads. Unfortunately, the high cost has thus far made any significant space manufacturing activity uneconomical. Similarly, solar power satellites have been considered for an extended period as a means of providing clean energy in nearly unlimited amounts to the Earth's surface [Glaser et al., 1998]. But this won't occur in any reasonable quantity until there is a reduction in the cost of launch by a factor of 5 to 10.

Commercial space activity is discussed by Haskell and Rycroft [1998], Peeters [2000], in a series of telecommunications books by Elbert [1990, 1992, 2003, 2008], and in a number of volumes listed in the web bibliography on communications satellites.

2.1.5 Science and Applications

Science and Applications Satellites monitor the Earth and space for both scientific studies and Earth applications such as weather prediction and monitoring global conditions on both land and oceans. Weather forecasting is now largely done with analytic models based on satellite data. The first photos of the aftermath of Hurricane Katrina were made by a small satellite from the Disaster Monitoring Constellation built by SSTL. Much of the scientific data about the Earth as a whole and the universe at large comes from scientific satellites. The science and applications community tends to break down further, either by what they are studying (such as weather, Earth resources, Mars, or asteroids) or by the wavelength they are working at (such as visible, IR, UV, or X-Ray). The NASA Great Observatories program, shown in Table 2-3, is intended to take advantage of the fact that satellites outside of the Earth's atmosphere can see the Universe without the distortions caused by the atmosphere and in wavelengths not visible from the Earth's surface. The most well-known of these is the Hubble Space Telescope that has both created enormous amounts of scientific data and sparked the human imagination with images of remarkable clarity of things and places we only thought we understood (Fig. 2-7).

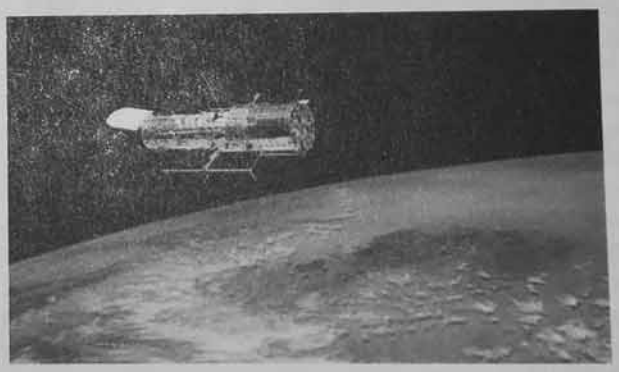


Fig. 2-7. Hubble Space Telescope after Release by the Final Shuttle Servicing Mission in 2009. (Photo courtesy of NASA)

In the future, both science and applications missions are likely to bifurcate in two distinct directions. Some of the science missions will continue to use very large, very expensive instruments and spacecraft as they continue to push the state of the art in terms of both size and sensitivity. However, continued budget pressure will likely stretch out the time and reduce the frequency of very large science observatories, at least until there is a continued human presence on the Moon or in orbit that can provide servicing for them. In the second branch, many observations and measurements will be taken over by much lower cost, smaller satellite missions that come about by the continued microminiaturization of electronics components and the development of radiation hardened, low-cost components.

The basic objectives of the science and applications community will not change—to provide continued, improved monitoring of the Earth and space and to stretch the boundaries of our knowledge of the Earth and its place in the universe. This includes, particularly, the search for life elsewhere in the universe and solutions to fundamental problems on Earth ranging from weather to environmental monitoring to earthquake prediction to disaster monitoring.

There are many volumes in the website bibliography on the results of both science and applications missions, including the COBE mission which resulted in the first Nobel Prize for NASA scientists Mather and Boslough [1996]. Among the volumes describing observing techniques and analysis are those by Elachi and van Zyl [2006], Conway [1997], Cruise et al. [2006], Lacomme et al. [2001], Lillesand et al. [2008], Olsen [2007], and Schowengerdt [2007].



Fig. 2-8. Jupiter's Moon Io Photographed by the Chandra X-Ray Observatory. (Photo courtesy of NASA)

2.1.6 Education

The goal of the *space educational community* is to provide the best possible educational environment at all levels—from primary and secondary education to the undergraduate and graduate level. Clearly a part of this can be done by looking at data and results from a wide variety of missions such as the Apollo program or the Mars Rovers. But the real issue for educators is to launch satellites built, and, if possible, operated by students so that they can get real, first-hand experience of the problems and issues involved. A literally shining example of this approach is the StarShine satellite developed by Gil Moore (Fig. 2-9). StarShine consisted of a 48-cm hollow aluminum sphere covered with 878 front surface mirrors that were polished by 25,000 children in 660 schools in 18 countries. StarShine 1 was launched from the Space Shuttle during a trip to the ISS in 1999 (2 others have been launched since then) and was visible to the naked eye before sunrise and after sunset. Young students tracked the satellite and older students determined its orbit, decay rate, and the atmospheric density during the 8-month flight.

Table 2-3◀, Fig. 2-10, Eq. 2-0



Fig. 2-9. Gil Moore holding a Mock-up of the StarShine Satellite. Right: sunshine reflecting off one of the mirrors. (Photo courtesy of Gil Moore)

Although there have been some successes, the most fundamental problem with all student satellites is finding a launch. NASA had determined that launching student satellites was not a part of its fundamental charter and most educational institutions couldn't afford to buy even a secondary launch opportunity for student payloads. Thus, it was nearly impossible to launch a student satellite while the student was still a student, or even while they were still in aerospace at all.

Much of this difficulty changed in 1999, with the invention of CubeSats by Bob Twiggs at Stanford University and Jordi Puig-Suari at California State Polytechnic University. (See Sec. 25.4 for a detailed discussion of CubeSats.) The purpose of the CubeSat was to find a way to drive down launch costs for student built satellites. This was done by building a standard launch container (named the Poly PicoSat Orbital Deployer, or P-POD, by the Cal Poly students) capable of holding and launching three 10 × 10 × 10 cm CubeSats. While CubeSats have greatly increased the potential for educational satellites, getting a ride to orbit remains a major problem, although it looks like solutions may occur within a few years. There are no books currently available exclusively on education satellites, largely because there have been very few education satellites in the past.



Fig. 2-10. An Early CubeSat 10 × 10 × 10 cm Satellite. (Photo courtesy of Pumpkin, Inc.)

2.1.7 Launch

There is a standing joke within the launch community that the fundamental purpose of spacecraft is to serve as ballast for testing and flying launch vehicles. Because of the cost and difficulty of launching to orbit, the launch community is largely a group apart from the spacecraft community, for which launch is simply a service to be provided. Launch vehicles and launch vehicle selection are discussed in detail in Chap. 26 and all of the modern launch vehicles available throughout the world are described by Isakowitz, et al. [2004].

Historically, launch vehicles have been about 90% reliable since not long after the opening of the space program. The most reliable, and lowest cost per pound to orbit, was the Saturn V (Fig. 2-11), developed for the Apollo program. Because there were only a small number of launches, the potential long-term reliability is not known and the low cost is due in part to being able to launch a small city to orbit (120,000 kg to LEO). One of the other vehicles with an excellent reliability record has been the Delta II, shown in Fig. 2-12, which had only 2 failures in its first 110 launches. The Russian (and former Soviet) Soyuz launcher had only 21 failures out of 704 launches (97% reliability) over more than 40 years, and most of those failures occurred during early development and testing.



Fig. 2-11. Saturn V Launch of Apollo 15 in 1971. The Saturn V was 110 m high, 10 m in diameter, and weighed 3,000,000 kg at liftoff. (Photo courtesy of NASA)

While launch is not the most expensive element of most large space missions, it tends to drive mission cost. It's not economically worthwhile to build a satellite for a few million dollars and launch it for $100 million. Therefore, much (but certainly not all) of the high cost of satellites can be attributed to the fact that it costs $10,000 to $20,000 per kg to get to low-Earth orbit and $80,000 to $150,000 per kg to get to the surface of the Moon.

It is clearly important to drive down launch costs in the future in order to create a robust space program that can be sustained over the long term. Unfortunately, it is not clear how to do this. With the exception of the Space Shuttle Orbiter and the Solid Rocket Boosters (SRBs) on the side of the Shuttle, all traditional launch vehicles have been *expendable*, i.e., used only once with the spent stages lost in the ocean or burned up on reentry. Although I do not agree with this conclusion, it is the consensus of the American launch community that the solution to the cost problem is to create a fully or partially *reusable launch vehicle*, in which portions of the launch vehicle are brought back to Earth, refurbished, and flown again. In the simplest terms, this argument is often expressed as "think how expensive air travel would be if you threw away a 747 every time you used it." (For the counter argument see, for example, Wertz [2000] or [2004].) Irrespective of what solutions might ultimately be successful, it is critical to the worldwide space program to significantly reduce the cost of launch for payloads of all sizes, from CubeSats and smallsats to large communications and reconnaissance satellites, to payloads going to the Moon or Mars. This problem has been discussed by a number of authors, including London [1994], Wertz [2000, 2004], and Hujsak [1994]. The standard reference for rocket propulsion is Sutton [2010]. An excellent cost model specifically for space transportation systems is provided by Koelle [2010].



Fig. 2-12. Delta II Launch of the Mars Rover Spirit in 2003. (Photo courtesy of NASA)

2.1.8 Smallsat

Smallsats are broadly defined as satellites weighing less than about 1000 lbs or 500 kg. However, with the advent of CubeSats and very small satellites, a variety of terms have come into use and it has become clear that there is a need for more well-defined terminology. The terminology now most widely adopted for small satellites is the following:

MiniSat	100 to 500 kg
MicroSat	10 to 100 kg
NanoSat	1 to 10 kg
PicoSat	0.1 to 1 kg

A *1U CubeSat* fits within a 10 cm × 10 cm × 10 cm cube and weighs at most 1 kg. Therefore, it's a large PicoSat. A common alternative configuration is a *3U CubeSat*, consisting of three 1U CubeSats in a line (i.e., 10 cm × 10 cm × 30 cm), which would be a NanoSat configuration.* (See Sec. 25.4 for a detailed discussion.)

Smallsats have been flown since the beginning of the space program, largely because of limitations on the size and mass capability of launch systems. Historically, about 20% of all satellites launched are less than 400 kg. Generally, smallsats are simpler than larger satellites, typically have only a single payload, and have a much more limited capability. (One of the principal reasons for building bigger, more expensive satellites is to obtain higher levels of performance.) However, in recent years the microminiaturization that has created iPhones, Blackberrys, and music players with nearly unlimited memory, has also allowed a dramatic increase in the capability of smallsats while maintaining very low cost. In addition, small satellites are ordinarily built in 1 to 3 years (largely because there isn't enough money to have a program last longer). This, and a higher tolerance for risk, allows small satellites to take advantage of new technology much faster than traditional satellite programs. Thus a 15 year old large satellite, built 20 years ago with technology from 25 to 30 years ago with largely analog electronics and mechanical systems, may or may not be more capable than a very small, low-cost satellite built a few years ago using the very latest computer technology and almost entirely software driven subsystems.

Like the satellites themselves, the basic objectives of the smallsat community are typically much less complex than those of traditional satellite systems. Ordinarily, traditional satellite programs will be requirements-driven with a very extensive list of potentially contradictory requirements and constraints from different potential users and multiple payloads that may or may not work well together. Smallsats, on the other hand will tend to have a single payload, many fewer requirements, and be much more oriented to providing whatever capability can be accomplished at modest cost. An example is the Alexis spacecraft, shown in Fig. 2-13. When the Chandra X-Ray telescope was being designed and built, there were extensive delays, due in part to cost overruns on Hubble. (Chandra was initially designed in the early 1970's and finally launched in 1999.) The community identified a need for continuing X-Ray observations and wanted whatever could be done quickly and at very low cost, rather than meet an extensive set of requirements. Alexis was built for approximately $17 million vs. approximately $2.75 billion for Chandra.

For many years, the world leader in low-cost small satellites has been Surrey Satellite Technology Limited, SSTL, started by Sir Martin Sweeting as an outgrowth of



Fig. 2-13. The Alexis Small X-ray Satellite was Launched in 1993. The satellite lasted 12 years on orbit in spite of a launch failure that broke off one of the solar arrays. (Photo courtesy of Comtech AeroAstro)

the space activity at the University of Surrey. SSTL started with small, low-cost communications satellites, and has since created a set of low-cost observation satellites, such as those that are a part of the Disaster Monitoring Constellation, DMC (Fig. 2-14). "Surrey Sats" often become the first satellite in a space program being started by one of the smaller space-faring nations.

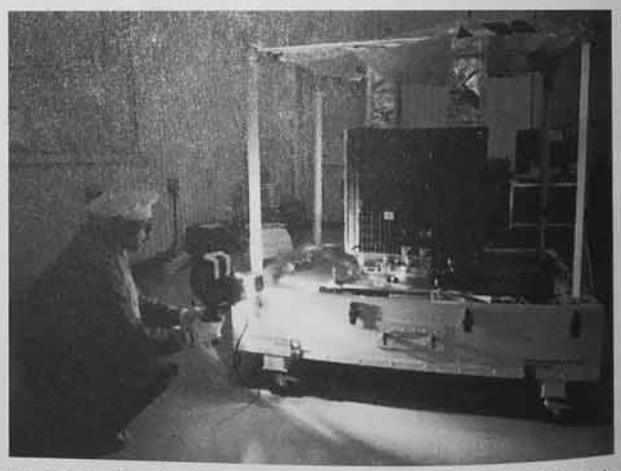


Fig. 2-14. Testing one of the SSTL Small Satellites that Make Up the Disaster Monitoring Constellation (DMC). NigeriaSat, a part of the DMC, took the first photos published in *Aviation Week and Space Technology* of the aftermath of Hurricane Katrina. (Photo courtesy of Surrey Satellite Technology)

For many smallsat programs, the basic objective is to find a reasonable compromise between what we would like and what we can afford, what is often called *trading on requirements*. This is one of several key characteristics that allow smallsats to be much more economical than their larger counterparts (See Chap. 13 and also Wertz and Larson [1996] and Wertz and Dawson [1996]). Smallsats typically are simpler, are built with much smaller teams, and in a shorter time than tra-

* For small satellites, this distinction by mass may or may not be useful. (There really is not much difference between a 1U and a 3U CubeSat.) For an alternative definition, see Swartout [2009].

ditional satellites, all of which contribute further to low cost. Intuition suggests that there is an irreducible "overhead" associated with building satellites, such that smaller satellites will become less efficient in terms of cost, but this does not appear to be the case. Figure 2-15 shows a verification plot for Version 7.4 of the *Small Satellite Cost Model* (SSCM), created by the Aerospace Corp. specifically for modeling the cost of small satellites, which always seem to be overestimated by the traditional large satellite cost models [Wertz and Larson, 1996]. The plot is actual vs. estimated cost only for the programs that make up the model, so by definition, the data will fit the model very well. What is interesting is the lower left hand corner of the plot in which the estimates from the model are all higher than the actual cost. At the very low cost end, even the SSCM overestimates satellite cost. The message here is that, in practice, there

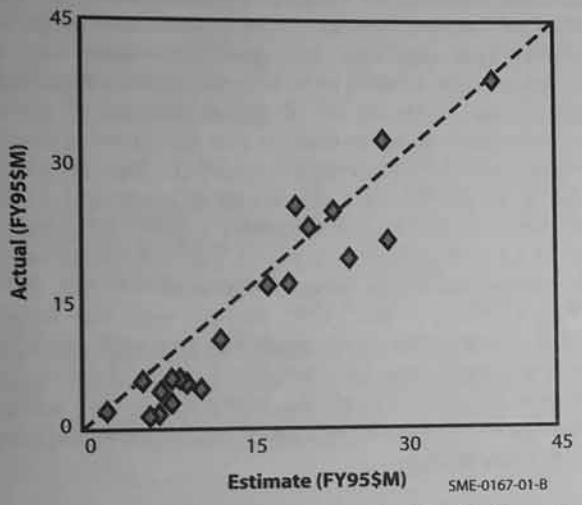


Fig. 2-15. Cost Estimates vs. Actual Cost for the Programs in the Aerospace Corporation Small Satellite Cost Model Database [Bearden, Boudreault, and Wertz, 1996].

is something about very low-cost systems that makes them even lower cost than we would expect. It may be that it is the simpler design, the shorter schedule, or the smaller number of people, but the effect appears to be real from at least this limited set of data. Interestingly, SSCM Version 7.4 was created before the invention of CubeSats and the CubeSat cost estimates would seem to continue this trend, although no statistical analysis has been done to verify this.

The future of smallsats and the smallsat community is uncertain. Smallsats will continue to be developed and flown and will be a significant fraction of the total satellite population. They will also become much more competent as both CubeSat technology and other elements of microminiaturization continue. What is less certain is whether they will continue to be used primarily for education and experimental purposes or whether they will play a much larger role in military, science, and applications programs. Smallsats themselves appear to be fully capable of taking on a much larger role and leading the way in driving down space system cost. However, for Smallsats to take on a true operational role, small, low-cost, launch-on-demand must become real.

There are a number of excellent books on Smallsats listed in the website bibliography. Ford [2010] provides effectively a handbook on how to build a communications satellite in your garage. Other standard and useful references are Sarsfield [1998], Fleeter [2000], and Wertz and Larson [1996], which includes a list of 90 representative smallsat programs over a 15-year period and detailed case studies of 10 of them.

2.1.9 Alt Space (Alternative Space)

The Alt Space (Alternative Space) community, or Spacers, is the community that has a long-term objective of fully utilizing space—i.e., colonizing and living on the Moon and planets, mining the solar system resources, and traveling to the stars.* They regard the current space program as simply the first step in a broad, long-term program of development and settlement in space. Many of the engineers, scientists, and program managers in the more traditional space program regard the Alt Space community as unrealistic dreamers that don't fully realize the cost or technical challenges associated with space exploration. On the other hand, the Alt Space community regards many of the more traditional communities as having too narrow a perspective to see the long-term potential and as being unwilling to develop the means to get there.

At the time of the Apollo 11 landing on the Moon, Wernher von Braun, the head of Marshall Space Flight Center that developed the Saturn V launch vehicle, described it as the most important event in history since fish came out of the sea onto the land [Mailer, 1970]. On the other hand, historian Gerard DeGroot [2006] described the Apollo program as "an expensive distraction of little scientific or cultural worth." These differing views of the significance of Apollo don't represent so much differing assessments of the Apollo mission, as they do differing views of the future. In the view of von Braun, people will, in the long term, leave Earth, settle the planets and moons of the Solar System, and create

* The *New-Space community* is a subset of the Alt Space community that is interested in the private spaceflight industry. To date, this has primarily involved the development of a form of space tourism involving suborbital flights using a rocket or rocket/plane combination to launch human passengers to an altitude of 100 km, often thought of as the "edge of space." This may serve to popularize space tourism, but in terms of energy required, and therefore the cost and complexity of accomplishing the mission, this is a far less difficult task than launching tourists into orbit, as has been done several times with visitors to the Space Station. The long-term NewSpace goals remain the same as the Alt Space community—to enable large numbers of people to go to orbit and visit the Moon and planets.

new civilizations away from Earth. In the view of DeGroot, people will always remain tied to the Earth and space will be used for the purpose of bringing back knowledge and possibly some materials for use on Earth, but not as a future home for mankind.



Fig. 2-16. Drawing of a Potential Human Mission to Mars. (Graphic courtesy of NASA)

A second objective of the Alt Space community is to overcome the limitations of life on Earth. The perception within this community is that the Earth is limited in its capacity to support continued human growth and development. In this view, there are serious threats to the continuation of modern civilization including ever more scarce resources, pollution, natural disasters, climate change, and nuclear or biological warfare. If civilization is to continue in spite of a major disaster or long-term slow degradation, then we need to develop a "home-away-from-home," much as the Europeans did in moving to North and South America. Therefore, they see space colonization and settlement as a means of potentially overcoming the physical limitations of sharing a planet that is rapidly becoming too small.

Although there may be great interest in the results of specific space missions, the Alt Space community as a whole is typically not as interested in the specific technical or scientific return from a particular mission as they are in whether it can further the long-term objective of space settlement. Some of the missions that would be seen as most furthering this objective are listed in Table 2-4. It is also important to note that there are probably far more people in the Alt Space community than in traditional space program development and they are vocal supporters of space exploration. While some space programs are privately funded, most space activity ultimately depends on spending public money for programs that are increasingly expensive. Therefore, the traditional programs in science, applications, and even military need the support and encouragement of the Alt Space community, just as the Alt Space community needs the technology developments that can only come about through real engineering, development, testing, and flying.

All of the missions listed in Table 2-4 are technically feasible, although the perception of the level of difficulty varies dramatically, depending on the point of view of those doing the evaluating. Similarly, the degree to which they will come about and in what time frame depends strongly on both the cost and perceived cost of the missions. A clear problem here is that using existing cost models for either human or unmanned spaceflight would indicate that any of these missions are in the 100's of billions of dollars or higher, and possibly much higher. However, current cost models were empirically developed based on entirely different missions with different rules, processes, requirements, and projected results.These traditional cost model estimates may be sufficiently far off so as to be effectively useless and may damage the prospects for advanced missions by giving the potentially false impression that the missions are too expensive to be economically justified. (See, for example, Wertz [1999] for a discussion of creating a 1,000-person Lunar colony for somewhat less than the cost of the ISS.) In the 1970's Gerard O'Neill had done extensive cost modeling for Space Settlements [O'Neill, 1974, 1975; O'Neill, et al., 1979], but the idea was largely killed when space cost modeling "experts" testified before Congress that his estimates were low by orders of magnitude, even though the professional cost models were based on data almost entirely unrelated to the Space Settlement process.



Fig. 2-17. O'Neill Space Colony. These space colonies were designed for the Earth/Moon L4 or L5 Lagrange points. (See Sec. 10.5.) (Graphic courtesy of NASA)

Because of its popular appeal, there are a great many books on the long-term exploration and settlement of space listed on the website bibliography. By far, the most technically detailed are the series of conference proceedings (9 volumes) on Space Settlements published by the Space Studies Institute founded by O'Neill. There are

Table 2-4. Representative Alt Space Missions. These are future missions that advance the prospects for the long term settlement and utilization of space. The fundamental requirement for nearly all of these missions is much lower cost launch to orbit.

Mission	Objective or Result
Satellite Solar Power	Reduce the reliance on fossil or nuclear fuels; "greener" energy.
He^3 Mining from the Lunar Regolith	Provide fuel for clean nuclear power that does not create long-term, toxic radioactive waste.
Space Tourism	Drives the economic engine of space development by significantly increasing the amount of money available for space.
Search for Extraterrestrial Life	If extraterrestrial life is found it could dramatically impact our perception of our place in the Universe and the motivation for space development, exploration, and settlement.
Human Mission to Mars	Advances popular interest in space exploration and settlement, particularly if any evidence of past life on Mars is found (Fig. 2-16).
Space Manufacturing	Provides processes and results not possible in a 1-g environment and serves as an economic driver for space development. (Pure chemicals, large crystals, ultra-sound ball bearings, "foam steel").
Asteroid Mining	Provides iron and other resources that can be used without further damaging Earth's environment.
Lunar Colonization and Settlement	Provides for science, exploration, and development away from Earth and drives the economics of space exploration.
Space Settlements	This is settlement in colonies in space itself (typically at the Earth/Moon Lagrange Point), not on the Moon or planets; originally conceptualized by Gerard O'Neill; provides benefits similar to Lunar colonization and settlement, but with a 1-g or nearly 1-g environment (Fig. 2-17).
Settlement of Mars, Other Planets, and Moons	This is a long-term goal of Alt Space, but not necessarily regarded as realistic in the near-term.
Interstellar Travel	The ultimate goal of the Alt Space community. Not realistic in the near term.

also many books on the broad, long-term justification for space exploration, including, for example, those by Berinstein [2002], Burrows [2006], Hardersen [1997], Harrison [2001], White [1998], Zimmerman [2010], and Zubrin[1999].

2.2 Differences and Similarities Between Communities

As described in Chap. 1, the basic goal of space mission engineering is to design a space mission to meet a set of broad, often poorly defined objectives at minimum cost and risk. The purpose of introducing and discussing the various space communities is to allow the reader to do that job better—either by coming up with new and innovative solutions or designing a mission with better performance, lower cost, reduced risk or some combination of these. This comes about in 3 distinct ways:

1. By having a better understanding of the basic objectives of the particular community (or communities) that represents the customer or end user of your mission.
2. By creating opportunities for multiple communities to work together for a common goal or to benefit the broader space enterprise.
3. To potentially apply technologies, processes, and methods developed in one space community to the benefit of other space communities.

All of these are potentially applicable to almost any space mission.

First, we need to recognize and work with the basic goals and objectives of the community we are in and, similarly, to recognize the differing goals and objectives of other communities. Nearly all of the authors of this volume spend most of their career within one of these communities. Yet it is important that the process and procedures that they and others describe be applicable across a broad range of communities or that the differences be clearly explained and identified.

Second, we want to create the opportunity for different communities to work together on a common problem or issue. For example, the science and military communities have relatively very little in common and frequently look only inward, within their own group or organization, for processes and technology. But they have a common interest in developing large space optics with high resolution and working at multiple wavelengths. Development funds are extremely scarce for inherently expensive large space systems. In the past, having two groups work together was often the "kiss of death" because both groups would bring firm requirements that were either incompatible or would drive up the cost to the point where the solution was unaffordable. If we can recognize the objectives of the other community and try to search out the overlap, rather than the differences, there is room for real progress at much lower cost.

A very large aperture telescope in GEO has been a long-term goal of the military and intelligence community for decades, such that it can provide continuous optical surveillance of a battlefield or other region of interest. The science community would like a very large aperture telescope for looking at the universe. Both tend to work within their respective communities to develop a set of agreed-to requirements that are then chiseled in stone and can become forever immutable. (There is also a certain amount of "not invented here" and intellectual ownership of the solution that are often involved within each community when looking at solutions invented by a different community.) The Smallsat community, however, has learned the importance of, and processes for, trading on requirements so as to meet broad objectives at minimum cost and risk. If the differences can be set aside, such that the communities and organizations can work together, it may be possible to make a major breakthrough to the benefit of both military and civilian space. If the military and intelligence community can contribute a significant share of the funding, and the science community can contribute its knowledge of large space telescopes (and, importantly, of the potential problems and failure modes), a major new development could be possible. In addition, building 2 or 3 such telescopes can significantly drive down the recurring cost for each and potentially allow the creation of a small production line that both further reduces cost and allows us to maintain the skill base that created the system.

Our third approach is to apply the technologies, processes, and approaches from one community to the space missions of another. For example, battery technology is always a problem for LEO space missions because of the large number of charge/discharge cycles and the long duration of many of these space missions. Battery technology in the commercial electronics arena is advancing very rapidly due to the increasing need for longer life in nearly all cordless consumer electronics. In the far distant past, phones were used mostly for making phone calls. Today they are used for multiple applications and may be in nearly continuous use throughout the day and, therefore, in need of smaller, lighter weight, more efficient, and longer-lasting batteries. These batteries are now being used on CubeSats and Smallsats, where there is much less prohibition against using commercial electronics. As these batteries show their performance characteristics (and, potentially, failure modes) on orbit, they can become more widely adopted across the space enterprise.

Finally, it is important to note that any one person or organization can reasonably be a member of multiple space communities. It is possible to be a member of the military space community working on small science satellites and a strong supporter of Alt Space objectives. (After all, your "day job" may not reflect what you believe about the long-term future of mankind.) There are substantial advantages to this being the case. That person or organization that crosses boundaries can serve both to bring the various communities closer together and to recognize the potential of using the technologies, processes, and methods from one area to benefit another.

2.3 Changing Missions

None of the space mission communities is a static entity. In Sec. 2.1, we describe the likely future evolution of these communities and the principal drivers that will bring that about. Here we will summarize the changing missions across the whole space enterprise and how this is likely to affect individual communities.

In the near term, the most obvious change is the rapid and continuing advance of microelectronics. At the same time, a number of organizations are working on methods for radiation hardening electronics produced by ordinary production processes. As we have discussed, battery technology is also advancing very rapidly because of the push of consumer electronics. The net effect of all of these changes is to make Smallsat evolution extremely rapid in terms of increased capability in an ever smaller package. This at least suggests a significant change in nearly all missions. If capabilities are advancing rapidly and being tested on orbit in Smallsats, then it becomes difficult to justify creating spacecraft that last 15 years using technology that will become obsolete in a few years and may even be so by the time the system is launched. Thus, in at least some areas, there may be a change from long-term missions to much shorter ones in order to take advantage of advancing technology.

Related to the ever smaller electronics is the ever more powerful on-board processor and the push for increasing amounts of information and data, available instantly. The mythical FireSat mission was developed some 20 years ago for the *Space Mission Analysis and Design* book [Wertz and Larson, 1991]. In FireSat, IR images of forested regions were sent to the ground for analysis, fires were identified, and maps or other data were then sent to the firefighter in the field. Today it may make more sense to have a fire detection algorithm on board the spacecraft, complete with a filter to eliminate large barbecues and 4th of July celebrations, and then simply send the coordinates and technical details of the fire to the firefighter in the field, so that they are displayed in his or her truck on a detailed map of the region provided by any of the GPS navigation map providers. In a sense, this isn't a change in mission objectives, only a change in methods of implementation. But, in a broader sense, it is a change in objectives to focus more on information and less on data.

As we have discussed, the most substantial change in space missions will come about if there is a reduction in launch cost for any reason by a factor of 2 to 10. This changes the list of possible missions that are economically feasible and, depending on the result, may bring about a major increase in the number of missions flown which will further reduce costs. This change would affect the entire space enterprise and even begin to open the possibility for more expanded space exploration in some of the Alt Space missions described in Sec. 2.1.9.

References

For annotated bibliography of space mission communities, see the book website.

Bearden, David, R. Boudreault, and J. R. Wertz. 1996. *Reducing Space Mission Cost*. El Segundo, CA: Microcosm Press.

Berinstein, Paula. 2002. *Making Space Happen: Private Space Ventures and the Visionaries Behind Them*. Medford, NJ: Plexus.

Brown, Charles D. 1998. *Spacecraft Mission Design. 2nd ed.* with software disk. Reston, VA: AIAA.

Burrows, William E. 2006. *The Survival Imperative: Using Space to Protect Earth*. New York: Forge.

Chafer, Charles M., and Cynthia S. Price. 2002. "Burial." *Macmillan Reference USA Science Library: Space Sciences*. Macmillan Reference USA.

Chun, Clayton K. S. and Chris Taylor. 2006. *Defending Space: US Anti-Satellite Warfare and Space Weaponry*. New York: Osprey Publishing, Ltd.

Churchill, Susanne E. 1997. *Fundamentals of Space Life Sciences*. Malabar, FL: Krieger Publishing.

Clement, Gilles. 2005. *Fundamentals of Space Medicine*. El Segundo, CA: Microcosm Press and Springer.

Clement, Gilles and Klaus Slenzka. 2006. *Fundamentals of Space Biology: Research on Cells, Animals, and Plants in Space*. El Segundo, CA: Microcosm Press and Springer.

Conway, Eric D. 1997. *An Introduction to Satellite Image Interpretation*. Baltimore, MD: Johns Hopkins University Press.

Cruise, A. M., J. A. Bowles, T. J. Patrick, and C. V. Goodall. 2006. *Principles of Space Instrument Design*. Cambridge, UK: Cambridge University Press.

DeGroot, Gerard J. 2006. *Dark Side of the Moon*. New York: New York University Press.

Eckart, Peter. 1996. *Spaceflight Life Support and Biospherics*. Torrance, CA: Microcosm Press and Springer.

Eckart, Peter. 2006. *The Lunar Base Handbook. 2nd ed.* Boston, MA: The McGraw-Hill Companies.

Elachi, Charles and Jakob van Zyl. 2006. *Introduction to the Physics and Techniques of Remote Sensing*. Hoboken, NJ: John Wiley and Sons

Elbert, B. R. 1990. *International Telecommunication Management*. Norwood, MA: Artech House.

Elbert, B. R. 1992. *Networking Strategies for Information Technology*. Norwood, MA: Artech House.

Elbert, B. R. 2003. *The Satellite Communication Applications Handbook*. Norwood, MA: Artech House.

Elbert, B. R. 2008. *Introduction to Satellite Communication. 3rd ed.* Norwood, MA: Artech House.

Fleeter, Rick. 2000. *The Logic of Microspace*. El Segundo, CA: Microcosm Press and Springer.

Ford, Steve. 2010. *The Radio Amateur's Satellite Handbook*. Newington, CT: American Radio Relay League.

Glaser, P. E., F. P. Davidson, and K. I. Csigi. 1998. *Solar Power Satellites: A Space Energy System for Earth*. Chichester, UK: John Wiley and Sons.

Hardersen, Paul S. 1997. *The Case for Space: Who Benefits from Explorations of the Last Frontier?* Shrewsbury, MA: ATL Press.

Harrison, Albert A. 2001. *Spacefaring: The Human Dimension*. Berkeley, CA: University of California Press.

Haskell, G. and M. J. Rycroft. 1998. *New Space Markets*. Dordrecht, The Netherlands: Kluwer Academic Publishers.

Hujsak, Edward. 1994. *The Future of U.S. Rocketry*. La Jolla, CA: Mina-Helwig.

Isakowitz, Steven J., Joshua B. Hopkins, and Joseph P. Hopkins Jr. 2004. *International Reference Guide to Space Launch Systems*. Reston, VA: AIAA.

Johnson, S. B. 2002. *The Secret of Apollo: Systems Management in American and European Space Programs*. Baltimore, MD: Johns Hopkins University Press.

Kanas, Nick and Dietrich Manzey. 2003. *Space Psychology and Psychiatry*. El Segundo, CA: Microcosm Press and Springer.

Kemble, Stephen. 2006. *Interplanetary Mission Analysis and Design*. Chichester, UK: Praxis and Springer.

Klein, John J. 2006. *Space Warfare: Strategy, Principles, and Policy*. New York: Routledge.

Koelle, Dietrich E. 2010. *Handbook of Cost Engineering for Space Transportation Systems (Rev. 3) with Transcost 8.0*. Ottobrunn, Germany: TransCostSystems.

Lacomme, P., J. P. Hardange, J. C. Marchais and E. Normant. 2001. *Air and Spaceborne Radar Systems*. Norwich, NY: William Andrew.

Lambeth, Benjamin S. 2003. *Mastering the Ultimate High Ground: Next Steps in the Military Uses of Space*. Santa Monica, CA: RAND.

Larson, Wiley J. and Linda K. Pranke. 1999. *Human Spaceflight: Mission Analysis and Design*. New York: The McGraw-Hill Companies.

Lillesand, Thomas M., Ralph W. Kiefer, and Jonathan W. Chipman. 2008. *Remote Sensing and Image Interpretation. 6th ed.* Hoboken, NJ: John Wiley and Sons.

London, John R. 1994. *LEO on the Cheap: Methods for Achieving Drastic Reductions in Space Launch Costs*. Maxwell AFB, AL: Air University Press.

Mailer, Norman. 1970. *Of a Fire on the Moon*. Boston, MA: Little, Brown and Co.

Mallove, Eugene F. and Gregory L. Matloff. 1989. *The Starflight Handbook: a Pioneer's Guide to Interstellar Travel*. Hoboken, NJ: John Wiley and Sons.

Mather, John C. and John Boslough. 1996. *The Very First Light: the True inside Story of the Scientific Journey Back to the Dawn of the Universe*. New York: Basic Books.

Olsen, R. C. 2007. *Remote Sensing from Air and Space*. Bellingham, WA: SPIE.

O'Neill, Gerard K. 1974. "The Colonization of Space." *Physics Today,* vol. 27, no. 9. Sept. pp. 32–40.

O'Neill, G. K. 1975. "Space Colonies and Energy Supply to the Earth." *Science,* vol. 190. Dec. 5. pp. 943–947.

O'Neill, G. K., J. Billingham, W. Gilbreath, and B. O'Leary. 1979. "Space Resource and Space Settlements." NASA SP-428. Washington, DC.

Parkinson, Bradford W. and James J. Spilker Jr. 1996. *The Global Positioning System: Theory and Applications*. Washington, DC: AIAA

Peeters, Walter A. R. 2000. *Space Marketing: A European Perspective*. Dordrecht, The Netherlands: Kluwer Academic Publishers and Microcosm Press.

Richelson, Jeffrey T. 2001. *America's Space Sentinels: DSP Satellites and National Security*. Lawrence, KS: University Press of Kansas.

Sarsfield, L. P. 1998. *The Cosmos on a Shoestring*. Santa Monica, CA: RAND.

Schowengerdt, Robert A. 2007. *Remote Sensing, Models, and Methods for Image Processing. 3rd ed.* Burlington, MA: Academic Press.

Sutton, George P. and Oscar Biblarz. 2010. *Rocket Propulsion Elements. 8th ed.* Hoboken, NJ: John Wiley and Sons.

Swartwout, M. A. 2009. "The First One Hundred University-Class Spacecraft, 1981–2008." *IEEE Aerospace and Electronic Systems Magazine,* vol 24, no 3.

Wertz, J. R. 1999. "Architecture for Developing an Economically Viable International, Large-Scale Lunar Colony." IAF Specialists Symposium on Novel Concepts for Smaller, Faster, Better Space Missions. Redondo Beach, CA. Apr. 19–21.

Wertz, J. R. 2000. "Economic Model of Reusable vs. Expendable Launch Vehicles." IAF Congress. Rio de Janeiro. Oct. 2–6.

Wertz, J. R. 2004. "Reusability: The Death Knell of Low-Cost, Responsive Access to Space?" *Space News*. Mar. 22. p. 11.

Wertz, J. R. 2009. *Orbit & Constellation Design and Management*. Hawthorne, CA: Microcosm Press and Springer.

Wertz, J. R. and Simon Dawson. 1996. "What's the Price of Low Cost?" 10th Annual AIAA/USU Conference on Small Satellites. Logan, UT. Sept.

Wertz, J. R. and Wiley J. Larson. 1996. *Reducing Space Mission Cost*. El Segundo, CA: Microcosm Press and Springer.

Wertz, J. R. and Wiley J. Larson. 1991. *Space Mission Analysis and Design. 1st ed.* Dordrecht, The Netherlands: Kluwer Academic Publishers and Microcosm Press.

Wertz, J. R., R. E. Van Allen, and T. Barcley. 2010. "NanoEye—Military Relevant Surveillance for Less Than $5 Million Total Recurring Mission Cost." 8th Responsive Space Conference. Los Angeles, CA. Mar. 8–11.

White, Frank. 1998. *The Overview Effect: Space Exploration and Human Evolution*. Reston, VA: AIAA.

Young, Anthony. 2007. *Lunar and Planetary Rovers: the Wheels of Apollo and the Quest for Mars*. Berlin, Germany: Springer.

Zimmerman, Robert. 2010. *The Universe in a Mirror: the Saga of the Hubble Telescope and the Visionaries Who Built It*. Princeton, NJ: Princeton University Press.

Zubrin, Robert. 1999. *Entering Space: Creating a Spacefaring Civilization*. New York: Putnam.

Table 2-4◀, Fig. 2-17◀, Eq. 2-0

3 Space Mission Engineering

James R. Wertz, *Microcosm, Inc./USC*

3.1 The Space Mission Engineering Process
The Need-Driven Process; The Capability-Driven Process; Finding the Right Balance
3.2 FireSat II and the Supplemental Communications System (SCS)
3.3 Mission Objectives and Constraints (Step 1)
3.4 Principal Players and Program Timescales (Steps 2 and 3)
Identifying the Principal Players; Defining the Space Mission Timeline
3.5 Preliminary Estimate of Mission Needs, Requirements, and Constraints (Step 4)

As discussed in Chap. 1,

> ***Space Mission Engineering* is the refinement of requirements and definition of mission parameters to meet the broad objectives of a space mission in a timely manner at minimum cost and risk.**

It begins with one or more broad objectives and constraints and then proceeds to define a space system that will meet them within an appropriate timeframe at the lowest possible cost and risk. Broad objectives and constraints are the key to this process. Procurement plans for space systems too often substitute detailed numerical requirements for broad mission objectives. To get the most performance for the money spent, we must require of the system only what it can reasonably achieve. Thus, while our overall objectives to communicate, navigate, or observe will generally remain the same, we will achieve these objectives differently as technology and our understanding of the process and problem evolve. This chapter summarizes, and the book as a whole details, this process of defining and refining both what is to be done and what mission concept will do it at the lowest cost and risk.

We have introduced the term "mission engineering" to distinguish it from *systems engineering*, which is concerned primarily with the formal requirements definition process and how to validate that all of the requirements will, in fact, be met. *Mission engineering* is a broader process and is concerned with whether the needs of the end user are satisfied. As we discussed in Sec. 2.1.4, a historical example of this distinction is the Iridium communications constellation. From a technical perspective, Iridium worked very well. The launch process had no failures and all of the satellites worked well on orbit. The only real problem was that the system lost money and Iridium itself declared bankruptcy in 1999. The backers of Iridium lost $5.5B. In my view, this is a mission failure. It didn't meet the fundamental objective of a commercial system—to make money for the people and organizations that paid for it. In a sense it is similar to a scientific satellite that works perfectly but fails to return data to the scientist, or a military satellite that is technically perfect, but is launched after the war is over. They haven't met the very fundamental objectives of the people that created them—to make money, to advance our knowledge of the Universe, or to save lives or otherwise impact the outcome of a war or battle. These are the things that a space mission must do to be useful, irrespective of whether it meets all of the technical requirements placed on it. It is the goal of space mission engineering to see that this "usefulness" criteria is met, as rapidly and at as low a cost as possible.

There are now a number of references available on the mission design process and the definition of mission objectives. Johnson [2002] does an excellent job of describing this process for Apollo—the largest space program ever undertaken. Maier and Rechtin [2009], Chesley, et al. [2008], and Larson, et al. [2009] describe the process of building large, formal space systems, largely for DoD. Shishko [2010] provides an overview from the NASA perspective. Ford [2010], Fleeter [2000], and Wertz and Larson [1996] discuss this process from the perspective of very low-cost missions, smallsats, and methods for dramatically reducing mission cost, respectively. Sandau [2006] and Sarsfield [1998] look specifically at low-cost Earth observation and science missions. Boden, Larson, and Squib [2006] discuss the mission engineering process specifically for mission operations. Finally, Kay [1995] examines the fundamental difficulty of doing technical trades within a democratic political environment. Chap. 13 describes the mission engineering process specifically oriented toward reducing mission cost.

3.1 The Space Mission Engineering Process

Table 3-1 summarizes our approach to the space mission engineering process. As discussed in Chap. 2, space missions range widely from communications, to planetary exploration, to proposals for space manufacturing, to burial in space. No single process can fully cover all contingencies, but the method in Table 3-1

Table 3-1. The Space Mission Engineering Process. This table is relevant for a *need-based mission*, i.e., one that is intended to fulfill a specific (though possibly not well-defined) set of mission needs, and is the most common type of space mission. The alternative is a *capability-driven system*, discussed in Sec. 3.1.2.

Typical Flow	Step	Where Discussed
	Define Objectives and Constraints	
	1. Define the Broad (Qualitative) Objectives and Constraints	Sec. 3.3
	2. Define the Principal Players	Sec. 3.4
	3. Define the Program Timescale	Sec. 3.4
	4. Estimate the Quantitative Needs, Requirements, and Constraints	Sec. 3.5
	Define Alternative Mission Concepts or Designs	
	5. Define Alternative Mission Architectures	Sec. 4.1
	6. Define Alternative Mission Concepts	Secs. 4.2, 4.3
	7. Define the Likely System Drivers and Key Requirements	Sec. 4.4
	Evaluate the Alternative Mission Concepts	
	8. Conduct Performance Assessments and System Trades	Sec. 5.3
	9. Evaluate Mission Utility	Sec. 5.4
	10. Define the Baseline Mission Concept and Architecture	Sec. 5.5
	11. Revise the Quantitative Requirements and Constraints	Sec. 5.5
	12. Iterate and Explore Other Alternatives	Sec. 5.5
	Define and Allocate System Requirements	
	13. Define System Requirements	Sec. 6.1
	14. Allocate the Requirements to System Elements	Sec. 6.2

summarizes a practical approach evolved over the first 50 years of space exploration.

The process we have defined here is based broadly on the space mission analysis and design process defined more than 20 years ago by Wertz and Larson [1991, 1999]. However, it has evolved in significant ways by asking us to identify the principal players in the space mission and their goals and objectives and the intended program timescale. A few other changes have been made as well to clarify how to go about getting to the mission concept that we want to achieve. In the end, it isn't so much the specific steps that matter, so much as having a process that we have carefully thought through to get us to a design concept that achieves our goal of meeting the end-users' objectives in a timely manner at minimum cost and risk.

Space is expensive. Cost is a fundamental limitation to nearly all space missions and is becoming more so. Consequently, Table 3-1 and subsequent ones reflect the assessment of each author on how things traditionally have been done **and** how they should be done differently, both to lower cost and to achieve the greatest return from the space investment.

Space mission engineering is an inherently iterative process, gradually refining both the requirements and methods of achieving them. Thus, we must repeat the broad process defined in Table 3-1 many times for each mission. The first several iterations may take only a day, but more detailed assessments will take far longer.

Successive iterations through Table 3-1 will usually lead to a more detailed, better-defined mission concept. In defining and refining the approach, there is strong pressure to proceed to ever greater detail, and never revise a decision once it has been made. But we must still return regularly to the broad mission objectives and search for ways to achieve them at a lower cost. Methods may change as a result of evolving technology, a new understanding of the problem, or simply fresh ideas and approaches as more individuals become involved.

Finally, we must document the results of this iterative process. If we wish to go back and reexamine decisions as new data becomes available, we must clearly understand and convey to others the reasons for each decision. We need this documentation for decisions based on detailed technical analyses, and, equally important, for those based on simplicity, ease of assessment, or political considerations.

3.1.1 The Need-Driven Process

Table 3-1 illustrates the process for a *need-based mission*, i.e., one in which we are trying to fulfill a specific set of mission objectives, such as identify and monitor forest fires, provide better communications to the warfighter, explore the Universe in the infrared region of the spectrum, or make money by providing direct broadcast of worldwide sporting events. (The alternative type of capability-based mission is discussed in Sec. 3.1.2.) To illustrate the broad process of Table 3-1, we will go through each of the top-level steps and indicate the type of information that needs to be developed:

In **Step 1**, we define what the mission needs to achieve. What are our qualitative goals, and why? This information should come largely from a mission statement of what the mission is attempting to achieve. (See Sec. 3.2.) We need to return to these broad goals over and over to verify whether we are achieving what we set out to do and whether we could do it better or at lower cost.

Step 2 identifies the principal players (often called *stakeholders*) and the space community that they are a part of. Is this a military mission intended to provide information to the warfighter in battle and funded by Congress? Is it the sports broadcasting mission intended to make money for the organizations or people that funded it? Notice that the end user and the people that have put up the money are typically not the same group and, in the end, it would be best to have both groups satisfied with the end results.

Step 3 defines the timeline over which the program needs to be executed to be useful. There is very little utility to the warfighter if the space system works exceptionally well, but isn't launched until after the war is over. If our sports broadcasting network has lined up 2 million customers and then there is a 3-year delay in the program, it is likely that most of those customers will have found other solutions and won't be there when the space system is finally launched.

Step 4 is significantly different from those above. It **quantifies** how well we wish to achieve the broad objectives, given our needs, applicable technology, who the users are, and cost and schedule constraints. These quantitative requirements should be subject to trade as we go along. A major error in many space-system procurements is to set requirements in concrete at a relatively early stage. For example, our sports broadcasting network might have set an initial goal of being able to broadcast 100 different sports channels, because of the marketing appeal that this would have. But it may turn out that this would require a larger spacecraft, or worse, a larger launch vehicle, than anyone currently has, and, therefore, may take far more time and money than anyone had initially planned for. It may be a much better choice to have 70 sports channels at low cost in the near future, than 100 sports channels at many times the cost and much further out in time. Space mission engineering provides the quantitative data needed to support such decisions.

Step 5 defines alternate combinations of mission elements or the *space mission architecture* to meet the mission objectives and requirements. We do this by looking at the individual mission elements, defined in Sec. 4.1, such as the payload that will do the work or the orbit that the spacecraft is in, and consider the principal alternatives for each of them that would best meet mission objectives.

In addition to identifying all of the pieces, we need to understand how they all work together to meet the needs of the end user. **Step 6** begins this process by developing alternative mission concepts. A *mission concept* or *concept of operations* is a broad statement of how the mission will work in practice. How do the various sporting events get broadcast and delivered to the satellite so they can be sent down to the sports fan sitting in her living room? Typically, the most effective mission concepts are ones that are totally transparent to the end user. Neither the warfighter getting shot at nor the sports fan in front of her TV knows or cares very much about space systems and typically don't want to. There is a long-standing joke in the space community that the warfighter on the ground will say that they don't need any space systems. All they need is their rifle, their communications gear, and their GPS. This is a good result. Neither the warfighter on the ground nor the driver in their car trying to figure out which way to turn needs to know anything at all about that remarkable array of atomic clocks and software at half-geosynchronous altitude, called the Global Positioning System, that can tell you where you are anywhere on the surface of the Earth to within a few meters. This indicates a truly successful mission concept that has been well executed.

In any real system, many things influence overall cost, performance, or the design of detailed components. However, these are influenced mainly by a relatively small number of key parameters or components, called *drivers*. Thus, there may be *cost, performance,* or *system drivers* which affect the design of the overall space system. In **Step 7**, we identify the principal cost and performance drivers for each alternative mission concept. Similarly, there are *critical requirements* that are those that have the greatest impact on cost, risk, and schedule. For most space missions, system drivers include the number of satellites, altitude, power, and instrument size and weight; critical requirements usually include lifetime, availability, coverage, and resolution or signal strength. (Section 4.4 gives a more detailed list of the most common system drivers and critical requirements.) By explicitly identifying the system drivers and the critical requirements, we can concentrate our effort on parameters having the most impact on the design and therefore on the cost and performance of the space mission. This improves our chances of getting the best possible design within the available budget. As discussed in Sec. 4.4, the system drivers and critical requirements are likely to be different for different mission concepts. What is hard in one approach to achieving our objectives may be simple in another approach and vice versa.

Step 8, conducting *performance assessments* and *system trades*, is typically the most involved step in mission design because it defines in detail what the system is and does. Here we determine the power, weight, and pointing budgets* and decide what to process on the ground or in space. Characterizing the mission and trading among alternatives is often the most costly step because it requires the expertise of many people. Developing detail is always comforting in managing any design process but, as noted earlier, we must take care not to overdo details while characterizing the mission. Achieving the mission objectives and meeting the needs of the end user must remain our primary focus.

---

* A *budget* is a numerical list of the components of any overall system parameter. Thus, the total spacecraft *mass budget* would consist of the masses assigned to the payload instruments, the various subsystems, the propellant required, and typically some margin for growth.

The above issues form the basis of *mission utility analysis*, **Step 9**, in which we quantify how well we are meeting both the broad objectives and the needs of the end user as a function of either cost or key system-design choices. We would like to provide the decision maker a single chart of how well the needs of the end user are satisfied vs. cost. More typically, we must settle for something less ideal, such as the number of hours per day that our hypothetical sports fan is expected to watch our network. Only the user or developer of the system can ultimately determine the goodness of these critical performance measures, called *Measures of Effectiveness* (MoEs) or *Figures of Merit* (FoMs). Consequently, mission definition must be, to some degree, a joint process between those who understand the mission engineering process and those who eventually must use the system or justify its cost.

Having evaluated alternative designs and done a preliminary assessment of mission utility, we select one or more baseline system designs in **Step 10**. A *baseline design* is a single consistent definition of the system that meets most or all of the mission objectives. A *consistent system definition* is a single set of values for all of the system parameters which are consistent with each other—e.g., number of channels and coverage area which correspond to the assigned altitude, antenna size, power, and resulting spacecraft weight. In actually designing a space system, many parameters are being defined and changed simultaneously. The baseline provides a temporary milestone against which to measure progress. It also allows us to limit the number of options which must be evaluated. Rather than looking at all possible combinations of altitude, antenna size, power, and spacecraft mass (a nearly impossible task), it is much more feasible to look at the impact of varying each of these individually relative to one or two baseline designs. As the system design matures, the baseline becomes more firm, and eventually becomes the system design. However, we should always remember that the baseline is only a starting point for the iterative trade process and should **not** be regarded as an ironclad definition of mission parameters.

Having determined the critical requirements and their impact on the mission design and mission utility, **Step 11** is to revise the system requirements and constraints consistent with what we have learned. We need to find the requirements that reflect what we want to achieve (our mission objectives) and what can be achieved with the tools, techniques, and hardware available to us. As a final step in concept exploration, **Step 12** is to go back and explore other alternatives and iterate on the design. As we will see in Sec. 3.1.3, the result we are looking for is a balance between what we would like to have and what we can achieve in a timely manner within a budget and level of risk we are willing to accept. When we start out, we don't know what this will be. This is the reason for exploring alternatives and looking for options. Is there another approach that achieves our objectives better or at lower cost? Or if we change the focus of our objectives a bit, can we get there at lower cost and less risk? For our sports network, should we consider two satellites of 50 or 35 channels each? Or should we change our objectives a bit to carry not only live sporting events, but also sports highlight shows and a sports-oriented shopping channel?

The first 12 steps are what are called *concept exploration*—looking at alternative requirements, constrains, and system options. Because builders of a space system typically work from specific requirements, we must translate the now better-defined objectives, constraints, and requirements into well-defined system requirements in **Step 13**. This is where the more traditional process of systems engineering begins. Finally, in **Step 14**, we *flow down* or allocate these numerical requirements to the components of the overall space mission in the same way that a budget allocates weight and power to the spacecraft's components. This will turn into a detailed list of requirements and, ultimately, into a space mission on orbit. The degree to which this meets our new set of mission objectives is a measure of how well we have done the mission engineering. Like GPS, we would typically like the end user to not know or care about space mission engineering or space at all. Success isn't measured by how well we have followed the steps outlined either here or in Chap. 1, but in whether the end user and the customer that put up all of the money to get there are fully satisfied by the result.

3.1.2 The Capability-Driven Process

Most missions follow the process outlined in the preceding section, i.e., a mission need is defined and then we start out to design a space mission that can meet that need at minimum cost and risk. However, most major breakthroughs in doing things better, faster, or cheaper come not from defining needs and how to fill them, but rather by identifying a new capability, or perhaps a new way to use an existing capability, and then finding a space mission that could use that capability. Some of the obvious potential breakthroughs could include, for example, a major reduction in launch cost or more computer capability or a new propulsion system that could provide far higher delta V. Indeed, much of space research is spent trying to find these breakthroughs, what are often called *game-changing technologies* or *disruptive technologies*.

An example of this type of game-changing technology is nuclear electric propulsion, proposed as a potential NASA program which has since been cancelled as being too expensive. This would have used nuclear power, essentially similar to that which has been safely used in submarines for decades, in space to provide 50 kW to 100 kW of power for large spacecraft for both electric propulsion and to provide high power for other mission aspects. This would most likely make no difference to some space missions, such as smallsats. However, there are many missions for which it could make a dramatic difference. Some examples of the things that might be enabled include:

- Synthetic aperture radar missions with continuous imaging that can see through clouds and at night to provide continuous surveillance, possibly from GEO.
- High delta V missions that could put spacecraft in orbit around planets and asteroids with high data rate, high-resolution images coming back from distant parts of the solar system.
- Reducing the round trip time to Mars from roughly 2.7 years to about 5 months due to cutting out one synodic period and, therefore, enabling near-term human missions to Mars.
- Allowing very high power transmissions from GEO such that our sports network defined in the previous section becomes a thousand channels of high-definition sports programs (which may or may not be a good thing, depending, perhaps, on whether your goal is to get a sports fanatic to get off the couch and mow the lawn).

While most breakthroughs in technology or ways of applying existing technology won't have effects that are this large, they may still have a major impact on what missions are possible and what level of mission performance can be achieved at what cost. This in turn, fundamentally changes how we go about defining a space mission. The major change is that we start out differently and ask ourselves, what would this new process or technology enable? If we had nearly unlimited power, or far more computing, or the ability to repair spacecraft at low cost, what would this let us do that we can't do today, either because it is impractical or far too expensive?

Step 2 in the capabilities-based process is the same as before, only asked for a different reason. Now we are looking for the principal players in order to explain the new technology to them and how it can enable missions that weren't previously possible or weren't economical to undertake. This is where things often get difficult. Previously, we had a customer or end user that wanted an answer and we went about finding the best possible answer. Now we have an answer and are looking for someone who wants to ask the question for which we have the answer and, more importantly, pay for the answer that we've found. This means we are likely to be looking for someone willing to pay for what may be a very expensive space mission and, therefore, we will probably need to be very convincing. For this reason, capability-driven missions are often very difficult to make happen. Nonetheless, they are also the real source of major breakthroughs.

Most of the remainder of the mission engineering process proceeds as defined in Table 3-1, with the caveat that we continually go back and review both the needs of the end user and the capabilities that are available. As with the need-driven process, the purpose is not to make the most use possible of whatever new capability or process we have found, as it is to use the new capability to allow us to meet a set of mission objectives and end-user needs better, more quickly, or at lower cost and risk.

3.1.3 Finding the Right Balance

A key issue in mission engineering is iterating the design a number of times to try to get to a system that does the job well. We want to convince ourselves that we have looked at the reasonable alternatives and found the solution that best meets the needs of the customer and the end user.

If we start with objectives, we must look at what capabilities are available and cost effective. If we start with a capability, we must look closely at the mission needs that it can fulfill and who the players are who could make use of that capability at the likely price. Ultimately, every practical mission must balance needs and capability. We are trying to meet our mission objectives by balancing needs and capabilities.

Going through the space mission engineering process, there are 3 possible adverse outcomes:

- We simply can't do what is needed (e.g., we don't have a launch vehicle with sufficient volume to launch our payload)
- We can do something but it doesn't have sufficient utility to the customer (e.g., the payload that we can build doesn't have enough resolution to be helpful to the problem we're trying to solve)
- We can do it and it's useful, but it isn't worth the cost or risk or can't be done in time (by the time we get it built, the comet we're trying to explore will have swung past the Sun and be on its way back to the Oort Cloud)

In order to proceed, the mission must be achievable and useful, but must also be worth the cost to whoever is putting up the money to do it.

If the space mission engineering process is successful, we will have defined a space mission that is sufficiently low-cost, low-risk, and near-term, with sufficient performance that the customer is willing and anxious to proceed to build the system. If we have done it correctly, we will be able to accomplish the mission within the time and cost that we have estimated (or at least reasonably close) and will be able to meet the ultimate objectives of the customer. We don't need more Iridiums. It is not sufficient that the system work—it has to work well enough to meet, or preferably exceed, the expectations of the customer and the end user, including cost and timeliness.

At its best, space mission engineering is a lot like determining what car to buy for an individual. We have some characteristics that we need, such as safety, reliability, and enough space for the people we expect to be riding in it. We also have a limited budget and need something that falls within, or reasonably close, to that budget. There are many different cars available in the marketplace, new ones with new features and capabilities are coming out all the time, and we can add features to almost any car we buy. We're looking for the right balance between what we want and what we can reasonably afford, so that we're happy with the new car (or space mission) that we buy for many years to come.

Table 3-1◀, Fig. 3-0, Eq. 3-0

3.2 FireSat II and the Supplemental Communications System (SCS)

In 1990, the concept of a hypothetical space mission called FireSat was introduced to illustrate the process of space mission design [Wertz and Larson, 1991]. Since then, FireSat has become a standard example mission throughout the astronautics community, but has never been funded and built, although space assets have certainly been used to photograph and monitor forest fires. This is probably related, at least in part, to the fact that the estimated cost of FireSat was $704M (inflated to FY10$) as a traditional pair of spacecraft and $85M as two smallsats. It simply wasn't worth that much money in order to achieve a relatively modest level of mission utility. Chapter 1 discusses how this is the case for many missions—they cost too much and take too long for what they are able to achieve. However, in the 20 years since FireSat was first proposed, substantial advances have been made in the capability of small satellites and an entirely new vocabulary has been created with minisats, microsats, nanosats, and picosats. (See Sec. 2.1.8.)

FIRESAT II
MISSION STATEMENT

Because forest fires pose an ever-increasing threat to lives and property, have a significant impact on recreation and commerce, and also have an even higher public visibility (largely because of the ability to transmit television images from nearly anywhere in real time), the United States needs a more effective system to identify and monitor them. In addition, it would be desirable (but not required) to monitor forest fires for other nations; collect statistical data on fire outbreaks, spread, speed, and duration; and provide other forest management data. This must be done at low cost to make the system affordable to the Forest Service and not give the perception of wasting money that could be better spent on fire-fighting equipment or personnel.

Ultimately, the Forest Service's fire-monitoring office, fire management officers in the field, and individual firefighters and rangers fighting the fire will use the data. Data flow and formats must meet the needs of all of the groups without specialized training and must allow them to respond promptly and efficiently to changing conditions.

Fig. 3-1. Mission Statement for the Hypothetical FireSat II Mission. FireSat II is used as one of two primary examples throughout this book.

Can these very small satellites, or other advances, such as improved on-board computing, finally make FireSat a practical system? To find out, we will introduce a new hypothetical mission, called *FireSat II*, for this volume. FireSat II will have broadly the same objectives as the original FireSat mission, i.e., to detect and monitor forest fires in the United States and Canada, but with an additional objective of being very low cost. This reflects the idea that the original FireSat concept was a buildable spacecraft, but not a practical one that achieved what the

Table 3-1◀, Fig. 3-2, Eq. 3-0

SUPPLEMENTAL COMMUNICATIONS SYSTEM (SCS)
MISSION STATEMENT

There is ever-increasing need for high-quality, real-time global communications to meet military needs to save lives, bring to bear the necessary resources, and avoid collateral damage. Large military communications systems currently exist, but these are often overloaded in a time of conflict and may not be available to an individual or small unit in a remote location. A secondary mission of equal importance is to aid in natural disaster relief, particularly in remote areas with severe damage where traditional communications systems may be destroyed, unavailable, or overloaded.

The end user for SCS is an individual or small group in a remote location with very limited resources. The user needs two-way, real-time communications with either a command post broadly in the same vicinity, or with planners and rescuers in CONUS or other locations scattered around the world. The communications content can range from brief text messages, to voice communications, to photos taken, for example, from a cell phone.

Fig. 3-2. Mission Statement for SCS. SCS will also be used as a principal example throughout the book.

end user needed at an acceptable cost and risk. We will use FireSat II as one of two example missions throughout the book and will look specifically at ways to implement the mission at very low cost in Chap. 13.

As a second example, we will use a small constellation, the *Supplemental Communications System* (SCS), that is intended to provide better communications to military or disaster relief personnel that are in remote areas where communications are either disrupted or didn't exist in the first place. While we will use many examples from real space missions for specific topics, we will use these two hypothetical missions throughout the book to follow the whole space mission engineering process. Our goal is to start with a "blank sheet of paper" (or, in this case, with a small text box in Fig. 3-1 for FireSat II and Fig. 3-2 for SCS), and define a space mission that meets this set of broad, not too well defined, mission objectives in a timely manner at minimum cost and risk.

The *mission statements* shown in the two figures define what is needed to be useful. They don't describe the orbit, or the required bandwidth, or the resolution that must be achieved, or the size of the spacecraft, or the cost, or timeframe. All of these are to be determined by the space mission engineering process. In a sense the process is simple. We do what is in the figure quickly, cheaply, and at low risk.

We want to stress that the parameters developed throughout the book are by no means the only possible set for FireSat II or SCS, nor are they necessarily the best. To show how solutions may vary, we will present alternatives along the way. Our example system simply illustrates the iterative process of space mission engineering. Different assumptions, requirements, or proposed solutions may lead to dramatically different results.

For both missions, we know that the system is technically feasible because they can be accomplished by existing, but relatively expensive, systems. What we don't know is whether they can be done sufficiently near term and at low enough cost and risk to be of interest to our potential customer. Therefore, our goal should not be a stringent set of requirements, but rather a strong system solution that can do the job economically and quickly such that the customer can gain benefit from the mission. Cost and timeline will be two key characteristics that determine whether our missions succeed or fail.

3.3 Mission Objectives and Constraints (Step 1)

The first step in the space mission engineering process is to define the *mission objectives*: the broad goals that the system must achieve to be productive. Figure 3-3 shows sample objectives for the FireSat II mission and Fig. 3-4 for SCS. We draw these qualitative mission objectives largely from the mission statements in Figs. 3-1 and 3-2. In contrast, the *mission requirements* and *constraints*, discussed in Sec. 3.5, are quantitative expressions of how well we achieve our objectives—balancing what we want against what the budget will allow. Thus, whereas we are likely to modify objectives only slightly or not at all during concept exploration, we will trade on requirements and revise our constraints many times throughout the process. For FireSat II to be FireSat II, it must detect, identify, monitor, and report forest fires. That is what the mission is about. How accurately or how often it does that depends on what is achievable at what cost. As we trade and design elements of the system during concept exploration, we must ensure that the system meets its fundamental objective.

Ordinarily, space missions have multiple objectives. Some are *secondary objectives* that can be met by the defined set of equipment, and some are additional objectives which may demand more equipment. Nearly all space missions have a *hidden agenda* that consists of secondary, often nontechnical, objectives. Frequently political, social, or cultural, they are equally real and equally important to satisfy. For example, a secondary objective for FireSat II could be to show the public a visible response to frequent forest fires. Third World nations produce satellites in part to show that their developing technology makes them important players in international politics. Of course, this political objective for space programs has been important for many years in the United States, the former Soviet Union, and now China as it seeks to establish itself as a major world power. If we are to meet all of a space mission's objectives, we must identify secondary and nontechnical objectives as well as primary ones. As discussed in Sec. 3.2, we have added the words "and at low cost" to the end of the primary mission objective to emphasize the imprecise, but nonetheless very real, need for a reduced-cost solution if FireSat II is to be funded.

FIRESAT II MISSION OBJECTIVES

***Primary Objective*:**

- To detect, identify, monitor, and report forest fires throughout the United States, including Alaska and Hawaii, in near real time and at low cost.

***Secondary Objectives*:**

- To demonstrate to the public that positive action is underway to contain forest fires.
- To collect statistical data on the outbreak and growth of forest fires.
- To monitor forest fires for other countries.
- To collect other forest management data.

Fig. 3-3. FireSat II Mission Objectives. Unlike *requirements*, which specify numerical levels of performance, the *mission objectives* are broad statements of what the system must do to be useful.

SCS MISSION OBJECTIVES

***Primary Objectives*:**

- To provide supplementary, continuous real-time communications at low cost to troops, including individual soldiers, in remote locations (at mid and low latitudes) using cell phone-size equipment, if possible, equipment that they already have.
- To provide the same or similar levels of communications to disaster relief workers in a remote location.

***Secondary Objectives*:**

- To support comparable communications worldwide, depending on the cost.
- To give the individual remote soldier or disaster relief worker the sense that they are a part of a team that will provide them the support they need.
- To provide similar support to allies and relief workers from other countries.
- To automatically and autonomously collect GPS or comparable data to monitor and track the location of system users in real time.

Fig. 3-4. SCS Mission Objectives. SCS has two primary objectives corresponding to the two missions defined for it in Fig. 3-2.

Multiple objectives also occur when we use a single satellite to meet different demands, as in the case of SCS. In the case of FireSat II, we might use its temperature-sensing instruments to monitor global changes in ocean temperatures as well. In this case, the secondary objectives could become as important as the primary ones. A second example would be adding a communications payload to FireSat II to permit better communications among the distributed groups who fight forest fires, or add a GPS receiver to the SCS unit to allow autonomous monitoring of the location of individuals or small groups on the ground to allow rescue or support of people who can no longer communicate. Although the primary objective usually will be quite stable, secondary objectives may shift to meet the users' needs and the redefined potential of the space mission concept.

Table 3-2. The Principal Players, Their Objectives, and What Might Change Those Objectives for Representative Systems. See text for discussion.

Mission	FireSat II	SCS	Safari Airplane
Primary Customer / Objective	US Forest Service / Detect and monitor forest fires in US and Canada using as little financial resources as possible	Both DoD and Disaster Relief Agency / Supplemental, continuous, real-time communications	African safari operator / Improve profits by moving people quickly between sites
Secondary Customer / Objective	Congress / Demonstrate to the public that action is being taken and government funds spent wisely	Congress / Satisfy military needs and, at the same time, respond to public perception of slow disaster relief support	National Bank of Kenya / Make money on a commercial loan; support increased safari business
Operator / Objective	FireSat II Mission Operations Office / Provide forest fire data to the end user in real time; archive data; monitor and maintain the health and safety of FireSat II	SCS Spacecraft Operations Office / Monitor and maintain the continuously operating communications link; monitor and maintain the health and safety of the satellites	2 pilots hired by the safari company / Provide rapid, safe interlocation transportation
End User / Objective	Forest fire field management or firefighter in truck / Track current location of fire relative to themselves	Individual soldier or relief worker in the field / Explain situation and get the right help where it's needed as quickly as possible	People on safari / Don't care about airplane—just want to see as much as they can in 10 days
What Might Change System Objectives	Change in administration; change in projected system cost (up or down); major fire that didn't have sufficient monitoring resources	Splitting procurement to buy 2 systems; major disaster where communications weren't available in a timely manner	Safari company sold to either much bigger or much smaller company; a new, very remote site for wildlife viewing is opened

As in the case of most top-level trades, we strongly recommend against numerical formulas that try to "score" how well a mission meets its objectives. We can compute probabilities for achieving some technical objectives, but trying to numerically combine the coverage characteristics of different constellations with the political or psychological impact of putting a system in place is too simplistic for effective decision-making. Instead, we must identify objectives separately so we can judge how to balance alternative objectives and mission concepts.

Good mission objectives incorporate user needs and, at least indirectly, the characteristics of space that we are exploiting to achieve them. (See Table 1-3 in Sec. 1.5 for a list of these characteristics.) As stated earlier, space is expensive. If our end objective does not use one of the fundamental space characteristics, it will likely cost more to do in space than on Earth. There is little reason to manufacture and distribute low-cost consumer goods or manage a restaurant chain from space.

3.4 Principal Players and Program Timescales (Steps 2 and 3)

3.4.1 Identifying the Principal Players

In a sense, we could design and build a passenger airplane by just looking at the specifications provided by the airline that wants to buy it and responding to all of those specifications. But we would do a much better job if we also looked at the needs and interests of both pilots and airplane passengers. What set of features meets the needs of the pilot to ensure safe and efficient operations? What do passengers want in choosing an airline? Can we help our hypothetical start-up airline by providing them information on what pilots complain about the most in small airplanes, what passengers normally ask about, or the trends in the growth of the width of passenger rear ends over the last few decades? This help would be even more important if, for example, our customer wasn't an airline, but an African safari company that was an expert in African wildlife, but had never bought an airplane before and doesn't have a group of pilots on its staff. Similarly, it might be helpful to understand what banks that finance the purchase of aircraft normally require to make the purchase go smoothly. We could do a better job for our customer by telling them that we work with banks that finance aircraft on a regular basis and what our experience has typically been.

This analogy works well in the space mission engineering arena. As we have said several times, our goal is to start with a set of broad, often poorly defined mission objectives and to meet those objectives in a timely manner at minimum cost and risk. This means understanding the real needs of our customer and his customers, both those who are supplying the money to build the mission and the operators and end users that will ultimately use it. Typically, there are four distinct groups, each with their own goals and objectives, as defined in Table 3-2. The *end user* is the set of people that will actually use the data, and typically don't care very much about space. Most GPS users don't really care about what frequency it uses or the algorithms used to compute the position. (Most users also don't care that GPS measures time really, really well or that this is an important part of the process.) For FireSat II, the end user is either the fire management boss in the field or the firefighter with a shovel in their truck. For SCS, it's the field commander, the soldier that needs help, or the rescue worker in a remote mountainous area. For each of these end users,

they have far more important things to think about than space systems. They really don't care very much about us or about this really neat space system we're designing for them, so long as it works. If it doesn't work they are likely to think about us a great deal in ways that we would probably prefer to not be thought about.

While the end user doesn't care about us or the space system, we very much care about them. Someone is going to be asked to spend lots of money satisfying at least one aspect of their needs and that means that we really want to understand those needs very well. The best mission engineering will be done by a combination of people that understand space missions very well and people that understand the end user very well. If we're going to do the very best possible job of creating a strong design for FireSat II, we need an ex-forest ranger or a former forest fire field commander on our team. They can't design the system because space mission engineering isn't their forte, but they can be sure that what we're designing and ultimately implementing will genuinely meet the needs of the end user.

In addition to the end user, there is the *operator* that will actually fly the spacecraft, i.e., the pilot in our airplane analogy. These are the people that care the most about the satellite, even more than the people that buy it, because they have to live with it day-to-day, potentially for many years. They are interested in the practical aspects of space systems. How do you command it? How do you get the data down; and how do you process it, analyze it, archive it, and get it to the end user? The operator is also interested in monitoring the health and safety of the spacecraft itself. How much fuel is left? What's the temperature of the payload? Will a spacecraft maneuver inadvertently get direct sunlight on a surface that is intended to always be in the shade? Often the most asked question by spacecraft operators is: How could an intelligent engineer have been so dumb as to design the spacecraft this way? This suggests that it might be a good idea to have a spacecraft operator on the initial design team and even pay attention to what they're saying from time-to-time.

As implied above, the end user and the operator are rarely the same as the customer for a space system. The *primary customer* or *sponsor* is the person or group, or possibly collection of groups, that actually buy the space system. Certainly, their goal is also to satisfy the needs of the end user, but they may have ancillary goals along the way as well. Our African safari team has a clear objective of making money and we want to understand this as well. If we sell them a magnificent airplane that costs so much that they can't afford to run safaris or if they have to charge a price that is too high to get any customers, then we haven't done a good job. For FireSat II, let's assume that our customer is the US Forest Service. While they're not interested in making a profit, if we sell them a FireSat II system that doesn't work well, costs so much that they can't afford other critical equipment, or takes so long that all of northern California burns down before the system was in place, then again, we haven't done a good job, irrespective of whether we have met all of the specifications that were put in place. This will likely be made very clear to the provider of FireSat II in future discussions with the Forest Service and with Congress.

The idea of explaining the results to Congress leads to the concept of a *secondary customer*, i.e., not the people or group that signs the contract, but the people or groups that they must satisfy and who may have put up the money for the project—and likely more money than they would have preferred to spend. For US Government programs such as FireSat II and SCS, this would be the US Congress. Once again, there may be additional issues in dealing with Congress or any financial backer. It is likely that Congress regards its role in FireSat II as different than that of the Forest Service. Both would probably regard themselves as the ultimate decision-maker. It is in our best interest to recognize the role that Congress plays and to be sure that the system we build meets those needs and objectives as well. Most formal proposals to the Government for large space systems include a several page executive summary that includes a map of the United States showing all of the states in which some part of this space system is going to be built. That is because FireSat II, SCS, and many space programs, both military and civilian, are ultimately funded by the people of the United States, and the American people would like to have the jobs that these programs create spread throughout the United States. It will be very hard to get Congressional funding for any **large** space program that is built exclusively in either California or Iowa.

Another complication arises when there are multiple customers or multiple stakeholders in any one space venture. The International Space Station (ISS) includes both funding and hardware from many countries. It is probable that the goals and interests of, for example, the United States, Canada, and Russia in the ISS are not all identical. It would be surprising if they were. As discussed in Sec. 3.3 above, most space programs have both technical and non-technical (potentially political), objectives. In order for the project to be funded to meet the needs of the end user, it also has to satisfy the needs and objectives of the multiple customers involved, including the non-technical objectives. It is unlikely that this will happen by chance. In order for the competent space mission engineer to ensure that these diverse goals and objectives are met, he or she must understand these goals and objectives; and therefore, must also understand who the principal players are, the community they come from, and their principal objectives. Suppose, for example, we are considering a major expansion of the ISS, but that expansion does not make any use of the robot arm created by Canada for the ISS and a major source of Canadian pride. It is almost certain that we wouldn't get any political support or any funding from Canada, and could well get substantial opposition to our plan, and possibly to other joint Canadian-US space programs as well.

Table 3-2◀, Fig. 3-4◀, Eq. 3-0

Table 3-3. Representative Space Mission Timeline. The names used here are generic, but fairly widely used. Each of the large space organizations have their own set of mission phases, typically defined by formal Milestone reviews that are required for "graduation" to the next phase.

Phase	End Defined By	Typical Duration (small program)	Typical Duration (major program)
CONCEPT EXPLORATION	Start of technology funding; preliminary requirements release	1–6 months	2–10 years
	DETAILED DEVELOPMENT		
Risk Reduction/Technology Development	Start of program funding	0–12 months	1–10 years
Detailed Design and Development	Formal requirements release	2–12 months	0.5–4 years
	PRODUCTION AND DEPLOYMENT		
Production	Ship to Launch Site	6–24 months	2–5 years
Launch	Lift-Off and Arrival in Low-Earth Orbit	1 month	3 – 12 months
On-Orbit Check-Out /Transfer to Operational Orbit	Start of operations	0–10 days	1–3 months in Earth orbit***
	OPERATIONS AND SUPPORT		
Operations	Spacecraft dies or decision to be put to sleep	1 month–5 years*	5–15 years*
Disposal	Re-entry or turn-off**	0–5 years	up to 25 years

* Upper limit shown is typical design life. In practice, on-orbit spacecraft rarely wear out and can live for many years after the formal end of their design life if they have enough propellant or other consumables on board.
** Data analysis may continue for an extended period. *** Up to 10 years for Interplanetary

The potential principal players in FireSat II, SCS, and our safari airplane system are identified in Table 3-2 above, along with the principal objectives that they will be trying to meet and potential changes in those objectives. Note that it is entirely possible for these principal players to change from time-to-time, either because a new administration comes into power or because some players drop out or others join in order to find sufficient resources or support to make the program occur. A famous example is the GPS program which did not originally have sufficient political support to be funded in Congress. The Nuclear Detection System (NDS) was added as a secondary payload to each of the GPS satellites in order to gain sufficient support to be fundable in Congress. It would be of no use to the many users of GPS if we had designed a remarkably capable system, but weren't able to make it achieve enough Congressional support to be funded.

3.4.2 Defining the Space Mission Timeline

Much the same logic applies to the program timeline as it does to the principal players. If the timeline we create and the space system that we build on that timeline don't support the needs of the users, we have done a bad job of space mission engineering. We can always argue (and often do) that this wonderful space system will be of great use in future forest fires or future wars or disasters, but that is of very little comfort to those who were killed or who lost their homes or loved ones while we were spending time arguing over specifications or how many reaction wheels are needed. In order to be useful, results must also be timely, although of course, the meaning of timeliness can change with the circumstances.

At a top level, the space mission life cycle goes through four broad phases:

- *Concept Exploration*, the initial study phase, which results in a broad definition of the space mission and its components, cost, and schedule.
- *Detailed Development*, the formal design phase, which results in a detailed definition of the system components and, in major programs, technology development and validation of test hardware and software.
- *Production and Deployment*, the construction of the ground and flight hardware and software and launch of the first satellite or constellation.
- *Operations and Support*, the day-to-day operation of the space system, its maintenance and support, and finally its deorbit at the end of the mission.

These four phases are broken down into a finer level of detail in Table 3-3 along with a definition of what typically defines the end of one phase and the beginning of the next and the typical duration for both small programs and major programs.

Essentially all programs go through the eight phases defined in Table 3-3, but the amount of time spent varies dramatically. For a small program, the time frames are measured in weeks or months and the overall mission will typically take from 1 to 3 years from a broad mission statement to launch. The design life for these small missions will typically be 1 to 5 years, although many of them have lived much longer because well-designed, simple spacecraft have little wear-out once they are on orbit.

Major programs operate on a very different time scale. Here each phase can last for years and the program can go for decades from concept inception to launch. For example, The Chandra X-ray telescope,* one of the Great Observatory missions (see Table 2-3), was initially proposed to NASA in 1976 by scientists at the Smithsonian

Astrophysical Observatory. It was finally launched in 1999 and is still operational as of 2010. Largely because of the high cost, major programs will typically have an on-orbit design life of 10 to 15 years in order to get as much utility as possible.

In most cases, the timeline for both small and large programs is dominated not by technology or the difficulty of achieving the results, but rather by financing and the ability to raise the money for the program. Chandra was delayed in large part due to the high cost of the Hubble Space Telescope. There simply wasn't enough room in the NASA budget for two major programs proceeding simultaneously at full speed. Small programs, by definition, are much less expensive than larger ones, but are typically financed by organizations or governments that have much smaller overall budgets and, therefore, are still major expenses for the group that is funding them. An extended timeline also opens the potential for political problems. Each new administration will review ongoing, expensive programs and may not be strongly supportive of programs begun under a previous administration and unlikely to generate results before the next administration.

Typically, the major space organizations, such as NASA, the Air Force, or ESA, have their own, very formal process and sequence for major programs. The NASA project life cycle is shown in Fig. 3-5 and the systems engineering to be done in each phase is shown in Fig. 3-6. The upper row of triangles in Fig. 3-5 represents the key decision points which are basically the hurdles to be overcome in order to proceed to the next step, with the potential for both schedule delays and funding adjustments (typically upward) at each. This by no means represents all of the formal reviews for a major program. The triangle labeled PDR is the Preliminary Design Review for the program as a whole, with a specific set of criteria to be met before getting through Key Decision Point C. (These are the things listed under the box called "virtual prototyping" in Fig. 3-6.) However, in addition to the PDR for the system as a whole, there is likely to be a PDR for each of the subsystems of the spacecraft and the ground segment and there may be a similar set for the launch segment. Most of the major space organizations have a process fundamentally similar to the NASA process outlined in Fig. 3-5. (See, for example, DAU [2010] or ESA BSSC [1991].)

Recall that our original goal in this process was to satisfy the needs of the customers, operator, end user, and ultimately the American people at low cost and in a timely manner. Today, American space programs take a minimum of 1 year to put in place and, quite often, many years. In the current paradigm, there are very few things faster than a spacecraft in orbit and very few things slower than the process of getting it there.

There are a number of alternatives that can dramatically speed up the space mission implementation process in the future.* The most straightforward of these is to build both spacecraft and launch vehicles to inventory and launch them from inventory within a few hours or days of an identified need. ICBMs are rockets that can be launched within minutes, although the cost of the system is very high. We would like to do the same thing for spacecraft at a much lower cost. The former Soviet Union, and now Russia, have had launch-on-demand from inventory for both spacecraft and launch vehicles for many years. (See Sec. 25.5.) For example, on Friday, April 2, 1982, war broke out between Argentina and the United Kingdom over the Falklands Islands in the South Atlantic east of Argentina, a region of the world where there was very limited satellite coverage. The war ended on June 14, 1982, when Argentina surrendered. During this 73-day period, the Soviets launched 29 payloads into orbit, most in direct response to the war.

The technology for launch-on-demand (i.e., within a day) has been available for many years, but requires that the cost come down to the point where it is economically appropriate to keep assets in inventory. The idea of low-cost, rapid missions led to the creation of the Responsive Space Conference in 2003 by the LA Section of the AIAA [Wertz and Kulpa, 2003], which has been held annually since that time. The technical papers from all of the Responsive Space Conferences are available online at no cost [Responsive Space Conference, 2010]. The push for responsive space has led to the creation by the Air Force of the Operationally Responsive Space Office (ORSO) in 2006 with the objective of providing a rapid response to urgent needs of Air Force commanders [Wertz, Van Allen, and Shelner, 2006; Doyne et al., 2007]. The ORSO has created the idea of a "6-day spacecraft" that could be assembled and tested in 6 days from components and subsystems that are stored at a single location.

Irrespective of whether we choose to build a spacecraft rapidly, or build it more slowly to inventory and then launch when needed, the basic idea remains the same—to satisfy the needs of the user community within a few hours or days of the need being identified. More traditional missions will continue to take a very long time to implement, as is appropriate for major developments in new technology. What we are trying to achieve in Responsive Space is an excellent product, at a low cost, in a timely manner.

3.5 Preliminary Estimate of Mission Needs, Requirements, and Constraints (Step 4)

Having defined the broad objectives that the space mission is to achieve, and who is trying to achieve them on what schedule, we wish to transform these objectives into preliminary sets of numerical requirements and

* Chandra went through the first 22 years of concept exploration as the Advanced X-Ray Astrophysics Facility (AXAF) and received its current name in 1998, a year before launch on board the Space Shuttle Columbia.

* For a detailed list of alternatives, see Chap. 13 or Wertz et al. [2011].

CDR Critical Design Review
CERR Critical Events Readiness Review
DR Decommisioning Review
FRR Flight Readiness Review
KDP Key Decision Point
MCR Mission Concept Review
MDR Mission Definition Review
ORR Operational Readiness Review
PDR Preliminary Design Review
PFAR Post-Flight Assessment Review
PIR Program Implementation Review
PLAR Post-Launch Assessment Review
PRR Production Readiness Review
P/SDR Program/System Definition Review
P/SRR Program/System Requirements Review
PSR Program Status Review
SAR System Acceptance Review
SDR System Definition Review
SIR System Integration Review
SRR System Requirements Review
TRR Test Readiness Review

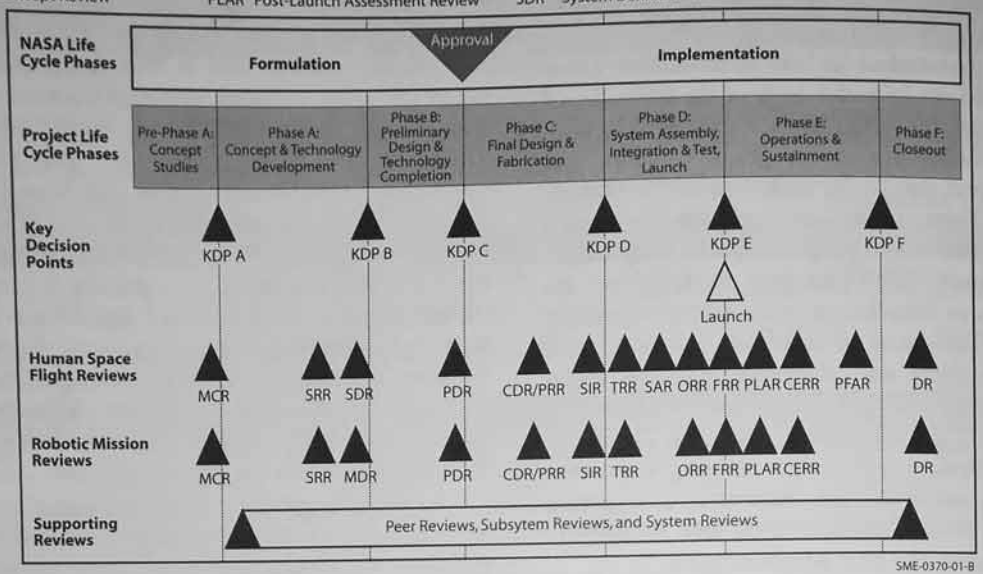


Fig. 3-5. NASA Project Life Cycle. Each of the triangles represents a major milestone. Courtesy of NASA [2007].

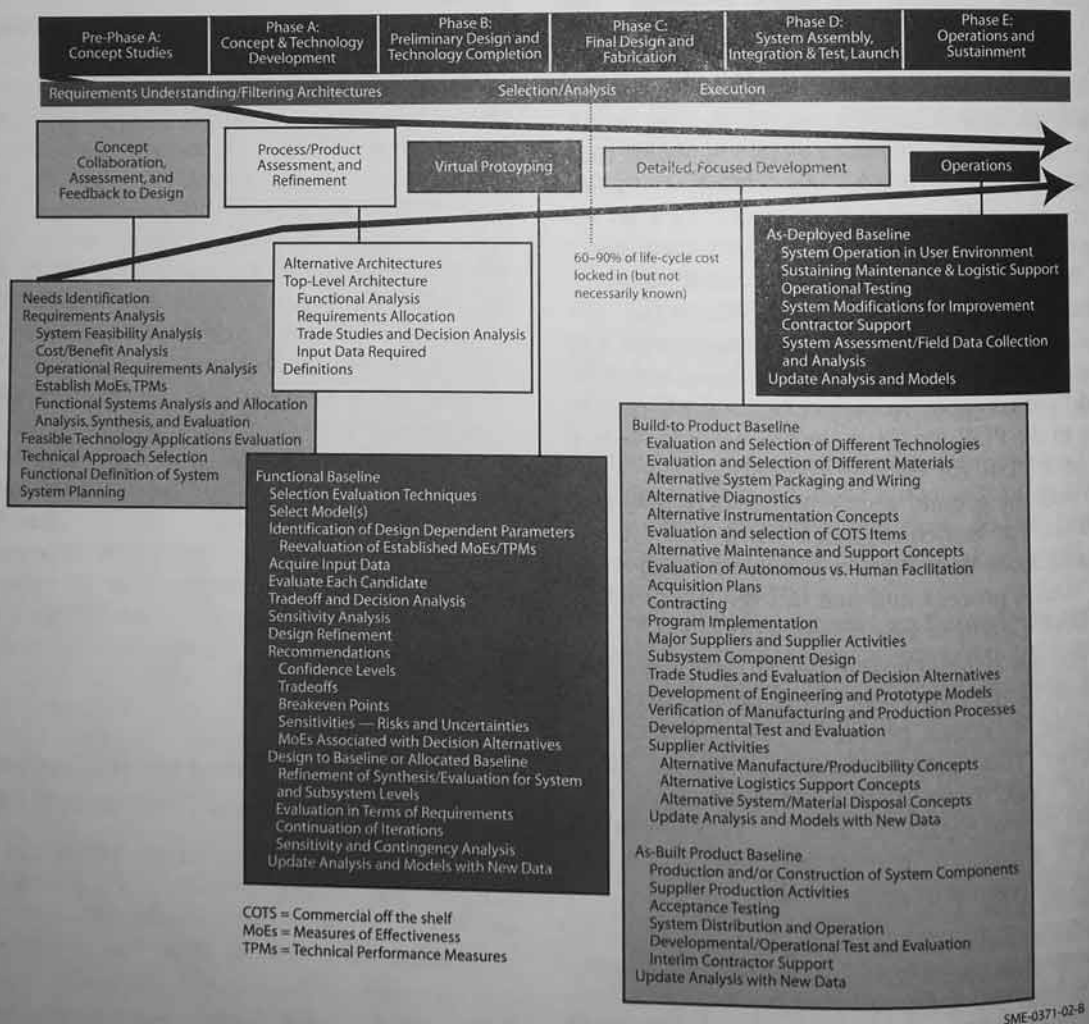


Fig. 3-6. System Engineering Activities in Each Phase of a Major NASA Mission. Courtesy of NASA [2007].

Table 3-3◀, Fig. 3-6, Eq. 3-0

constraints on the space mission's development, performance, and operation. These requirements and constraints will largely establish the operational concepts that will meet our objectives. Thus, we must develop requirements which truly reflect the mission objectives and be willing to change them as we more clearly define the space system.

To transform mission objectives into requirements,* we must look at three broad areas:

- *Functional Requirements*, which define how well the system must perform to meet its objectives.
- *Operational Requirements,* which determine how the system operates and how users interact with it to meet their specific needs.
- *Constraints,* which limit cost, schedule, and implementation techniques available to the system designer.

The needs, requirements, and constraints for any specific mission will depend upon the mission itself and how we implement it. For example, the mission may be a commercial venture, a government scientific program, or a crash emergency program responding to a dire need. Still, most space missions develop their initial requirements in areas similar to those shown in Table 3-4.

Establishing top-level mission requirements is extremely difficult, depending on mission needs and on the perceived complexity or cost of meeting them. Therefore, contrary to frequent practice, we should iterate the numerical requirements many times in the design process. The first estimate of mission requirements typically comes from the goals and objectives combined with some view of what is feasible. Often, we can reiterate or slightly modify requirements and specifications from previous missions, thus carrying over information known from those missions. Of course, we must be prepared to trade these requirements as we develop the mission concept, thereby avoiding the problem of keeping old and inappropriate requirements.

The next step in setting up preliminary mission requirements is to look for the "hidden agenda" discussed in Sec. 3.3. This agenda contains the implicit goals and constraints of all of the principal players identified in Sec. 3.4. For example, the FireSat II mission may need to be perceived as responding quickly to a public demand for action and SCS may need to be built entirely in the US to be acceptable to Congress. Thus, an extended R&D program to develop the most appropriate FireSat II satellite may not be acceptable and using a communications payload from Europe on SCS would not let the program go forward without a fight, even though the European payload might be better, lower cost, or both.

As discussed further in Sec. 12.3, we must recognize that developing a space mission depends on political, legal, and economic constraints, as well as technology. Thus, the most appropriate solution must meet mission technical requirements **and** the political and economic goals of everyone involved. For example, satellite systems for a small nation may use components built in that nation or develop some new components locally, even though they would cost less if bought in other countries. In this case, we would spend more money to meet a political constraint: using the space mission to develop and promote national engineering resources. The technical community often sets aside nontechnical considerations and regards them as less important or less real than technical constraints. But a successful mission design must include **all** requirements and constraints placed on the system, even if they are not necessarily written in a specifications document.

Finally, we reiterate that preliminary mission requirements should be subject to later trades. Mission designers often simply try to meet the procuring group's requirements and constraints, because not meeting them appears to put them at a strong competitive disadvantage. Consequently, designers may not modify the initial requirements, even if changes could make the system cost less or perform better for a given cost. Chapter 13 provides the details of this process of *trading on requirements* to maximize performance vs. cost.

As an example, we consider the requirement for mission design life.† This exemplifies the difficulty of establishing requirements. The length of the mission is often indefinite, as is the case with both FireSat II and SCS. We want to identify and monitor forest fires and provide communications for natural disasters continuously at a reasonable cost per year. In practice, however, we must develop a system that meets this need and then deploy it with an established design life and, perhaps, a replenishment philosophy. The design life of the individual FireSat II or SCS spacecraft will strongly affect cost and will determine the level of redundancy, propellant budgets, and other key system parameters. In principle, we would like to obtain a graph of spacecraft cost vs. design life as shown in Fig 3-7. We could then easily compute the total expected cost per year for different design lives, as shown by the heavy dark line, and the design life that results in the minimum spacecraft cost per

* The alternative to starting with the question of what the system must do to meet the mission needs is the Capabilities-Driven Process, defined in Sec. 3.1.2. In this case, we would start our requirements definition by writing down what the system can do and asking whether that has utility. For example, our primary requirement might be to achieve 1 m resolution on the Earth, simply because a new breakthrough allows this to be done at very, very low cost. We then ask what the missions are that this new capability would support and what are the secondary functional requirements, operational requirements, and constraints that would support the mission or missions using this new capability.

† The *design life* is the intended minimum operational lifetime on orbit. Most spacecraft will last much longer. The design life is used for things such as propellant budgets, redundancy requirements, solar array sizing (because of degradation due to radiation), and any component with a predicable wear-out.

Table 3-4. Representative Initial Requirements. These should be adjusted over time to balance what is desired with what is affordable.

Requirement	Where Discussed	Factors which Normally Impact the Requirement	FireSat II Example	SCS Example
Functional Requirements				
Performance	Chaps. 15, 16, 17	User needs, payload aperture, orbit	Work through light clouds 50 m resolution 1 km geolocation accuracy	300 simultaneous users Continuous link to CONUS 65 kbps bandwidth per user
Coverage	Secs.10.2, 10.6	Orbit altitude and inclination, swath width, number of satellites, scheduling process	Coverage of specified forest areas within the US at least twice daily	Primary: ± 50 deg lat. band Polar: All areas outside of ±40 deg latitude band
Interpretation	Sec. 5.4	Cloud cover, image quality, manual or automated interpretation	Identify an emerging forest fire within 8 hours; with less than 10% false positives	Not applicable
Timeliness	Sec. 10.6	Interpretation, communications, processing, operations	Interpreted data to end user within 5 min	Less than 150 ms total lag
Secondary Missions	Secs. 3.1, 3.3	As above	Monitor changes in mean forest temp. to ±2 C	Automatic geolocation data
Operational Requirements				
Commanding	Sec. 21.1	Who will do commanding, tasking from the field, need for real-time schedule changes	Commandable within 3 min of events; download units of stored coverage areas	Entire system commanded by primary or back-up ground stations in CONUS
Mission Design Life	Secs. 3.4, 3.5, 14.2	Duration of need, level of redundancy, altitude	8 years	10 years
System Availability	Sec. 24.1	Level of redundancy, where processing and interpretation occur	95% excluding weather, 24 hour maximum downtime	99% availability, irrespective of weather
Survivability	Chap. 7	Orbit, hardening	Natural environment only; not in radiation belts	10 year survival in the Van Allen belts
Data Distribution	Chaps. 16, 29; Sec. 21.1	User needs, communications architecture; ancillary communications channels	Up to 500 fire-monitoring offices + 2,000 rangers worldwide (max. of 100 simultaneous users)	Can send to any user worldwide on the system and to any user with comm. connectivity via operations
Data Content, Form, and Format	Chaps. 4, 14, 20	User equipment, available bandwidth, level and place of processing	Location and extent in lat/long for local plotting, avg. temp. for each 40 m^2	Voice, text, images
User Equipment	Chap. 16	Mass, size, and power; existing equipment or new, user interface	10 × 20 cm display with zoom and touch controls; built-in GPS quality map	Hand-held communications transceiver less than 500 g or add-on to cell phone
Constraints				
Cost	Chaps. 11, 12, 13	Manned flight, number of space-craft, size and complexity, orbit	Non-recurring < $10M Recurring < $3M/year	Non-recurring < $30M Recurring < $10M/year
Schedule	Sec. 3.4	Financing, technical readiness	Operational within 3 years	Operational within 4 years
Risk	Sec. 24.2	Primary and secondary customers, schedule, cost	Probability of success > 90%	Probability of success > 95%
Regulations	Sec. 12.3	Law and policy	Orbital debris, civil program regulations	DoD and civil program regulations
Political	Sec. 12.3	Sponsor, whether international program, debris removal	Responsive to public demand for action	Public demand for better disaster response
Environment	Chap. 7	Orbit, lifetime	Natural	Natural, jamming, weather
Interfaces	Chaps. 28, 29	Level of user and operator infrastructure	Interoperable through NOAA ground stations	Simple to use
Development Constraints	Chap. 3	Primary and secondary customers, schedule, cost	None	Must use all American components

year. For the case of Fig. 3-7, we would choose a design life of 6 years. We could also assess technological obsolescence, or the point at which we wish to replace the spacecraft because of better or cheaper technology.

In practice, figures such as 3-7 are almost never created or, at best, are done qualitatively. The mission design life is normally assigned rather arbitrarily with a general perception of cost per year. Thus, there may be a push to produce spacecraft lasting 5 or 10 years because people believe that these will be more economical than ones lasting only a few years. No matter how we choose the design life requirement, we would like to go through the process described above for decisions about mission lifetime. If at all possible, it would be desirable to create a graph similar to Fig. 3-7 based on even crude estimates of spacecraft cost. Doing so provides a much stronger basis for establishing mission requirements and, e.g., determining whether we should push harder for a longer spacecraft lifetime or back off on this requirement to reduce spacecraft cost.

Having made a preliminary estimate of mission requirements and constraints, we proceed in Chap. 4 to define and characterize one or more baseline mission concepts. The issue of refining requirements, revising our baseline, and assessing how well objectives can be met is discussed in Chap. 5. However, we always want to return to our basic goals:

- Meeting the mission objectives in a timely manner at minimum cost and risk, and
- Making all of the principal players delighted with the resulting space system by doing a better job, quicker, and at lower cost than they had expected

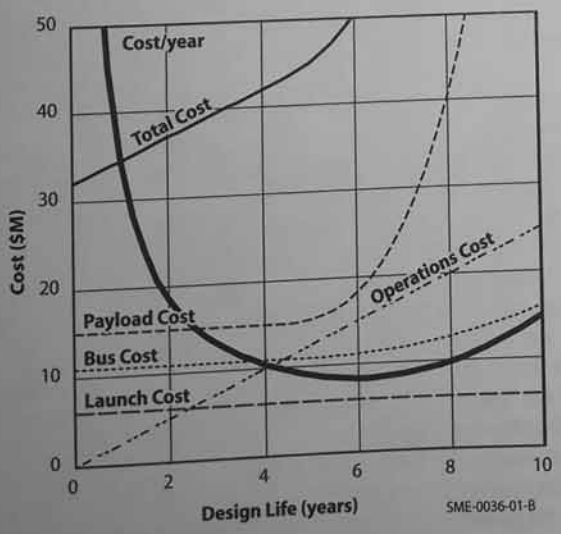


Fig. 3-7. Hypothetical Curve of Cost vs. Spacecraft Design Life. The cost per year is the total cost divided by the design life. In principle, we should use such curves to set the Spacecraft Design Life requirement. In practice, they rarely exist.

References

For annotated bibliography of space mission engineering, see the book website.

Boden, D. G., Gael Squibb, and Wiley G. Larson. 2006. *Cost Effective Space Mission Operations. 2nd ed.* Boston, MA: The McGraw-Hill Companies.

Chesley, J., W. J. Larson, M. McQuade, and R. J. Menrad. 2008. *Applied Project Management for Space Systems*. New York: The McGraw-Hill Companies.

Defense Acquisition University. 2010. "Integrated Defense Acquisition, Technology, and Logistics Life Cycle Management System."

Doyne, T., P. Wegner, C. Olmedo, G. Moretti, M. Hurley, M. Johnson, T. Duffey, and C.Huffine. 2007. "ORS and TacSat Activities Including the Emerging ORS Enterprise." 5th Responsive Space Conference. RS5-2007-4001. Los Angeles, CA. Apr. 23–26.

ESSC. 1991. ESA Board for Software Standardisation and Control. *ESA Software Engineering Standards*.

Fleeter, Rick. 2000. *The Logic of Microspace*. Torrance, CA: Microcosm Press and Springer.

Ford, Steve. 2010. *The ARRL Satellite Handbook*. Newington, CT: American Radio Relay League.

Johnson, S. B. 2002. *The Secret of Apollo: Systems Management in American and European Space Programs*. Baltimore, MD: The Johns Hopkins University Press.

Kay, W. D. 1995. *Can Democracies Fly in Space?: the Challenge of Revitalizing the U.S. Space Program*. Westport, CT: Praeger.

Larson, W. J., D. Kirkpatrick, J. J. Sellers, L. D. Thomas, D. Verma. 2009. *Applied Space Systems Engineering*. New York: The McGraw-Hill Companies.

Maier, M. W. and E. Rechtin. 2009. *The Art of Systems Architecting. 3rd ed.* New York: CRC Press.

NASA. 2007. *Systems Engineering Handbook*. Hanover, MD: NASA.

Responsive Space. 2010. Website.

Sandau, T. 2006. *International Study on Cost-Effective Earth Observation Missions*. Leiden, The Netherlands: Taylor & Francis.

Sarsfield, L. P. 1998. *The Cosmos on a Shoestring: Small Spacecraft for Space and Earth Science*. Santa Monica, CA: RAND.

Shishko, Robert. 2010. *NASA Systems Engineering Handbook*. Memphis, TN: General Books.

Wertz, James R. and Wiley J. Larson. 1991 *Space Mission Analysis and Design. 1st ed.* Dordrecht, The Netherlands: Kluwer Academic Publishers and Microcosm Press.

Wertz, J. R. and W. J. Larson. 1996. *Reducing Space Mission Cost.* Torrance, CA and Dordrecht, The Netherlands: Microcosm Press and Kluwer Academic Publishers.

Wertz, J. R. and W. J. Larson. 1999. *Space Mission Analysis and Design. 3rd ed.* Hawthorne, CA and New York: Microcosm Press and Springer.

Wertz, J. R. and J. Kulpa. 2003. "Responsive Launch with the Scorpius Family of Low-Cost Expendable Launch Vehicles." 1st Responsive Space Conference. RSI-2003-5001 Redondo Beach, CA. Apr. 1–3

Wertz, J. R., R. Van Allen, and C. J. Shelner. 2006. "Aggressive Surveillance as a Key Application Area for Responsive Space." 4th Responsive Space Conference. RS4-2006-1004 Los Angeles, CA. Apr. 24–27.

Wertz, J. R., et al. 2011. "Methods for Achieving Dramatic Reductions in Space Mission Cost." Reinventing Space Conference. AIAA-RS-2011-500. Los Angeles, CA. May 2–5.

4 Mission Concept Definition and Exploration

James R. Wertz, *Microcosm, Inc./USC*

4.1 Defining Alternative Mission Architectures (Step 5)—Choosing the Pieces

4.2 Defining Alternative Mission Concepts (Step 6)—How the Pieces Work Together
Data Delivery; Tasking, Scheduling, and Control; Mission Timeline

4.3 Introduction to Concept Exploration

4.4 Defining System Drivers and Critical Requirements (Step 7)

In Chap. 3, we defined the broad mission objectives, the principal players in the development of the mission, the program timescale that we have to develop it on, and a preliminary list of requirements and constraints. In this chapter, we begin the process of defining alternative mission concepts that will achieve these objectives. Of course, our goal is to look at a number of these concepts to find out how we can meet our mission objectives in a timely manner at minimum cost and risk, such that the various players in the process will be delighted and amazed at the end product we have produced. Sadly, in modern space missions, that does not happen as often as we would like, and it is up to the new generation of space mission engineers to change that.

Most of the references in the introduction to Chap. 3 also discuss the process of concept definition and exploration. In addition, Fortescue, et al. [2003], Griffin and French [2004], and Pisacane and Moore [2005] provide a broad discussion for general missions. Elbert [2003, 2008], Gordon and Morgan [1993], and Maral and Bousquet [2010] provide excellent discussions for geosynchronous communications missions. Kemble [2006] discusses concept exploration for interplanetary missions. Chapter 13 discusses the major differences for low-cost missions.

In common use, "mission architecture," "mission concept," and "concept of operations" are frequently interchangeable and, at best, vaguely defined. Throughout this chapter, we wish to clearly distinguish between them. The *mission architecture*, introduced in Sec. 4.1, consists of the definition of each of the eight major components of the mission, i.e., the elements that make up the mission. The *mission concept*, discussed in Sec. 4.2, is a broad statement of how the mission will work in practice—how all the pieces fit together to meet the needs of all of the players involved. This should not be confused with *mission operations*, which provides the details of how people operate and control the mission.

Another way to think about the overall architecture is by an analogy with a swift water rescue team being set up by the local fire department. In this case, the Mission Architecture is all of the equipment and people that we need to make it happen—the boat, the fire truck that can carry the boat, the roads we can use, the radio equipment to keep in touch with home base, and the boat launching locations that we will use to get into the water. The Mission Concept is broadly how we will use these resources to rescue people. We'll use the truck to get as close to the people needing help as possible. We'll launch the boat, but stay in close contact on the radio and be searching for other options if the boat can't get to where it needs to be. We'll get the people that need help into the boat and assess whether we need the helicopter to get them to a hospital. Finally, the Mission Operations is the detailed process of getting the boat through the swift water, to the people that need help, and getting them the help they need. The Concept Definition and Exploration Process is the planning and thinking very carefully about all of this in advance (and the simulations and dry runs) to make sure that when the call for help arrives, we're genuinely ready to go. Similarly with FireSat II—when forest fires occur, we want to provide the right data to the right people in time to help them do whatever needs to be done.

4.1 Defining Alternative Mission Architectures (Step 5)—Choosing the Pieces

All space missions consist of a set of 8 *elements* or *components* as shown in Fig. 4-1. The collection of these elements form a *space mission architecture*. The elements themselves are defined in Table 4-1 and discussed below.

The *subject* of the mission is the thing that is sensed by or interacts with the space payload. There are two broad types of subjects. *Passive subjects* are ones that the mission designer can't control, such as the fire for FireSat II; moisture content, atmospheric temperature, or pressure for weather missions; types of vegetation, water, or geological formations for Earth-sensing missions; or a rocket or intercontinental ballistic missile for space defense missions. Although mission designers can't control the subject itself, they can control how to sense the subject. For FireSat II, it is likely that we would sense the heat from the fire. But we could also sense the light, the smoke, the increased CO_2 level, the reduction in oxygen, or some combination of all of those. (Home fire detection systems often use both smoke detection and CO_2 detection, for example.)

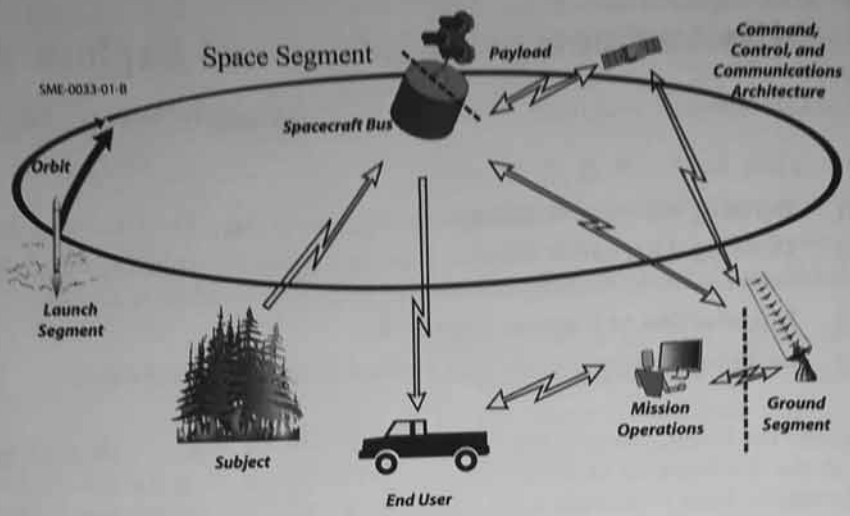


Fig. 4-1. Space Mission Architecture. All space missions include these basic elements to some degree. See Table 4-1 for definitions and the text for discussion. Requirements for the system flow from the operator, end user, and developer and are normally allocated to the various mission elements.

Table 4-1. The Space Mission Architecture Consists of Eight Elements or Components Plus the Mission Concept and End User. All of these must work together to meet the mission objectives and needs of the end user at a reasonable cost, risk, and schedule.

Segment	Description	Where Discussed
1. Subject	What the spacecraft observes (***passive subject***) or interacts with (***active*** or ***controllable subject***).	Sec. 15.2
Space Segment or Spacecraft Consisting of:		
2. Payload	Spacecraft hardware and software that observe or interact with the subject.	Chaps. 15, 16, 17
3. Spacecraft Bus	The other spacecraft subsystems needed to support the payload.	Chap. 14
4. Ground Segment	The communications equipment and facilities that communicate with and control the spacecraft.	Chap. 28
5. Mission Operations	The people and software that run the space mission on a day-to-day basis.	Chaps. 29, 30
6. Command, Control, and Communications Architecture	How all of the parts of the space mission communicate with each other.	Secs. 16.1, 21.1
7. Orbit	The path of the spacecraft during its operational mission. If there is more than one spacecraft in coordinated orbits, it's a ***constellation***.	Chaps. 9, 10
8. Launch Segment	How the spacecraft gets into orbit; may include ***upper stages*** or ***integral propulsion***.	Chaps. 26, 27
Mission Concept	The ***end users*** are the people or equipment that actually make use of the data generated or transmitted by the spacecraft. The ***mission concept*** is the definition of how the mission elements work together to meet the needs of the end user.	Secs. 1.1, 3.4, 4.3

In contrast to the passive subjects, *active* or *controllable subjects* are ones for which the design and basic parameters can be controlled by mission designers. These include GPS receivers, direct broadcast television dishes, and the user equipment for SCS. Here the mission designer has the opportunity to trade between putting more capability at the user end or more capability on the spacecraft. Since there are normally lots of users and few spacecraft, it would seem to make sense to put as much capability as possible on board the spacecraft and allow the user equipment to be simple and low cost. But memory and logic are remarkably cheap these days, so, for example, all of the map data on which your GPS coordinates are displayed are in the receiver that is with the end user, not sent down on the message from the GPS satellite.

The *payload* consists of the hardware and software that sense or interact with the subject. Typically, we trade off and combine several sensors and experiments to form the payload, which largely determines the mission's cost, complexity, and effectiveness. In most cases, multiple payloads significantly increase cost.

The *spacecraft bus* consists of a set of subsystems, defined in Chap. 14, that support the payload by providing orbit and attitude maintenance, power, command, telemetry and data handling, structure and rigidity, and

temperature control. The payload and spacecraft bus together are called the *spacecraft*, *space segment*, or *launch vehicle payload*.

The *launch system* includes the launch facility, launch vehicle, and any upper stage required to place the spacecraft in orbit, as well as interfaces, payload fairing, and associated ground support equipment and facilities. The selected launch system constrains the size, shape, and mass of the spacecraft.

The *orbit* is the spacecraft's path or trajectory in space. Typically, there is a separate initial parking orbit, transfer orbit, and final mission orbit. There may also be an end-of-life or disposal orbit. The mission orbit significantly influences every element of the mission and provides many options for trades in mission architecture.

The *command, control, and communications (C3) architecture* is the arrangement of components that satisfy the mission's communication, command, and control requirements. It depends strongly on the amount of data and timing requirements for the data to be transferred, as well as the number, location, availability, and communicating ability of the space and ground assets.

The *ground system* consists of the facilities and communications equipment associated with both fixed and mobile ground stations around the world, connected by various data links. They allow us to command and track the spacecraft, receive and process telemetry and mission data, and distribute the information to the operators and end users.

Mission operations consist of the people, computers, and software that execute the mission operations concept and attendant policies, procedures, and data flows.

Finally, the *mission concept* is how all of these elements work together to satisfy the needs of the key players in the system. For FireSat II, how do firefighters in the truck get the information that they need to do their job? Most likely a simple image from space isn't going to do them much good. What they need is the interpreted results—what's a forest fire? Where is the fire and which way is it going? We need to put that information on a map and deliver it to firefighters in their truck in near-real-time in order to be useful. For SCS, our basic goal is to let disaster relief workers in a remote location communicate with the people that can provide the help and guidance they need to reduce suffering and save lives and property. These are the things that our mission concept needs to tell us.

Our goal is to arrive at a set of candidate architectures for further evaluation large enough to encompass all approaches offering significant advantages, but small enough to make the more detailed definition and evaluation manageable. Table 4-2 summarizes the mechanism for doing this.

Step A. Identify mission elements subject to trade. We begin by examining our basic mission concept and each of the eight mission elements in light of the preliminary requirements and constraints from Sec. 3.5 to determine which have more than one option. Usually this step greatly reduces the number of tradeable elements.

Table 4-2. Process Summary for Identifying Alternative Mission Architectures. This highly creative endeavor can have a significant impact on mission cost and complexity.

Step	Where Discussed
A. Identify the mission elements subject to trade.	Table 4-3
B. Identify the main options for each tradeable element.	Table 4-4
C. Look for options that work well together or don't work well together.	Table 4-5
D. Look for other alternatives that could substantially influence how we do the mission.	Table 4-15, Chap. 13

Table 4-3 summarizes this process for FireSat II and SCS. Both missions have multiple options that will affect not only cost, but also performance, flexibility, and long-term mission utility. Thus, we should carry through several different options so the decision-making audience can understand the main alternatives.

Table 4-3 lists one of the options as "cost only," meaning that the trade depends mainly on cost and only secondarily on how or how well the mission is accomplished. An example would be the launch system, for which the main concern normally is what launch vehicle will get the spacecraft into orbit at the lowest cost. Still, these trades may be important in selecting the mission concept. For example, a major increase in the launch cost may outweigh being able to use a smaller number of satellites in a higher orbit.

Step B. Identify the main options for each tradeable element. Although in theory we have almost an unlimited number of options, we normally draw them from a limited set such as those listed in Table 4-4. Thus, we first choose options that apply to our mission and then look for special circumstances which may lead us to consider alternatives not listed in the table.

Step C. Look for options that work well together or don't work well together. We don't want to carry all possible options forward, because we simply won't have the resources to examine them in detail. We need to find an intelligent way to limit our choices somewhat. The first step in reducing the number of options is to identify the system drivers, as discussed in Sec. 4.4. The *system drivers* are parameters or characteristics that largely determine the system's cost and performance. We look at these options first because they normally dominate the design process and mandate our choices for other elements, thus greatly reducing our options.

Table 4-5 shows two major options for both FireSat II and SCS. Here the major options or system drivers are shown in bold. For FireSat II, there are two major drivers—whether the orbit is LEO or GEO and how to determine what a forest fire is. These are trades that we will have to carry with us for a long time and determine how they impact the overall cost, performance, risk, and schedule. These will determine many of our choices in

Table 4-3. Selecting FireSat II and SCS Elements that can be Traded. Not all options are compatible with each other and there may be options we haven't thought of yet. See text for discussion.

	FireSat II		SCS	
Element of Mission Architecture	Can Be Traded	Reason	Can Be Traded	Reason
Mission Concept	Yes	Want to remain open to alternative approaches	No	Mission objective is to be a communications system
Subject	No	Passive subject is well defined (could detect something besides heat, but unlikely)	Yes	Active subject—can consider how much we want the receiver to do
Payload	Yes	Can select complexity and wavelength	Yes	Power and frequency are the key issues
Spacecraft Bus	Yes	Multiple options based on scan mechanism and power	Yes	Multiple options based largely on launch capacity
Launch System	Cost only	Choose minimum cost for selected orbit	Yes	Find low-cost launch and use that to drive the spacecraft
Orbit	Yes	Options are low, medium, or high altitude with varying number of satellites	Yes	Classic orbit is elliptical, but high altitude equatorial, circular orbit may be better
Ground System	Yes	Could share NOAA control facility, use dedicated FireSat II facility, or direct downlink to users	Yes	Could have dedicated system or use commercial service-provided system
Communications Architecture	Yes	Depends on where image interpretation is done	No	Largely defined by mission concept
Mission Operations	Yes	Can adjust level of automation	Not much	Mostly automated operations, may have some level of control

other areas. If the spacecraft is in LEO, it can be a small spacecraft, possibly with solar arrays simply mounted on the spacecraft body, and it will be launched by a small launch vehicle. On the other hand, if FireSat II is in GEO, it will have to be a large spacecraft with ample power, 1- or 2-axis solar arrays, and a medium to large launch vehicle. As discussed in Sec. 4.3, our other major trade is, how do we determine what a forest fire is? Is it a manual process, or one that is automated on board? How do we distinguish a forest fire from a really large barbeque?

Similarly, we have two major system drivers for SCS. Again the orbit is key because it determines how large the Earth appears and how far away it is. In a Molniya orbit (Sec. 9.5.4) both the angular size and distance vary greatly over the orbit, which makes our spacecraft and payload design more complicated. Here our other major trade element is the subject. Will it be possible to work with an ordinary cell phone, possibly with some wire antenna attached? Will we have to equip our soldiers and disaster workers with a special phone in advance? What are the conditions under which we can communicate and how much data can we send?

The next step in reducing options is to look for trades that are at least somewhat independent of the overall concept definition or which will be determined by the selection of other elements. For example, the spacecraft bus ordinarily has many options. However, once we have defined the orbit and payload, we can select the spacecraft bus that meets the mission requirements at the lowest cost. Although bus options may not be a recognized part of the trade process, they may play a key role in selecting workable mission concepts because of cost, risk, or schedule.

The third technique for reducing options is to examine the combinations we have available and retain only sensible combinations. For example, nearly any launch vehicle above a minimum size will work for a given orbit and spacecraft. Because cost is the main launch-vehicle trade, we should retain in the option list only the lowest-cost launch vehicle that will fulfill the mission. This does not mean that we will ultimately launch the spacecraft on that vehicle. Instead, we should design the spacecraft to be compatible with as many launch vehicles as possible and then select the vehicle based on cost (which may well have changed) when we are deciding about initial deployment.

Step D. Look for other alternatives that could substantially influence how we do the mission. Finally, we want to look for alternative ways to do the mission. We'll return to this in Chap. 13 as we look at ways to reduce mission cost, but we want to be aware from the very beginning that looking for major alternatives is the principal way to reduce cost, improve performance, and have a customer and end user that are delighted with what we have accomplished. For example, can data from FireSat II be combined with data from another source to provide better, more frequent monitoring than would be possible with a satellite system alone? Can we use FireSat II to track volcanic activity so as to spread the cost over a larger number of users? The key point is that alternatives nearly always exist. We must look carefully for them and be willing to revise normal requirements and constraints to meet our fundamental mission objectives. This is the step where real creativity is important.

Table 4-4. Common Alternatives for Mission Elements. This table provides a broad checklist for identifying the main alternatives for elements of the mission architecture.

Mission Element [Where Discussed]	Option Area	Most Common Options	FireSat II Options	SCS Options
Mission Concept [Sec. 4.2]	Data delivery	Direct downlink to user, automated ground processing, man-in-the-loop processing and transmission	Direct downlink or through mission control	Choice of sending voice, text, images, or video
	Tasking	Ground commanding, autonomous tasking, simple operations (no tasking required)	Simple operation or ground commands	Simple operation or ground commands
Controllable Subject [Sec. 15.2]	Selection	Standard ground stations, private TV receivers, ship or aircraft transceivers, special purpose equipment	N/A [See Sec. 9.3]	Through modified cell phone or special equipment
	Performance	EIRP, G/T (See Sec. 13.3 for definitions)		Make do with whatever system can provide
	Steering	Fixed or tracking		Limited or no pointing
Passive Subject [Sec. 15.2]	What is to be sensed	Subject itself, thermal environment, emitted radiation, contrast with surroundings	Heat or visible light; chemical composition; particles	N/A
Payload [Chaps.15, 16, 17] (some items may not apply, depending on mission type)	Frequency	Communications: normal bands Observations: IR, visible, microwave Radar: L, C, S bands, UHF Single or multiple frequency bands, single or multiple instruments	IR, visible Single or multiple bands	Selection of communications band and output power
	Size vs. sensitivity	Small aperture with high power (or sensitivity) or vice versa	Aperture	Aperture and power
Spacecraft Bus [Chap. 14]	Propulsion	Whether needed; cold gas, monopropellant, bipropellant	Monopropellant, bipropellant	Monopropellant, bipropellant
	Orbit control	Whether needed, onboard vs. ground	Needed, onboard or ground	Needed, onboard or ground
	Navigation	Onboard (GPS or other) vs. ground-based	Probably onboard	On-board or ground
	Attitude determination and control	None, spinning, 3-axis; articulated payload vs. spacecraft pointing; actuators and sensors	3-axis, articulated payload or Earth-fixed	3-axis, Earth fixed
	Power	Solar vs. nuclear or other; body-mounted vs. 1- or 2-axis pointed arrays	Body mounted solar array	Solar array, body mounted or 1-axis
Launch System [Chap. 27]	Launch vehicle	SSLV Atlas, Delta, STS, Titan, Pegasus, Ariane, other foreign	Small launch vehicle, probably has to be US	Small or medium launch vehicle, probably has to be US
	Upper stage	Pam-D, IUS, TOS, Centaur, integral propulsion, other foreign	Not needed	Small upper stage or integral propulsion
	Launch site	Kennedy, Vandenberg, Wallops, Kwajalein, Kourou, other foreign	KSC, Wallops, or Kwajalein	KSC, Wallops, or Kwajalein
Orbit [Chaps. 9, 10]	Special orbits	None, geosynchronous, Sun-synchronous, frozen, Molniya, repeating ground track	Single GEO satellite, low-Earth constellation	Molniya; MEO circular, equatorial or polar
	Altitude	Low-Earth orbit, mid-altitude,geosynchronous	LEO	Mid-altitude or elliptical
	Inclination	0°, 28.5°, 57°, 63.4°, 90°, 98°	55 deg	0 deg or 90 deg
	Constellation configuration	Number of satellites; Walker pattern, other patterns; number of orbit planes	2 to 6 satellites	3 or 4 satellites, could grow
Ground System [Chap. 28]	Existing or dedicated	AFSCN, NASA control center, USN or other shared system, dedicated system	Shared NOAA system, dedicated system	Shared military system, USN, or internet node
Communications Architecture [Chap. 17]	Timeliness	Store and dump, real-time link	Either option	Real time
	Control and data dissemination	Single or multiple ground stations, direct to user, user commanding, commercial links	1 ground station; commercial or direct data transfer	Direct user-to-user
	Relay mechanism	TDRSS, satellite-to-satellite crosslinks, commercial communications relay	TDRS or commercial	Crosslinks or transfer via ground
Mission Operations [Chap. 29]	Automation level	Fully automated ground stations, part-time operations, full-time (24-hr) operations	Any of the options listed	Nearly fully automated
	Autonomy level	Full ground command and control, partial autonomy, full autonomy (not yet readily available)	Any, need some ground control	Any

Table 4-5. Two of the Principal Options for both FireSat II and SCS. While, in principle, we could evaluate all possible combinations, we try to find options that work together to meet mission objectives. The key design choices that determine most of the other elements are in bold.

	FireSat II		SCS	
Element	Option A	Option B	Option A	Option B
Mission Concept	**Automated fire detection on board the spacecraft with direct downlink**	**Manual fire detection at the ground station with results relayed via FireSat II**	Continuous voice, image, text, and slow video between remote user and local or CONUS command	Continuous voice, image, text, and slow video between remote user and local or CONUS command
Subject	Heat from forest fire	Heat from forest fire	**Phone built just for SCS users**	**Existing hand-held cell phone or equivalent**
Payload	Small-aperture IR	Large-aperture IR	Mid-sized comm., must accommodate variable range and angular size of Earth	Mid-size comm., fixed antenna, fixed power
Spacecraft Bus	Small, 3-axis, Earth-pointing	Mid-large size, 3-axis, Earth-pointing	Mid-size, 3-axis	Mid-size, 3-axis, Earth-pointing
Launch System	Small launch vehicle	Large launch vehicle	Mid-sized launch vehicle	Mid-sized launch vehicle
Orbit	**LEO, 2 satellites, 55 deg**	**Geosynchronous, 1 satellite centered over west coast of U.S.**	**3 or 4 satellites in Molniya orbits**	**3 or 4 satellites in mid-altitude, circular equatorial orbits**
Ground System	Single, dedicated ground station	Single, dedicated ground station	Multiple shared to ground station network	Single, dedicated ground station in CONUS
Communications Architecture	TDRS data downlink; commercial links to users	Direct to station; results relayed to users via FireSat II	Direct link to local command center; Use ground stations as relays for data return to CONUS	Direct link to local command center; Fixed crosslinks between all satellites gives redundant link to CONUS
Mission Operations	Continuous during fire season, partial otherwise	Continuous during fire season, partial otherwise	Continuous, nearly fully automated	Continuous, nearly fully automated

4.2 Defining Alternative Mission Concepts (Step 6)—How the Pieces Work Together

The broad *mission concept* is the most fundamental statement of how the mission will work—that is, how it gets its data or carries out the mission to satisfy the end users' needs. The mission concept as we are using it here consists of the four principal elements in Table 4-6. Notice that most of these elements are associated with data or information. Except for manufacturing in space and a small number of other space payloads, most space missions are concerned fundamentally with the generation or flow of information. Thus, FireSat II's mission is to generate and communicate to an end user information about forest fires. SCS and other communications satellites move data and information from one place to another. Weather, surveillance, and navigation satellites are all concerned with generating and communicating information. Thus, data flow is central to most space missions. How will FireSat II determine where a fire is and how big it is? How will the system communicate that information to the firefighter in a truck or plane? Once we answer these broad questions, we begin to understand FireSat II's abilities and limits.

As Table 4-7 shows, defining the mission concept consists of defining the various options that are available and then selecting the most appropriate. Section 4.3 describes how we define options and take a first cut at the broad choices available to us. The process of selecting among them, described in Sec. 5.4, is called *system trades*. Here we are interested in what these trades are and what some of the broader alternatives are to generate and transmit data. The process of defining how to transmit the data between the spacecraft and various users and controllers on the ground is called the *communications architecture* and is discussed in Chap. 16.

The program timeline differs from other elements of the mission concept in Table 4-7. It represents the overall schedule for developing, planning, and carrying out the mission. This defines whether it is a one-time only scientific experiment or a long-term operational activity that will require us to replace and update satellites. In either case, we must decide whether the need for the mission is immediate or long term. Should we give high priority to near-term schedules or allow more extensive planning and technology development for the mission? Of course, much of this has to do with the funding for the mission: whether money is available immediately or will be available only over time as we begin to demonstrate the mission's usefulness.

4.2.1 Data Delivery

Space missions involve two distinct types of data—mission data and housekeeping data. *Mission data* is generated, transmitted, or received by the mission payload. This is the basic information that is central to what the mission is all about. For FireSat II, this data starts out as infrared images on a focal plane and ends up as the latitude, longitude, and basic characteristics of a forest fire transmitted to a firefighter on the ground. The mission

Table 4-6. Elements of the Mission Concept. See Table 4-7 for a list of key trades and where discussed. Note that we discuss communications architecture in Chap. 16.

Element	Definition	FireSat II Example	SCS Example
Data Delivery	How mission and housekeeping data are generated or collected, distributed, and used	How is imagery collected? How are forest fires identified? How are the results transmitted to the firefighter in the field?	How do we get data to and from relief worker to the command post, either local or in CONUS?
Tasking, Scheduling, and Control	How the system decides what to do in the long term and short term	What sensors are active and when is data being transmitted and processed? Which forested areas are receiving attention this month?	Who controls SCS and sets its priorities?
Communications Architecture	How the various components of the system talk to each other	What communications network is used to transmit forest fire data to the users in the field?	How do the satellites talk to each other and to the ground?
Program Timeline	The overall schedule for planning, building, deployment, operations, replacement, and end-of-life	When will the first FireSat II become operational? What is the schedule for satellite replenishment?	When will the first SCS constellation become operational? What is the schedule for constellation augmentation and replenishment?

Table 4-7. Process for Defining the Mission Concept. See Table 4-6 for definitions and examples.

Step	Key Trades	Where Discussed
A. Define data delivery process for Mission and housekeeping data	Space vs. ground processing Level of autonomy Central vs. distributed processing	Sec. 4.2.1
B. Define tasking, scheduling, and control for Mission and housekeeping data for long term and short term	Level of autonomy Central vs. distributed control	Sec. 4.2.2
C. Define communications architecture for Mission and housekeeping data	Data rates bandwidth Timeliness of communications	Chap. 16
D. Define preliminary program timeline for Concept development Production and deployment Operations and end-of-life	Replenishment and end-of-life options Deployment strategy for multiple satellites Level of timeline flexibility	Sec. 4.2.3
E. Iterate and document	N/A	N/A

data has potentially very high data rates associated with it. However, the need for this data may be sporadic. Thus, FireSat II may generate huge quantities of raw data during periods of time that it is passing over the forests, but there is little need for this same level of data when it is over the poles or the oceans. For SCS the mission data is the communications content (voice, images, text, or data) between the soldier or relief worker and the local or CONUS command post. While the communications channel is intended to be continuously available, the usage will change greatly depending on the circumstances at the time.

Ultimately, the processed mission data may go directly to the end users or through ground stations and communication networks associated with mission operations. This will, of course, have a fundamental effect on how the mission works. In the first case, FireSat II would process its imagery and send the forest fire information as it is being observed to the firefighters in the field. In the second case, data would go instead to an operations center, where a computer system or human operators would evaluate it, compare it with previous data, and determine the location and characteristics of a forest fire. Then, the operations center would transmit this information to the firefighters in the field. The result is about the same in both cases, but the system's abilities, limits, characteristics, and costs may be very different. For SCS we are simply interested in moving data from place to place. Here, the ground station has relatively little to do with the data, unless it is serving as a relay from one place to another.

In contrast to the mission data, *housekeeping data* is the information used to support the mission itself—the spacecraft orbit and attitude, the battery temperature and state of charge, and the status and condition of all the spacecraft subsystems. Although FireSat II and SCS are very different missions, the housekeeping data is about the same for both of them, although there may be differences in the details, such as the accuracy required for various elements. Unlike the mission data, which is typically sporadic and may have huge data rates, the housekeeping data is usually continuous and at a low data rate. Continuously monitoring system performance does not require much information transfer by modern standards. In addition, rather than going to the end user, housekeeping data goes to the system monitoring and control activity within mission operations. Although housekeeping and mission data are distinct, we often need housekeeping data to

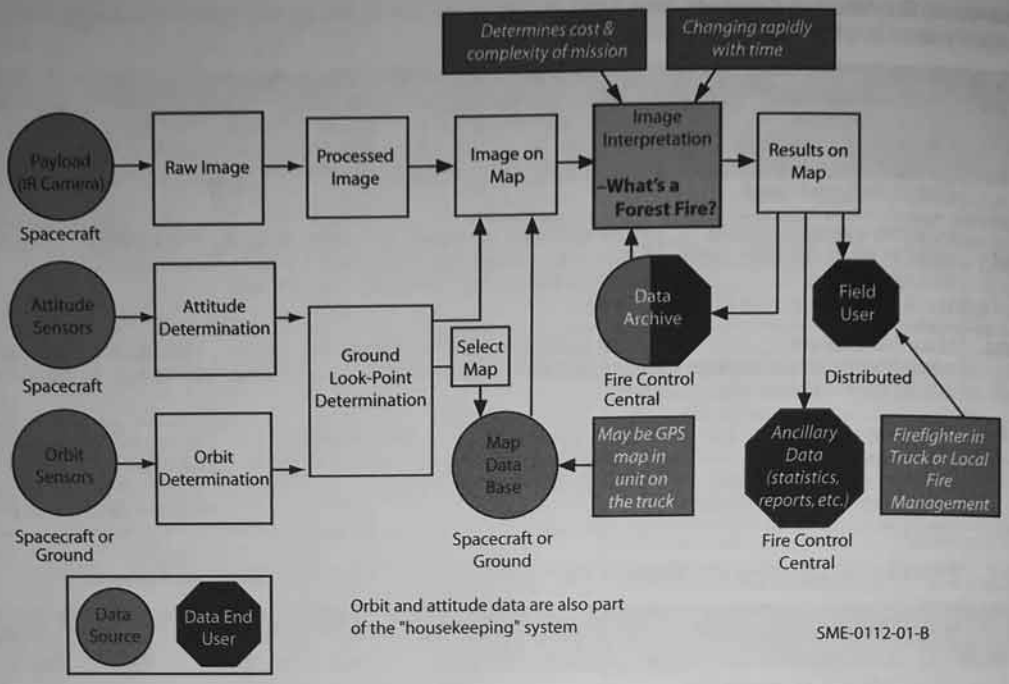


Fig. 4-2. FireSat II Data-Flow Diagram. The purpose of the data flow analysis is to view the space mission from a data-oriented perspective. We want to know where the data comes from, what processing must be done, and where the results are used. Our principal mission objective is to provide the necessary data to the end user in a timely manner at minimum cost and risk.

make the mission data useful. For example, we must know the spacecraft's orbit and attitude to determine the ground lookpoint of the payload sensors and thereby locate the fire. For SCS, we need to know which of the spacecraft can see which end users at any given time.

For both mission and housekeeping data, the data delivery system should be an information management-oriented process. We want to take a large amount of raw data, frequently from a variety of sensors or receivers, and either move that data to another user or efficiently transform it into the information the users need and provide it to them in a timely manner. For FireSat II, we do not know at first whether sending data directly to the field or sending it first to a mission operations center for interpretation and analysis is the best approach. But we do know our choice will dramatically affect how well the mission works and whether or not it is an efficient and effective system.

The principal trades associated with data delivery are:

- **Space vs. Ground**—how much of the data processing occurs on board the spacecraft vs. how much is done at mission operations or by the end user?
- **Central vs. Distributed Processing**—is one computer talking to another computer, or does one large central computer on the spacecraft or on the ground process everything?
- **Level of Autonomy***—how much do people need to intervene in order to provide intelligent analysis and minimize costs?

These trades are both strongly interrelated and are changing dramatically as the amount of processing available at low cost continues to increase. For example, autonomy is important by itself, but is also a key element of the space vs. ground trade. If human intervention is required (i.e., it can't be done autonomously), then the process must be done on the ground—or it must be a very large spacecraft. We will discuss each of these trades below after we have looked at the data delivery process as a whole. Autonomy is discussed in Sec. 4.2.2, because it is also critical to tasking and control.

The best way to start looking at the data-delivery problem is a *data-flow analysis*. This defines where the data originates and what has to happen to it before it gets to where it needs to go. To examine the data flow we use a *data-flow diagram* as shown in Fig. 4-2 for the FireSat II mission. A data-flow diagram lets us outline the tasks that we need to do, even though we don't understand yet how we will do most of them. For FireSat II we know that we need some type of information collection, probably a camera or imager or some other mechanism for detecting forest fires. As shown across the top row of Fig. 4-2 this imaging information must be digitized, probably filtered in some fashion, and its geographic coordinates determined. We must then interpret the image to identify whether a fire exists, put the results on the map, and distribute the map to the end user, i.e., the fire management team in the field or the individual firefighter in a truck out in the forest.

* The language here can be confusing. An *autonomous operation* runs without human intervention. An *autonomous spacecraft* runs without intervention from outside the spacecraft.

Even this very top-level flow suggests one of the major trades—where's the map? It could be at the mission operations center, it could be on the spacecraft, or it could be in the receiver in the truck, as is the case with GPS receivers. If the user's fire-monitoring receiver includes a GPS receiver and the associated maps, then all FireSat II would need to send down would be lat/long coordinates and perhaps some fire data, such as intensity or temperature. This could significantly decrease the bandwidth required to transmit data to the user and, therefore, lead to a lower cost, simpler FireSat II. (Recall from Sec. 3.2 that we created FireSat II to see if more modern technology could lead to a lower cost FireSat II that was more likely to be built than the original FireSat.) We don't know yet what the right answer is, but we want to watch as we go along for alternative ways to do things better, quicker, or at lower cost or risk.

To put the image on a map or determine its geographic coordinates, we need to determine the spacecraft's orbit and attitude. The attitude will almost certainly be determined on board. The orbit may be determined either on board or by observations from the ground. In either case, the orbit and attitude information are combined to determine where on the ground the sensor is looking.

Even though we are not certain yet how the data will be used, we can be fairly sure that our end data from FireSat II will have several applications other than immediate use by the firefighters. We will want to archive it in some central location for record keeping and improving our capacity to analyze and interpret future data. Finally, we will sort out a large amount of ancillary data, such as statistics, reports, and forest-management information, and use it over an extended period. The need for this data does not have the real-time demand of the fire data itself.

The importance of the data-flow diagram is that it lets us see what has to happen in order to make our mission work. For FireSat II, we need to combine the mission sensor with orbit and attitude information in order to make our system work in real time. However, the most difficult step is probably the one labeled "Image Interpretation—**What's a Forest Fire?**" Can an automated system quickly detect forest fires and send information directly to the user, or do we need extensive interpretation and analysis by trained experts in mission operations? What type of experiments or information must we have to determine which of these is best? Even after we have selected an approach, we should revisit it regularly to see that it still makes sense. If we decide that FireSat II's real-time requirements demand data processing in a computer on board the spacecraft, we may dramatically drive up the cost because on-board processing and software is expensive. Our mission analysis may result in an automated FireSat II that costs several times the annual budget of the Forest Service. If so, we need to reconsider whether it would be more economical to have an analyst on the ground interpret the data and then simply phone the results to an appropriate fire station. On the other hand, onboard processing has become dramatically cheaper and getting the data to a mission operations center in Philadelphia may be both complex and expensive if the fire is in California. The data-flow diagram is valuable in helping to identify and track these central issues.

SCS is a very different mission than FireSat II and this is reflected in the SCS data flow diagram in Fig. 4-3. Here, our goal is not to create data, but to get it back and forth continuously and reliably between a command post that can have good communications equipment, and an end user in the middle of a jungle, desert, or remote island hit by a tsunami, who almost certainly doesn't have good equipment and may have very little more than their cell phone. Here, it is likely that whatever equipment the end user has and the communications link that can be set up between them and the satellite will determine how well the job can be done. From our perspective of providing communications, the rest of the problem is relatively easy and we can concentrate most of our effort on what makes the problem the most workable with the equipment that the end user has or can get.

We will now look at two of the three principal trades associated with data delivery: space vs. ground processing and central vs. distributed processing. Section 4.2.2 discusses the level of autonomy.

Space vs. Ground Processing Trades. In most earlier space missions, ground stations processed nearly all of the data because spaceborne processors couldn't do much. Chapter 20 describes several reasons on-board processing lags ground processing. But many on-board processors are now available with greatly increased capacity, and in many cases, on-broad processing can be as capable as ground processing. Consequently, a major trade for future missions is how much to process data on board the spacecraft vs. on the ground, either at a mission-operations facility or with the end user.

Section 5.2 describes how we undertake these and other system trades and compare the results. The main issues in the space vs. ground trade are as follows:

Autonomy—How independent do we want the system to be of analysis and control by a mission operator? If evaluation by people is critical, we will likely do much of the data processing on the ground. If autonomous processing is appropriate, it can be done on board the spacecraft, at a central ground facility, or among the end users, as is the case with a GPS receiver. The level of autonomy is both a key trade in its own right and an element of the space vs. ground trade.

Data Latency—How late can the data get to the end user? If we are allowed only fractions of a second, we must go to automated processes, probably on board the spacecraft. For FireSat II, although we need the data in "near real time," the delays associated with sending the data to the ground for processing are not critical.

Communications Bandwidth—How much data needs to be transmitted? If we have large amounts of data from a sensor, we should process and compress it as near the source as possible. Bringing down all of the FireSat II imaging data and then deciding what to pro-

cess further on the ground will cause an enormous communications problem and will probably drive up the FireSat II mission's cost needlessly. For SCS our problem is likely to be how large a data bandwidth can we achieve with the limited equipment available to the end user.

Single vs. Multiple Users—If there are a large number of end users, as would be the case for FireSat II, we may be able to save considerable money by doing a high level of processing on board the spacecraft and sending the results directly down to the individual users. However, processing in the end-user equipment is becoming very low cost. It may be more economical to simply let the user equipment do a good part of the work, such as plotting the forest data on a map.

Location of End User—Is the "end user" for any particular data element on the ground or in space? In a space-to-space relay or a system for providing automatic orbit maintenance, the end application is in space. In this case, sending data to the ground for processing and then returning the results to the space system is likely to be complex and costly. On the ground, the complexity of the system is strongly affected by whether there is one end user at the mission operations center or multiple, scattered users, as in the case of FireSat II and SCS.

Upgradability—In older space missions one of the reasons for doing processing on the ground was to be able to make changes and upgrades from time-to-time. However, software is the one piece of the spacecraft that we can get at and upgrade after launch. It may be cheaper and easier to upgrade software on board the spacecraft than to try to find and change the software in many ground units for either FireSat II or SCS.

Even if we choose to process data mostly in space, the basic system design should allow us to obtain or recreate selected raw data for analysis on the ground. A fully automated FireSat II should have some means to record or broadcast the raw imaging data, so mission planners and analysts can evaluate how well the system is working, fix problems, and plan alternative and better techniques for later missions.

In the past, space software has been much more expensive than ground software. Therefore, in many cases processing on the ground was generally lower cost than processing on board the spacecraft. However, this is changing rapidly and, therefore, software cost should not be a major trade element in the space vs. ground processing trade. Today, the cost of software is a function of what is done and how reliable we need to make it, rather than where it is done.

The space vs. ground processing trade will be a key issue for many future missions. For short-lived, non-time-critical missions, such as a student payload to measure material properties in 0-g, it will probably be more economical to work on the ground with little automation. For long-lived missions, or time-critical applications, we will have to automate the processing and then do space vs. ground trades to minimize the operations and end-user costs. In any case, we wish to use the data flow analysis to evaluate where the data is coming from and where it will be used. If possible, we would like to minimize the communication requirements and *associate* data (e.g., attach time or position tags) as early as possible after the relevant data has been created.

For FireSat II the payload sensor generates an enormous amount of data, most of which will not be useful. One way to effectively deal with large amounts of raw data on board the spacecraft is to *compress* the data (i.e., reduce the amount of data to be stored or transmitted) prior to transmitting it to the ground. The data is then recreated on the ground using *decompression* algorithms. There are a variety of methods for compressing data, both lossless and lossy. *Lossless data compression* implies that no information is lost due to compression while *lossy compression* has some "acceptable" level of loss. Lossless compression can achieve about a 5 to 1 ratio whereas lossy compression can achieve up to 80 to 1 reduction in data. Many of the methods of data compres-

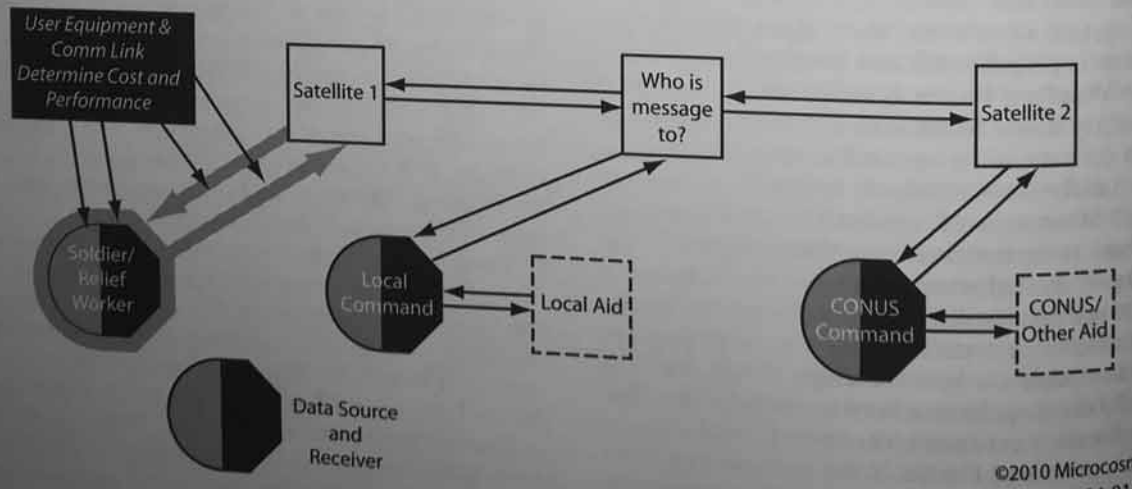


Fig. 4-3. SCS Data-Flow Diagram. For SCS, the problem is establishing reliable two-way communications and our principal problem area is communicating with the end user in the field.

Table 4-7◀, Fig. 4-3, Eq. 4-0

sion store data only when the value changes. Other approaches are based on quantization where a range of values is compressed using mathematical algorithms or fractal mathematics. By using these methods, we can compress the data to a smaller data set that is transmitted to the ground and the image is recreated based on the decompression algorithm. With the use of fractals, we can even interpolate a higher resolution solution than we started with by running the fractal for an extended period of time [Lu, 1997]. We select a method for data compression based on its strengths and weaknesses, the critical nature of the data, and the need to recreate it exactly [Sayood, 2005].

When we transmit housekeeping data we will generally use lossless compression for several reasons. First, raw housekeeping data is not typically voluminous. Second, it is important that none of the data is lost due to compression. However, when we transmit an image we might easily use lossy compression. We could either preview the image using lossy compression or we could say that the recovered image is "good enough." Alternatively, given the advances in on-board storage, we may choose to take very high-resolution imagery and send it to the ground in a low-resolution version. Either the operator or end user could select those images for which they wanted more detail, transmit that request to the spacecraft, and get high-resolution imagery only of those data frames where it was needed. For FireSat II, if the forest fire identification is done on board, we might choose to send to the ground only high-resolution imagery of the area within, say, 0.5 km of an identified fire.

This process is indicative of a broader trade across the space community. On one hand, there is an ever-increasing demand for more bandwidth. Thousands of years ago, when dinosaurs roamed the plains of North America, cell phones were used only for phone calls. Now they are used for sending full-color images. We may anticipate that they will soon be used for 3-D, high-resolution, video. On the other hand, while our firefighter in the field may want to see a high-resolution picture so that they can interpret for themselves what actions are most appropriate, it is unlikely that they really want a continuous 3-D movie of the fire. They're pretty busy doing other things. Somehow we need to determine what the right amount of information is and how to get it to the end user.

Although the problem is very different, much the same analysis applies to SCS. Here we are not creating data, only transmitting it. What we need to transmit is the amount of data that the end user and the command post want to send, which may vary greatly, depending on the circumstances.

Central vs. Distributed Processing Trade. This is a relatively new issue, because most older spacecraft did not have sufficient processing capability to make this a meaningful trade. However, as discussed above, the situation has changed. The common question now is, "how many computers should the spacecraft have?" Typically, weight and parts-count-conscious engineers would like to avoid distributed processing. However, centralized processing can make integration and test extremely difficult. Because integration and test of both software and hardware may drive cost and schedule, we must seriously consider them as part of the processing trade.

Our principal recommendations in evaluating central vs. distributed processing are:

- Group like-functions together
- Group functions where timing is critical in a single computer
- Look for potentially incompatible functions before assigning multiple functions to one computer
- Maintain the interface between groups and areas of responsibility outside of the computer
- Give serious consideration to integration and test before grouping multiple functions in a single computer

Grouping like-functions has substantial advantages. For example, attitude determination and attitude control may well reside in the same computer. They use much of the same data, share common algorithms, and may have time-critical elements. Similarly, orbit determination and control could reasonably reside in a single navigation computer, together with attitude determination and control. These hardware and software elements are likely to be the responsibility of a single group and will tend to undergo common integration and testing.

In contrast, adding payload processing to the computer doing the orbit and attitude activities could create major problems. We can't fully integrate software and hardware until after we have integrated the payload and spacecraft bus. In addition, two different groups usually handle the payload and spacecraft bus activities. The design and manufacture of hardware and software may well occur in different locations following different approaches. Putting these functions together in a single computer greatly increases cost and risk during the integration and test process, at a time when schedule delays are extremely expensive.

Another problem which can arise from time-to-time is incompatible functions, that is, activities which do not work well together. One example would be sporadic, computationally-intensive functions which demand resources at the same time. Another example occurs when the initial processing of either spacecraft bus or payload sensors may well be an interrupt-driven activity in which the computer is spending most of its time servicing interrupts to bring in observation data. This could make it difficult for the same computer to handle computationally-intensive processing associated with higher-level activities. This can be accommodated either by having the functions handled in separate computers or using a separate I/O processor to queue data from the process with a large number of interrupts.

Finally, we must consider the groups who oversee different activities. Integration and test of any computer and its associated software will be much more difficult if two

distinct groups develop software for the same computer. In this case, significant delays and risks can occur. This does not necessarily mean, however, that elements controlled by different groups cannot be accommodated in the same computer. One approach might be to have two engineering groups be responsible for development of specifications and ultimately for testing. The detailed specifications are then handed over to a single programming group, which then implements them in a single computer. This allows a single group to be responsible for control of computer resources. Thus, for example, the orbit control and attitude control functions may be specified and tested by different analysis groups. However, it may be reasonable to implement both functions in a single computer by a single group of programmers.

4.2.2 Tasking, Scheduling, and Control

Tasking, scheduling, and control is the other end of the data-delivery problem. If the purpose of our mission is to provide data or information, how do we decide what information to supply, whom to send it to, and where to obtain it? Many of the issues are the same as in data delivery but with several key differences. Usually, tasking and control involve very low data rates and substantial decision-making. Thus, we should emphasize how planning and control decisions are made rather than data management.

Tasking and scheduling typically occur in two distinct time frames. *Short-term tasking* addresses what the spacecraft should be doing at this moment. Should SCS be recharging its batteries, sending data to a ground station, turning to look at a disaster relief area, or simply looking at the World below waiting for someone to start sending data? In contrast, *long-term planning* establishes general tasks the system should do. For example, in some way the FireSat II system must decide whether to concentrate its resources on northwestern Pacific forests for several weeks before beginning to look systematically at forests in Brazil. During concept exploration, we don't need to know precisely how these decisions are made. We simply want to identify them and know broadly how they will occur.

On the data distribution side, direct downlink of data works well. We can process data on board, send it simultaneously to various users on the ground, and provide a low-cost, effective system. On the other hand, direct distributed control raises serious problems of tasking, resource allocation, and responsibility. The military community wants distributed control so a battlefield commander can control resources to meet mission objectives. For FireSat II and SCS, this would translate into the local commanders deciding how much resource to apply in a particular area. The two problems here are the limited availability of resources in space and broad geographic coverage. For example, both FireSat II and SCS may have limited power or data rates. In either case, if one regional office controls the system for a time, they may use most or all of that resource, such that other users would have very little left. Also, both systems could be in a position to cover different areas of interest at the same time. Distributed control can create conflicts that are in need of a process for resolution.

For most space systems, some level of centralized control is probably necessary to determine how to allocate space resources among various tasks. Within this broad resource allocation, however, we may have room for distributed decisions on what data to collect and make available, as well as how to process it. For example, the remote fire station may be interested in information from a particular spectral band that could provide clues on the characteristics of a particular fire. If this is an appropriate option, the system must determine how to feed that request back to the satellite. We could use a direct command, or, more likely, send a request for specific data to mission operations, which then carries out the request.

Note that this problem with local control arises largely from the fact that space resources are inherently expensive, so we are trying to optimize the use of the resource. It is rather like asking who should be allowed to use the government's car. That's a reasonable question if each car costs a significant fraction of the budget of a government organization. If cars become more economical, it's reasonable to have a car assigned to a particular person, although others may use it from time to time. If spacecraft become less expensive then we could reasonably assign different space resources to individual military commanders who can then depend on them being available to meet their mission needs. In this model, distributed control by individual commanders may be a good choice.

Spacecraft Autonomy. Usually, high levels of autonomy and independent operations occur in the cheapest and most expensive systems. The less costly systems have minimal tasking and control simply because they cannot afford the operations cost of deciding what needs to be done. Most often, they continuously carry on one of a few activities, such as receiving and relaying radio messages or continuously transmitting an image of what is directly under the spacecraft. What is done is determined automatically on board to save money. In contrast, the most expensive systems have autonomy for technical reasons, such as the need for a very rapid response (missile detection systems), or a problem of very long communications delays (interplanetary missions). Typically, autonomy of this type is extremely expensive because the system must make complex, reliable decisions and respond appropriately to changes.

Autonomy can also be a critical issue for long missions and for constellations, in which cost and reliability are key considerations. For example, long-duration orbit maneuvers may use electric propulsion, which is highly efficient, but slow. (See Chap. 18 for details.) Thruster firings are ordinarily controlled and monitored from the ground, but electric propulsion maneuvers may take several months. Because monitoring and controlling long thruster burns would cost too much, electric propulsion requires some level of autonomy.

Table 4-7◀, Fig. 4-3◀, Eq. 4-0

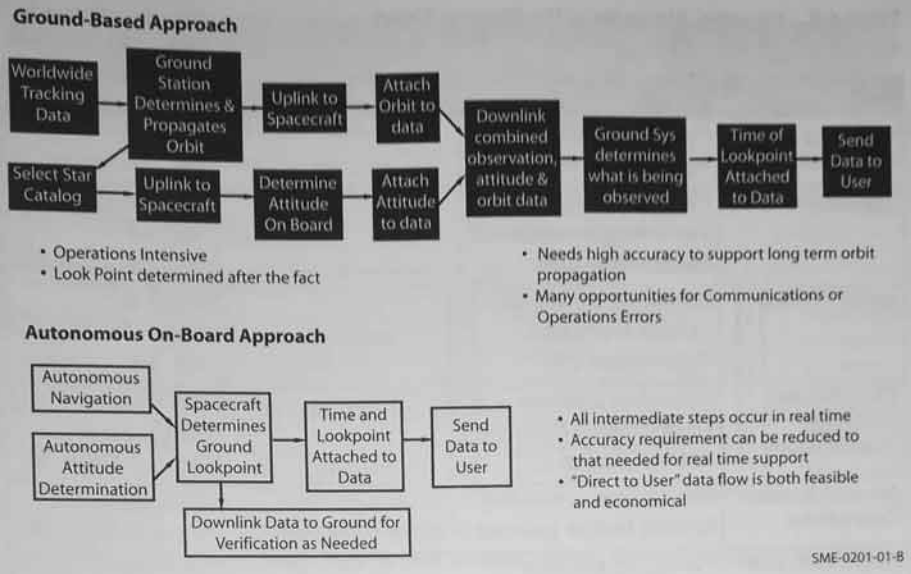


Fig. 4-4. Comparison of Traditional vs. Autonomous Approach to Satellite Navigation. In some circumstances, use of autonomous operations may significantly reduce mission complexity and thereby increase reliability and reduce cost.

As shown in Fig. 4-4, autonomy can add to mission reliability simply by reducing the complexity of mission operations. We may need to automate large constellations for higher reliability and lower mission operations costs. Maintaining the relative positions between the satellites in a constellation is routine but requires many computations. Thus, it is at least possible that on-board automation—with monitoring and operator override if necessary—will give us the best results.

With the increased level of on-board processing available, it is clearly possible to create fully autonomous satellites. The question is, should we do so or should we continue to control satellites predominantly from the ground?

Three main functions are associated with spacecraft control: controlling the payload, controlling the attitude of the spacecraft and its appendages, and controlling the spacecraft orbit. Many space payloads and bus systems do not require real-time control except for changing mode or handling anomalies. Thus, both the FireSat II and SCS payloads will probably fly rather autonomously until a command changes a mode or an anomaly forces the payload to make a change or generate a warning. Autonomous, or at least semi-autonomous, payloads are reasonable for many satellites. There are, of course, exceptions such as Hubble Space Telescope, which is an ongoing series of experiments being run by different principal investigators from around the world. In this case, operators control the payload, although we would still use some automatic operations to save money or to make the operator's job easier.

Controlling the attitude of the spacecraft and its appendages is done autonomously on board for nearly all satellites. Controlling the attitude from the ground is too expensive and too risky. The attitude control system on board most spacecraft provides various attitude control modes and can work over extended periods with little or no intervention from the ground.

Ground control has remained strongest in orbit maintenance and control, in which virtually all thruster firings intended to change the orbit are set up and enabled by ground command. This ground control will probably continue whenever large rocket engines produce orbit maneuvers such as when a kick stage moves the satellite from a parking orbit into a geosynchronous transfer orbit. Once in their operational orbit, however, many satellites either leave the orbit entirely uncontrolled or simply maintain the orbit at a given altitude or within a given window. In this case, low-thrust propulsion is both feasible and desirable because it is much less disturbing to the normal spacecraft environment. Low-thrust orbit maneuvers have been used on geosynchronous spacecraft for a long time so normal satellite operations can continue during the course of these stationkeeping maneuvers.

With low-thrust propulsion and current technology, such as GPS, for autonomous navigation, autonomous orbit control is cheap, easy, and inherently less risky than autonomous attitude control. If the attitude control system stops working for even a short time, the spacecraft can have serious problems, including loss of power, loss of control, and pointing of sensitive payloads toward the Sun. In contrast, if we lose low-thrust orbit control for a while, nothing disastrous happens to the spacecraft. The spacecraft proceeds in its orbit drifting slowly out of its assigned position. This is easily detected and corrected by the ground, assuming that the orbit control system didn't fail completely. (For a more extended discussion of autonomous orbit control, see Wertz and Gurevich [2001a, 2001b], and Plam et al. [2008].)

Current satellite technology allows us to have fully autonomous, low-cost satellites. Autonomy can reduce cost and risk while enabling mission operations people to

Table 4-8. Principal Elements of the Mission Timeline. Key milestones in the mission or project timeline can have a significant effect on how the space system is designed and operated.

Element	Typically Driven By	Where Discussed
Planning and Development	Funding constraints System need date	Sec. 3.5 Sec. 3.4
Production	Funding constraints Technology development System need date	Sec. 3.5, Chap. 14 Sec. 1.3 Sec. 3.4
Initial Launch	Spacecraft completion Launch availability System need date	Chap. 14 Chaps. 26, 27 Sec. 3.4
Constellation Build-up (if applicable)	Production schedule Launch availability Satellite lifetime	Sec. 23.5 Chaps. 26, 27 Sec. 3.4, Chap. 30
Normal Mission Operations	Planned operational life Satellite lifetime (planned or failure constrained)	Chap. 29 Sec. 3.4, Chap. 30
Replenishment	Production schedule Failure of on-orbit satellites Launch availability Satellite lifetime (planned or failure constrained)	Sec. 23.5 Chaps. 23, 24 Chaps. 26, 27 Sec. 3.4, Chap. 30
End-of-Mission Disposal	Legal and political constraints Danger to other spacecraft	Chap. 30 Sec. 7.5

do what they do best—solve problems, handle anomalies, and make long-term decisions. I believe that fully autonomous satellites, including autonomous orbit maintenance, will come about over the next decade as lower costs and risks, validated by on-orbit experiments, begin to outweigh the value of tradition.

4.2.3 Mission Timeline

The *mission timeline* is the overall schedule from concept definition through production, operations, and ultimately replenishment, and end-of-mission. It covers individual satellites and the whole system. Table 4-8 shows the mission timeline's main parts and where they are discussed. Notice that two distinct, potentially conflicting, demands can drive planning and production. One is the demand for a particular schedule or time by which the system must be operational. Thus, a mission to Halley's Comet depends on launching a satellite in time to rendezvous with the comet. On the other hand, funding constraints frequently slow the mission and cause schedule gaps that add both cost and further delays. Of course, funding constraints can affect much more than timelines. They can determine whether we will do a mission, as well as its scope.

If the mission involves a constellation of satellites, a key timeline driver is the need to have the full constellation up for most of the satellite's lifetime. If a single satellite will last 5 years and we need a constellation of 50, we'll never get a full constellation with a launch rate of 5 per year. If having the full constellation is important, we must deploy the initial constellation within 20–25% of an individual satellite's lifetime. This schedule allows some margin for almost inevitable delays as difficulties arise during the mission. If the constellation must remain complete, we need to plan for regular replenishment of satellites. We can replenish on a predefined timeline, as satellites wear out or become technically obsolete, or we can respond to on-orbit failures, collisions, or other catastrophic events that "kill" a particular satellite.

Two areas of the mission timeline typically do not receive adequate attention during concept exploration: performance with less than a full set of satellites while building up the constellation, and end-of-life disposal. In a constellation of satellites we would like to increase performance levels as we add satellites. If FireSat II or SCS are constellations, we want to achieve some level of utility with the first satellite launch and more with each added launch until all satellites are in place. As described further in Sec. 10.6, designers of constellations often concentrate only on the full constellation's performance. However, the period of time before the constellation is fully in place can frequently be long and may well be a critical phase, since a large fraction of the funding has been spent, but full capability has not been achieved. Thus, it is particularly important for constellation design to take into account the problem of performance with less than a full set of satellites. In addition, we want graceful degradation, so a satellite failure will still allow strong performance until we replace the failed satellite. These issues are important during concept exploration because they may significantly affect the design of the constellation and the entire system.

There is now growing concern with disposal of satellites after their useful life in orbit. We have already recognized this need for geosynchronous satellites because the geosynchronous ring is rapidly filling. But it

Table 4-8, Fig. 4-4◀, Eq. 4-0

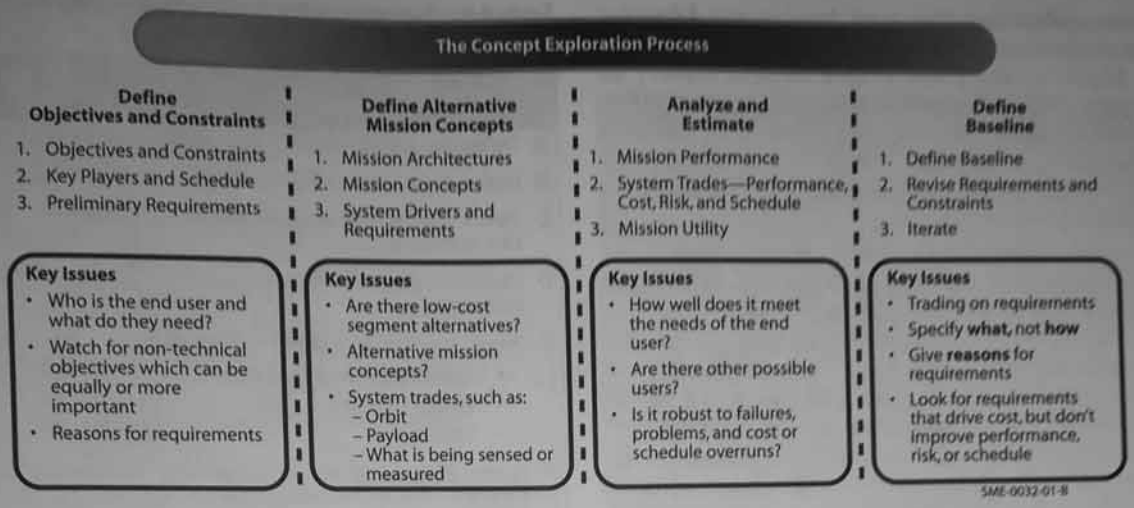


Fig. 4-5. Concept Exploration is the Early Part of the Overall Space Mission Engineering Process shown in Table 3-1. See text for discussion.

is also very important for low-Earth orbit constellations in which debris and spent satellites left in the pattern can threaten the remaining constellation, as occured with the collision between Iridium 33 and Cosmos 2251 in February, 2009. Again, we must address this issue early in concept definition.

4.3 Introduction to Concept Exploration

Concept exploration is the early part of the overall space mission engineering process in which we look at different ways to meet the needs of both the customer and end user. As summarized in Fig. 4-5, this process can be broadly divided into four parts:

- **Defining the objectives and key players** (Chap. 3). Here the key issues are who the key players are, what do they need to accomplish, and what are the reasons for the various requirements that we will create along the way?
- **Defining the alternative mission concepts** (Chap. 4). Here we ask what elements we will bring together to do the mission (the *mission architecture*) and how they will work together to meet the user's needs (the *mission concept*). The key questions here are looking at alternative mission concepts, finding the system drivers and critical requirements, and trying to find low-cost solutions that could do the job better at lower cost.
- **Analyzing the mission and estimating how well it works** (Secs. 5.1 to 5.4). This gets to the heart of the matter by asking in a quantitative way how well we can do what we set out to do. How many lives and how much property can be saved by building FireSat II or how much better can we be at responding to disasters by building SCS?
- **Defining the baseline approach and evaluating alternatives** (Secs. 5.5 and 5.6). Here we want to go back and re-examine the requirements and constraints to see if these are really the way to best meet our overall mission objectives. We also want to look for requirements that drive cost, but don't improve performance or reduce risk or schedule.

Throughout concept exploration a key step is to document the reasons for various choices. The purpose here is to allow ourselves or others to re-examine these choices at a later date. Space mission engineering and concept exploration are highly iterative. We go back many times to look at alternatives and try to find better ways of achieving our objectives. If we don't document the reasons for choices then we have to either assume that they were written on stone tablets and are defined to be correct or redo the analysis on which they were based, neither of which is a good solution.

You may sense a trend in this discussion that we are looking for alternatives, for new ways of doing things better, faster, cheaper, or all three. That's what concept exploration is all about. We may find that our first approach turns out to be the very best way to do the mission. Or we may find alternative approaches or new technologies that can provide a better approach. Airplanes and automobiles are continuously evolving—much of it based on the continuing advances in performance and reduction in mass and cost of electronics. Large numbers of electronic systems started primarily in expensive cars to allow them to do things with more precision and, in many cases, simply wow the buyer or passenger. Now electronic systems can make the car perform better and safer and be lighter and lower cost. Can we use these same electronic systems or approaches to allow a 100 kg spacecraft to do what a 1,000 kg spacecraft used to do? That, of course, depends on what we're asking the spacecraft to do. But most systems built today, from cameras to telephones to automobiles and airplanes are smaller, lower cost, and

more competent than they were decades ago. Advances in microelectronics led to a whole new approach in aviation. Maybe we don't need to have the pilot actually on the airplane. What if we could have the pilot in an office somewhere flying an airplane via remote control? Unmanned aerial vehicles, UAVs, are now commonplace and likely to become much more so as a new idea in aviation gains acceptance.

It isn't just more use of microelectronics, but all aspects of the space mission, that we want to look at and challenge. What can SCS do to make it easier for the soldier in a remote location or a disaster relief worker to get their job done safer and more quickly? Can FireSat II detect forest fires by looking at something other than the heat from the fire? Can it look at the light or smoke or increase in CO_2? Is there a better way to communicate the information to the people who need it? Can we combine the space asset with other assets to do the job better, faster, and cheaper? All of these are a part of concept exploration.

Once we have established alternative mission concepts, architectures, and system drivers, we must further define the mission concepts in enough detail to allow a meaningful comparison of the alternatives. For concept exploration, the steps in this process correspond to the space mission elements. Figure 4-5 above illustrates the sequence of activities. The definition of objectives and constraints was discussed in Chap. 3. Each of the steps in defining alternative mission concepts is described below and summarized in Table 4-9. **Our goal is to develop just enough detail to allow quantitative comparisons between the alternatives.**

A. Define the Preliminary Mission Concept (Chap. 4)

As described in Sec. 4.2, the key elements are data delivery, tasking, scheduling, and control, communications architecture, and mission timeline. We begin with a broad concept and refine this concept as we define the various mission elements in the steps below.

B. Define the Subject Characteristics (Sec. 15.2)

We can divide space missions into two broad categories. Missions with *active* or *controllable subjects* service other system elements, typically run by the operators or users, such as Comsat ground stations or GPS navigation receivers. Missions with *passive* subjects sense elements that are not a part of the mission system, such as the galaxies observed by a space telescope. Our first step in defining the system elements is to determine the subject's key characteristics. If a mission interacts with user equipment, we must define the subject characteristics either from known information for well-established services or by a trade study involving the rest of the system. The parameters for specifying passive subjects are largely the same as those for specifying user elements, except that we don't have a "receiver" to characterize, and the effective isotropic radiated power (EIRP) specification for the transmitter is replaced by the definition of the object's emission intensity as a function of the wavelength. Table 4-10 summarizes the characteristics of both types of elements.

Table 4-10, Fig. 4-5◀, Eq. 4-0

Table 4-9. Summary of the Process for Defining Alternative Mission Concepts. See text for details.

Step	Where Discussed
A Define the preliminary mission concept	Chap. 4
B Define the subject characteristics	Sec. 15.2
C Determine the orbit or constellation characteristics	Chap. 10
D Determine payload size and performance	Chaps. 15, 16, 17
E Select the mission operations approach	
• Communications architecture	Sec. 16.2
• Ground system	Chap. 28
• Operations	Chap. 29
F Design the spacecraft bus to meet payload, orbit, and communications requirements	Chaps. 14, 18–23
G Select a launch and orbit transfer system	Chap. 26
H Determine deployment, logistics, and end-of-life strategies	Chaps. 23, 27, 30
I Provide costing support	Chap. 11
J Look for major alternatives	Chaps. 13, 25
K Document and iterate	Sec. 6.4

C. Determine the Orbit and Constellation Characteristics (Chap. 10)

The mission orbit influences every part of space mission development and operation. Combined with the number of spacecraft, it determines all aspects of space-to-ground and ground-to-space sensor and communication coverage. For the most part, the orbit determines sensor resolution, transmitter power, and data rate. The orbit determines the spacecraft environment and, for military spacecraft, strongly influences survivability. Finally, the orbit determines the size and cost of the launch and delivery system. Chapter 10 gives detailed directions for orbit design. As Table 4-11 shows, the design should include parameters for the mission and transfer orbits, propellant requirements, and constella-

Table 4-10. Characteristics of Space Mission Subjects. See Chap. 16 for definitions of communications parameters.

Controllable Subjects	Passive Subjects
1. Quantity	1. Quantity and Distribution
2. Location or range	2. Location or range
3. Transmitter EIRP	3. Emission intensity (W/sr) as a function of frequency or spectral band
4. Receiver G/T	4. Needed resolution
5. Frequency and bandwidth	5. Needed temporal coverage (duty cycle)
6. Duty cycle	
7. Processing capability	

Table 4-11. Summary of Orbit and Constellation Characteristics for Earth Orbit. See Chaps. 9 and 10 for discussion. See Sec. 10.7 for interplanetary orbits.

1	Altitude	} These are the most important
2	Inclination	
3	Eccentricity	
4	Argument of perigee for noncircular orbits	
5	ΔV budget for orbit transfer	
6	ΔV budget for orbit maintenance	
7	Whether orbit will be controlled or uncontrolled	
8	Number and relative orientation of orbit planes (constellations)	
9	Number and spacing of spacecraft per orbit plane (constellations)	
10	Collision avoidance and debris mitigation strategy	

tion characteristics. As space becomes more crowded, a key, but often overlooked, element is designing constellations for collision avoidance and debris mitigation. The problem is discussed in detail in Sec. 7.5 and possible solutions are discussed in Sec. 13.1.1 (See Fig. 13-2).

D. Determine the Payload Size and Performance (Chaps. 15, 16, and 17)

We next use the subject characteristics from Step B and orbit characteristics from Step C to create a mission payload concept. We can divide most mission payloads into six broad categories: observation or sensing, communications, navigation, in situ sampling and observations, sample return, and crew life support and transportation. More than 90% of current space-system payloads observe, sense, or communicate. Even navigation payloads are basically communications payloads with ancillary data processing and stable time-base equipment to provide the navigation signal. Detailed directions for sizing and definition appear in Chap. 16 for communications payloads and in Chap. 17 for observation payloads. Table 4-12 summarizes the key parameters we need to specify.

System-level payload trades typically involve the user element, selecting a mission orbit, and allocating pointing and tracking functions between the payload and spacecraft elements. User element trades involve balancing the performance of the payload and user equipment to get the lowest overall system cost for a given orbit and constellation design. As an example, if a single geosynchronous spacecraft must service thousands of ground stations, such as direct broadcast TV, we would minimize the system cost by selecting a large, powerful spacecraft that can broadcast to simple and inexpensive ground terminals. A system designed for high bandwidth communication between half a dozen ground stations uses more complex and capable ground systems and saves cost with a simpler spacecraft.

Payload vs. orbit trades typically try to balance the resolution advantages of low altitude against the fewer

Table 4-12. Summary of Typical Payload Characteristics. For multiple payloads, we must determine parameters for each payload.

1. ***Physical Parameters***
   - 1.1 Envelope dimensions
   - 1.2 Mass properties
   - 1.3 Moving parts
2. ***Viewing and Pointing***
   - 2.1 Aperture size and shape
   - 2.2 Size and orientation of clear field of view required
   - 2.3 Primary pointing direction*
   - 2.4 Pointing direction range and accuracy required
   - 2.5 Tracking or scanning rate
   - 2.6 Pointing or tracking duration
3. ***Electrical Power***
   - 3.1 Voltage
   - 3.2 Average and peak power
   - 3.3 Standby, operating
4. ***Operations***
   - 4.1 Operating modes
   - 4.2 Duty cycle
   - 4.3 Need for eclipse operations
5. ***Telemetry and Commands***
   - 5.1 Number of command and telemetry channels
   - 5.2 Command memory size and time resolution
   - 5.3 Data rates or quantity of data
6. ***Thermal Control***
   - 6.1 Temperature limits (operating/non-operating)
   - 6.2 Heat rejection to space (average and peak, duty cycle)

*e.g., Sun, stars, nadir, ground target, another spacecraft

spacecraft needed for the same coverage at higher altitudes. At a higher altitude, we need a sensor with a larger aperture and better sensitivity to obtain the same resolution; the more capable sensor both costs more and needs a larger spacecraft and launch system.

Payload vs. spacecraft trades usually try to meet pointing and tracking requirements at the lowest cost. At one extreme, the payload does all the pointing independently of the spacecraft attitude; an example is the use of gimballed scan platforms on the Mariner MK-II spacecraft. At the opposite extreme, Space Telescope and Chandra X-Ray Observatory point the entire spacecraft with the required level of accuracy. An intermediate approach is to point the entire spacecraft to a lower level of accuracy, allowing the payload to do fine pointing over a limited field of regard.

E. Select the Mission Operations Approach: Communications Architecture, Ground System, and Operations (Sec. 16.2, Chaps. 28 and 29)

We next select and size the elements needed to support communications and control of the spacecraft and

payload. Table 4-13 gives the key parameters. Typically a *mission operations control center* commands and controls the spacecraft and delivers data to the user. With rare exceptions, we would choose an existing control center, based on the user's needs, downlink data rates, and, in some cases, security considerations. Both NASA and the Air Force have existing systems. Some institutions, such as Intelsat, use custom systems. Most commercial operators employ system-specific control centers. If needed, we can interconnect most systems with different options for relaying communications.

Table 4-13. Summary of Mission Operations Characteristics.

1. ***Communications Architecture***
   - 1.1 Number and distribution of ground stations
   - 1.2 Downlink and uplink path design
   - 1.3 Crosslink characteristics, if used
   - 1.4 Relay satellites used
   - 1.5 Communications link budget
   - 1.6 Space-to-ground data rates
2. ***Ground System***
   - 2.1 Use of existing or dedicated facilities
   - 2.2 Use of facilities
   - 2.3 Required transmit and receive characteristics
   - 2.4 Required data handling
3. ***Operations***
   - 3.1 Level of automation
   - 3.2 Software to be created
   - 3.3 Full-time or part-time staffing
   - 3.4 Number of personnel
   - 3.5 Amount of commanding required
   - 3.6 Getting the data to the end user
   - 3.7 Timeliness of data distribution

The *communications architecture* transfers the required mission data (payload and housekeeping data) from the spacecraft down to the mission operations control center. In addition, we must send commands back to the spacecraft, and meet other requirements such as encryption. Thus, we select the communications relay elements along with the mission control system after most payload and orbit trades are complete. Typical options are SGLS for Air Force missions or TDRSS, DSN, or STDN with the NASA mission control centers. Custom systems are required for some applications and are commonly used for commercial missions in geosynchronous orbit. Chapter 16 describes communications architectures, and Chap. 29 treats operations.

F. Design the Spacecraft Bus to Meet Payload, Orbit, and Communications Requirements (Chaps. 14, 18 to 23)

The spacecraft and its subsystems support the payload in the mission orbit—point it and supply power, commands, data handling, and thermal control. They must be compatible with the communications architecture and mission-operations concept. These elements, along with the launch system, drive the spacecraft design. We usually choose the launch system that costs the least to place the minimum required weight in the mission or transfer orbit. Once we make this selection, the spacecraft's stowed configuration is constrained by the shroud volume of the selected vehicle or vehicles. Table 4-14 summarizes the items we need to specify while defining the spacecraft. Chapter 14 covers how we synthesize spacecraft concepts and their definition and sizing.

A key spacecraft vs. launch system trade is the use of integral propulsion. Many commercial spacecraft ride the launch system to transfer orbit and then insert themselves into the mission orbit using integral propulsion or an internal stage. Some spacecraft depend on a launch system with an upper stage for insertion directly into the mission orbit. They do not carry large integral propulsion subsystems.

Table 4-14. Summary of Spacecraft Characteristics. See Chapter 14 for discussion.

1. **General arrangement including payload fields of view (deployed and stowed)**
2. **Functional block diagram**
3. **Mass properties, by mission phase (mass and moments of inertia)**
4. **Summary of subsystem characteristics**
   - 4.1 ***Power:*** conversion approach; array and battery size; orbit average and peak payload power; orbit average and peak overall spacecraft power
   - 4.2 ***Attitude control:*** attitude determination and control components; operating modes; ranges and accuracy
   - 4.3 ***Navigation and orbit control:*** accessing requirements; use of GPS; on-board vs. ground
   - 4.4 ***Telemetry and command:*** command/telemetry format; command and time resolution; telemetry storage capacity; number of channels by type
   - 4.5 ***Computer:*** speed and memory; data architecture
   - 4.6 ***Propulsion:*** amount and type of propellant; thruster or motor sizes
   - 4.7 ***Communications:*** link margins for all links; command uplink data rate; telemetry downlink data rates
   - 4.8 ***Primary structure and deployables***
   - 4.9 ***Unique thermal requirements***
   - 4.10 ***Timing:*** resolution and accuracy
5. **System parameters**
   - 5.1 Lifetime and reliability
   - 5.2 Level of autonomy
   - 5.3 End of life and system disposal

Another trade between the spacecraft and launch system involves guidance of the upper stage. Often, the spacecraft control system can guide the upper stage, which may allow deletion of equipment from that stage, thereby increasing performance and lowering cost. This trade is particularly important when using 3-axis-stabilized stages.

G. Select a Launch and Orbit Transfer System (Chap. 26)

The launch system and its upper stage need to deliver the spacecraft to the mission orbit or to a transfer orbit from which the spacecraft can reach the mission orbit on its own. The chosen launch system usually determines the launch site. The launch site organization provides prelaunch processing, checkout, and installation to the launch vehicle, often on the launch pad as described in Chap. 27.

Launch vehicles and upper stages may be combined in many ways to match almost any combination of payload and mission orbit. Chapter 26 details the characteristics and selection of launch systems. Selecting a launch system typically involves the trades with the spacecraft discussed above. In addition, we must decide between a single spacecraft launch and manifesting two or more spacecraft in a shared launch. In general, multiple manifesting costs less, but constrains the schedule. Finally, we should bring some launch system parameters to the system level design process: type of vehicle, cost per launch, and flow times for processing and prelaunch activities at the launch site.

H. Determine Logistics, Deployment, Replenishment, and Spacecraft Disposal Strategies (Chaps. 23, 27, and 30)

Logistics is the process of planning to supply and maintain the space mission over time. Whereas only military missions typically demand formal plans, the process described in Chap. 23 can strongly affect costs for any multi-year mission requiring extended support. Historically, most life-cycle costs have been locked in by the end of concept exploration, so we must evaluate operations, support, and replenishment during this phase.

Planners often overlook the sequence for building up and maintaining satellite constellations. To deploy a constellation effectively, we must create performance plateaus which allow us to deploy in stages and to degrade the system gracefully if individual satellites fail. These performance plateaus develop from the constellation design, as described in Sec. 10.6.

Section 7.5 describes the ever-increasing problem associated with *orbital debris*, consisting of defunct satellites and associated parts. Because of this problem, all new satellite designs should plan for de-orbiting or otherwise disposing of satellites at the end of their useful life, as discussed in Chap. 30. In particular, satellites must be removed from areas such as the geostationary ring, where they would seriously threaten other spacecraft, and from any low-Earth orbit constellation.

I. Provide Costing Support for the Concept-Definition Activity (Chap. 11)

Developing costs for system elements is vital to two objectives: finding the best individual mission architecture and comparing mission architectures at the system level. Chapter 11 describes parametric, analogous, and bottoms-up methods for costing. Typically, for concept exploration, we use only the first two because we lack a detailed definition of the design. At this level, we simply want relative comparisons rather than absolute estimates, so we can accept the greater uncertainty in these methods.

J. Look for Major Alternatives (Chaps. 13 and 25)

As discussed previously in Step D, a key characteristic of reducing mission cost is to look for major alternatives for one of the principal mission elements. Iridium, for example, was the first communications satellite system to choose a low Earth orbit, where all previous communications satellites had been in GEO or in a Molniya highly elliptical orbit. The GPS constellation changed the nature of navigation by having the computational work done, along with the maps, in the GPS receiver that the user has. Table 4-15 gives a few examples of potential alternatives for the mission elements. It is in this step that the creativity of the mission designer is the most important and where innovative solutions to improving performance or reducing cost, risk, or schedule are created.

Table 4-15. Representative Examples of Major Alternatives for the Mission Elements. See Chap. 13 for a detailed discussion.

	Mission Element	Representative Alternatives
A	*Mission Concept*	Combine with ground information or another space mission, use suborbital flight, use data from an existing mission, process data on board and send only the results.
B	*Controllable Subject*	Use existing equipment not intended for this mission (cell phone or GPS receiver).
C	*Passive Subject*	Look for alternative features to sense or observe. Use secondary features or data from another system.
D	*Payload*	Visible, IR, multi-spectral. Use existing equipment built for other purposes.
E	*Spacecraft Bus*	Use small spacecraft or CubeSat components. Use a fractionated spacecraft.
F	*Launch Vehicle*	Use integral propulsion or low thrust to fly the spacecraft to its final orbit.
G	*Orbit*	Use very low altitude LEO, medium altitude, or Molniya.
H	*Ground System*	Use commercial ground station or AMSAT for science missions.
I	*Communications Architecture*	Use commercial network. Use Internet for both data delivery and commanding.
J	*Mission Operations*	PC-based. Omit standard ground operations.

K. Document and Iterate.
As always we need to document our conclusions (and the reasons for them) and then go back and look again.

4.4 Defining System Drivers and Critical Requirements (Step 7)

System drivers are the principal mission parameters or characteristics which influence performance, cost, risk, or schedule **and** which the user or designer can control. For example, the temperature at which a forest fire burns may heavily influence how easily it can be detected; however, this is beyond our control and, therefore, is not a system driver. Correctly identifying the system drivers is a critical step in mission analysis and design. Misidentifying system drivers is one of the most common causes of mission analysis error. For example, we may focus a great deal of time and effort on getting the most coverage for an orbit when the system's ultimate performance depends mainly on data rates or cloud cover.

Table 4-16 lists the most common system drivers for space missions, along with what limits them, what they limit, and where they are discussed. The table helps us ensure that we do not overlook system drivers. In identifying these drivers we must clearly determine whether we are looking for drivers of *performance*, *cost*, *risk*, or *schedule*. These may be the same or different. To identify system drivers, we:

1. **Identify the Area of Interest.** Explicitly identify the area of interest, typically performance, cost, risk, or schedule.
2. **Identify Parameters Which Measure the Area of Interest.** Define numerical parameters which measure the identified area of interest. (See Sec. 5.3 on measures of effectiveness and performance parameters for more details on how to do this.) The important point is to find parameters that genuinely measure the goal rather than ones which are simply easy to compute.
3. **Develop First-Order Algorithms.** Develop a formula or algorithm to express the first-order estimate for the value of the parameter identified above. This could include either system algorithms as defined in Sec. 5.3.2 (Table 5-5), or unique algorithms for the identified parameter. (See Table 4-17 for the FireSat II example.)
4. **Examine the Factors.** Examine each of the factors in the expression identified in Step 3. Those which can be adjusted and which have the strongest effect on results are the system drivers.
5. **Look for Possible "Hidden Drivers."** Examine each of the first-order algorithms for implicit variables or for factors affecting more than one characteristic. For example, altitude will influence the ground resolution of a given instrument, the area covered by the field of view, and the spacecraft's velocity relative to the Earth. Therefore, it will more strongly influence effective area search rates than a single formula may show.

The way we have defined our particular problem, or which parameters are available to us, may affect our list of system drivers. Thus, defining system drivers depends in part on the physical and technical nature of the problem and in part on the constraints imposed on the mission design and the solution we have chosen. Usually, we

Table 4-16. Common System Drivers. System drivers can frequently be identified by examining the parameters in this list.

Driver	What Limits Driver	What Driver Limits	Where Discussed
Size	Shroud size, available weight, aerodynamic drag	Payload size (frequently antenna diameter or aperture)	Chaps. 14, 15
On-orbit Weight	Altitude, inclination, launch vehicle	Payload weight, largely determines design and manufacturing cost, propellant load	Chaps. 14, 15
Power	Size, weight (control is secondary problem)	Payload & bus design, system sensitivity, on-orbit life, data rates	Secs. 14.5, 21.2
Propellant	Available mass	Delta V, flexibility, on-orbit responsiveness	Secs. 14.5, 18.1
Data Rate	Storage, power, antenna sizes, limits of existing systems	Information sent to user, can push demand for on-board processing	Secs. 16.3, 21.1
Communications	Coverage, availability of ground stations or relay satellites	Timeliness, ability to command	Sec. 21.1, Chap. 16
Pointing	Cost, weight	Resolution, geolocation, overall system accuracy; pushes spacecraft cost	Secs. 14.5, 19.1
Number of Spacecraft	Cost	Coverage frequency and overlap	Sec. 10.6
Altitude	Launch vehicle, performance demands, weight, radiation environment	Performance, survivability, coverage (instantaneous and rate), communications, resolution	Chap. 10
Coverage (geometry and timing)	Orbit, scheduling, payload field of view and observation time	Data frequency and continuity, maneuver requirements	Sec. 10.2
Scheduling	Timeline and operations, decision making, communications	Coverage, responsiveness, mission utility	Secs. 3.4, 4.2, Chap. 29
Operations	Cost, crew size, communications	Frequently principal cost driver, principal error source, pushes demand for autonomy; can also save "lost" missions	Chap. 29

Table 4-17. Identification of Performance Drivers for FireSat II. First-order algorithms are given to allow us to estimate the performance drivers. Definition of performance drivers may change as we create more detailed definitions of the system and system algorithms. Comparison of columns two and three shows that the performance drivers may depend on the mission concept used.

Key Parameters	First Order Algorithm (Low-Earth Orbit)	First Order Algorithm (Geosynchronous)	Performance Drivers
Observation Frequency	(Number of spacecraft)/12 hr	Scan frequency	Number of spacecraft for low orbit
Time Late	Onboard storage delay + processing time	Communications + processing time	Storage delay (if applicable)
Resolution	Distance × [(wavelength/ aperture) + control error]	Distance × [(wavelength/ aperture) + control error]	Altitude, aperture, control accuracy
Observation Gap	Cloud cover interval or coverage gap	Cloud cover interval or coverage gap	None (weather dominated)

want to make these constraints explicit, so we will know which variables are available for adjustment and which are assumed to be given. Table 4-17 shows the major performance drivers for FireSat II.

System drivers are the defining mission parameters that most strongly affect performance, cost, risk, and schedule. Similarly, *Critical requirements* (also called *driving requirements* or *key requirements*) are the mission requirements that dominate the space mission's overall design and, therefore, most strongly affect performance, cost, risk, or schedule. For a human mission to Mars, the critical requirements will be clear: get to Mars all of the required mass to explore the planet and return, and maintain crew safety for a long mission in widely varying environments. For less ambitious space missions, we cannot establish the critical requirements so easily. Because we want to achieve the best performance at minimum cost, we need to identify these key requirements as early as possible so they can be a part of the trade process. The goal of mission engineering is to adjust both the critical requirements (e.g., coverage and resolution) and the system drivers (e.g., altitude and aperture) to satisfy the mission objectives at minimum cost and risk.

Table 4-18 lists the most common critical requirements, the areas they typically affect, and where they are discussed. There is no single mechanism to find the critical requirements for any particular mission. Like the system drivers discussed above, they may be a function of the mission concept selected. Consequently, once we establish the alternative mission concepts, we usually can determine the critical requirements by inspection. Often, concept exploration itself exposes the requirements that dominate the system's design, performance, and cost. One approach to identification of critical requirements is as follows:

1. **Look at the Principal Performance Requirements.** In most cases, the principal performance requirement will be one of the key critical requirements. Thus, for FireSat II, the requirements on how well it must detect and monitor forest fires would normally be principal drivers of the system design. For SCS, the need to work with existing equipment will likely be a critical requirement.

2. **Examine Table 4-18.** The next step is to look at the requirements list in Table 4-18 and determine which of these entries drive the system design, performance, or cost.

Table 4-18. Most Common Critical Requirements. See text for discussion.

Requirement	What it Affects	Where Discussed
Coverage or Response Time	Number of satellites, altitude, inclination, communications architecture, payload field of view, scheduling, staffing requirements	Secs. 10.2, 10.6, Chap. 29
Resolution	Instrument size, altitude, attitude control	Sec. 17.1
Sensitivity	Payload size and complexity, processing, thermal control, altitude	Chaps. 16, 17
Mapping Accuracy	Attitude control, orbit and attitude knowledge, mechanical alignments, payload precision, processing	Sec. 8.6
Signal Strength	Payload size and power, altitude, transmit power	Chap. 16
On-orbit Lifetime	Redundancy, weight, power and propulsion budgets, component selection	Secs. 3.4, 9.6, Chap. 30
Survivability	Altitude, weight, power, component selection, design of space and ground system, number of satellites, number of ground stations, communications architecture, system redundancy	Sec. 7.5

3. **Look at Top-level Requirements.** Examine each of the top-level requirements established when we defined the mission objectives (Sec. 3.3) and determine how we will meet them. For each, ask whether or not meeting that requirement fundamentally limits the system's design, cost, performance, risk, or schedule.
4. **Look for Hidden Requirements.** In some cases, hidden requirements such as the need to use particular technologies or systems may dominate the mission design.

For most FireSat II approaches, resolution and coverage are the principal critical requirements, and we could find them by any of the first three approaches listed above. The critical requirements depend on a specific mission concept. For the low-cost FireSat II of Chap. 13, they are coverage and sensitivity. For an alternative system design, that may not be the case. For SCS, we will be dominated by either the desired bandwidth or the ground equipment that we're forced to deal with to make the system workable on the ground.

References



For annotated bibliography of mission concept definition and exploration, see the book website.

Elbert, Bruce R. 2003. *The Satellite Communication Applications Handbook.* Norwood, MA: Artech House.

Elbert, Bruce R. 2008. *Introduction to Satellite Communication. 3rd ed.* Norwood, MA: Artech House.

Fortescue, Peter W., John Stark, and Graham Swinerd. 2003. *Spacecraft Systems Engineering. 3rd ed.* Chichester, UK: John Wiley and Sons.

Gordon, Gary D. and Walter L. Morgan. 1993. *Principles of Communication Satellites.* New York, NY: Wiley Interscience.

Griffin, M. and J. R. French. 2004. *Space Vehicle Design.* Washington, DC: AIAA.

Kemble, Stephen. 2006. *Interplanetary Mission Analysis and Design*, with software disk. Chichester, UK: Praxis Publishing, Ltd. And Springer.

Lu, Ning. 1997. *Fractal Imaging.* San Diego, CA: Academic Press.

Maral, Gerard, and Michel Bousquet. 2010. *Satellite Communications Systems: Systems, Techniques and Technology. 5th ed.* Chichester, UK: John Wiley and Sons.

Pisacane, Vincent L. and Robert C. Moore. 2005. *Fundamentals of Space Systems. 2nd ed.* Oxford: Oxford University Press.

Plam, Yegor, Richard Van Allen, James R. Wertz, and Thomas Bauer. 2008. "Autonomous Orbit Control Experience on TacSat-2 using Microcosm's Orbit Control Kit (OCK)." 31st Annual AAS Guidance and Control Conference. AAS 08-008. Breckenridge, CO. Feb. 1–6.

Sayood, Khalid. 2005. *Introduction to Data Compression. 3rd ed.* San Francisco, CA: Morgan Kaufman Publishers.

Wertz, James R. and Gwynne E. Gurevich. 2001a. "Applications of Autonomous On-Board Orbit Control." 11th AAS/AIAA Space Flight Mechanics Meeting. AAS 01-238. Santa Barbara, CA. Feb. 11–15.

Wertz, James R. and Gwynne E. Gurevich. 2001b. "Autonomous On-board Orbit Control: Flight Results and Cost Reduction." JHU/APL Symposium on Autonomous Ground Systems for 2001 and Beyond. Laurel, MD. Apr. 25–27.

5 Mission Analysis and Mission Utility

James R. Wertz, *Microcosm, Inc./USC*

5.1 Introduction to Mission Analysis
The Mission Analysis Hierarchy

5.2 Studies with Limited Scope
Feasibility Assessment; Sizing Estimate; Point Design

5.3 System Trade Studies and Performance Assessments (Step 8)
System Trade Studies; Performance Assessments

5.4 Mission Utility and Figures of Merit—Is the Mission Worthwhile? (Step 9)
Performance Parameters and Measures of Effectiveness; Evaluating Mission Utility

5.5 Defining the Baseline Mission Concept, Revising Requirements, and Evaluating Alternatives (Steps 10–12)

5.6 Examples: FireSat II and SCS

5.7 Deciding Whether a Mission Should Proceed

Chapter 4 discussed defining and characterizing alternative concepts and architectures for space missions. This chapter shows how we evaluate the ability of these options to meet fundamental mission objectives. We address *mission analysis*—how to quantify mission performance, *mission utility*—how to evaluate its utility to the customer or end user, and *mission selection*—how to select one or more concepts for further development or to decide that we cannot achieve the mission within current constraints or technology.

The narrow purpose of mission evaluation is to determine whether the system meets all of the requirements and specifications set for it. This is critically important (particularly to Congress or a group of investors who are putting up the large amounts of money that space missions normally require) and is largely the subject of Chap. 6. However, in the broad context of mission engineering, we are really concerned with two issues:

1. How well does the system perform (*performance analysis*)
2. How well does it meet the customer's or end user's objectives (*mission utility analysis*)

The distinction between performance and utility can be exceptionally important. This is clearly illustrated by two historical examples—the Space Shuttle and the Iridium LEO Communications Constellation.

The original goal of the Space Shuttle was to achieve "airplane-like" access to space and to drive down the launch cost to less than $10 million/flight in FY1972$, corresponding to roughly $50 million/launch in FY2010$. The actual cost per launch is not well-defined, but is likely well over $1 billion/launch, not counting any amortization of the original development cost. Irrespective of the exact costs, it is clear that the Shuttle was never able to achieve the dramatic cost reductions that were promised. As discussed in Sec. 2.1.7 and Chap. 13, high launch costs remain a major impediment to the large-scale expansion of the exploration and use of space.

Similarly, the Iridium LEO communications constellation declared bankruptcy in 2000, representing a loss of approximately $5.5 billion—most likely from a group of people that would have preferred to not lose that money. As discussed in Sec. 2.1.4, this represented a loss for the astronautics community as a whole. It indirectly increased costs for nearly all other programs and made it difficult or impossible to fund other constellations.

The important issue is that both Iridium and the Space Shuttle largely met their performance requirements.* (The Shuttle program had two failures resulting in loss of life, which represents a major and significant programmatic failure, even though we know that human spaceflight carries risks that can not realistically be avoided.) Both programs were exceptionally well-engineered, and were built by some of the best engineers and space organizations in the world. Nonetheless, while the performance assessment was very good, they failed to meet their fundamental mission objectives, i.e., to achieve an adequate level of mission utility. We, as a community, did not do what we should have done. If a surgeon performs a remarkable operation, but the patient dies, they should not take comfort in the great work that

* For the Space Shuttle the performance requirements are things like payload mass to orbit, injection accuracy, and loiter time on orbit, whereas the mission objective was to achieve "airplane-like" performance and dramatically drive down launch costs. For Iridium the performance requirements were availability, number of telephone calls supported, and ability to successfully complete a call. The objective was simple—to make money for the people who put up the initial investment.